



US 20250283112A1

(19) **United States**

(12) **Patent Application Publication**
Hsu et al.

(10) **Pub. No.: US 2025/0283112 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **COMPOSITIONS AND METHODS FOR RNA EDITING**

Publication Classification

(71) Applicant: **The Regents of the University of California, Oakland, CA (US)**

(72) Inventors: **Patrick Hsu, Berkeley, CA (US); Sita Chandrasekaran, Berkeley, CA (US); Cyrus Tau, Berkeley, CA (US)**

(21) Appl. No.: **18/863,599**

(22) PCT Filed: **May 11, 2023**

(86) PCT No.: **PCT/US2023/066872**

§ 371 (c)(1),

(2) Date: **Nov. 6, 2024**

Related U.S. Application Data

(60) Provisional application No. 63/341,823, filed on May 13, 2022.

(51) **Int. Cl.**

C12N 15/87 (2006.01)

C12N 9/22 (2006.01)

C12N 15/113 (2010.01)

(52) **U.S. Cl.**

CPC ***C12N 15/87*** (2013.01); ***C12N 9/226*** (2025.05); ***C12N 15/113*** (2013.01); ***C12N 2310/14*** (2013.01); ***C12N 2310/20*** (2017.05); ***C12N 2310/3519*** (2013.01)

(57)

ABSTRACT

The present disclosure provides a hybrid RNA molecule comprising a targeting region and a donor RNA, and compositions comprising the hybrid RNA molecule. A hybrid RNA molecule of the present disclosure is useful in methods of modifying a target RNA, which methods are also provided.

Specification includes a Sequence Listing.



Gene



Transcription



pre-mRNA



cis Splicing



mRNA

FIG. 1A-1B

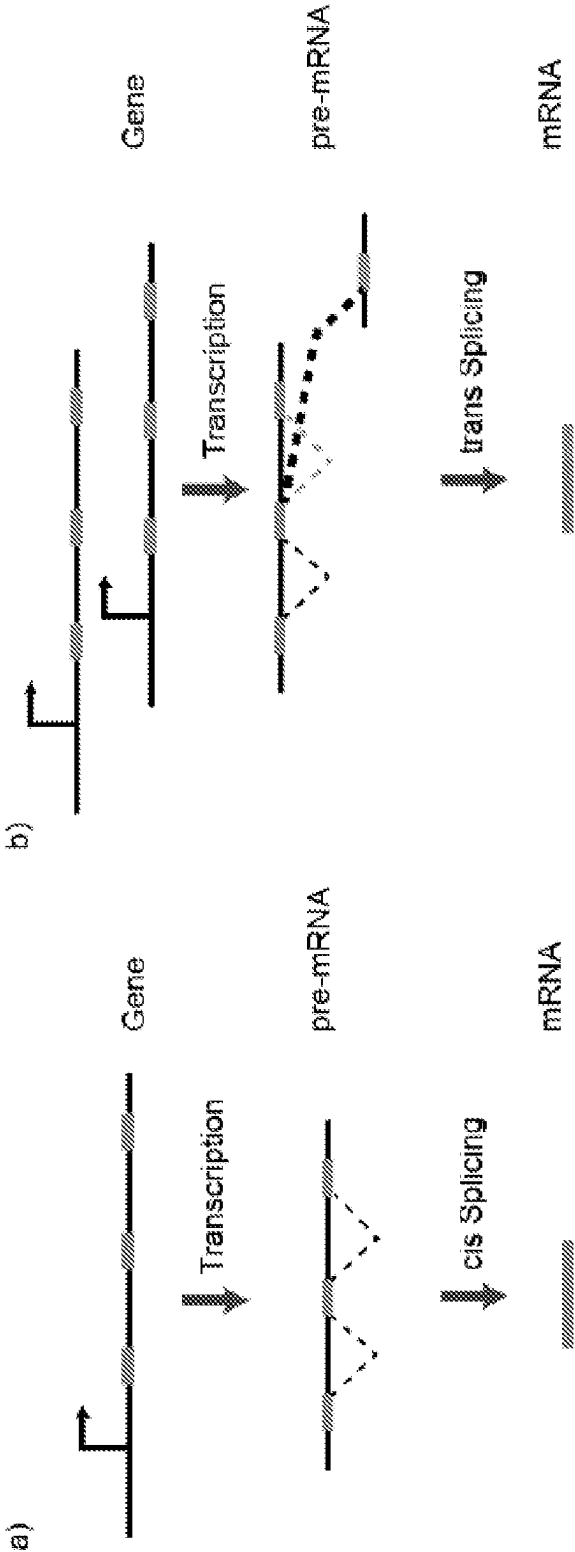
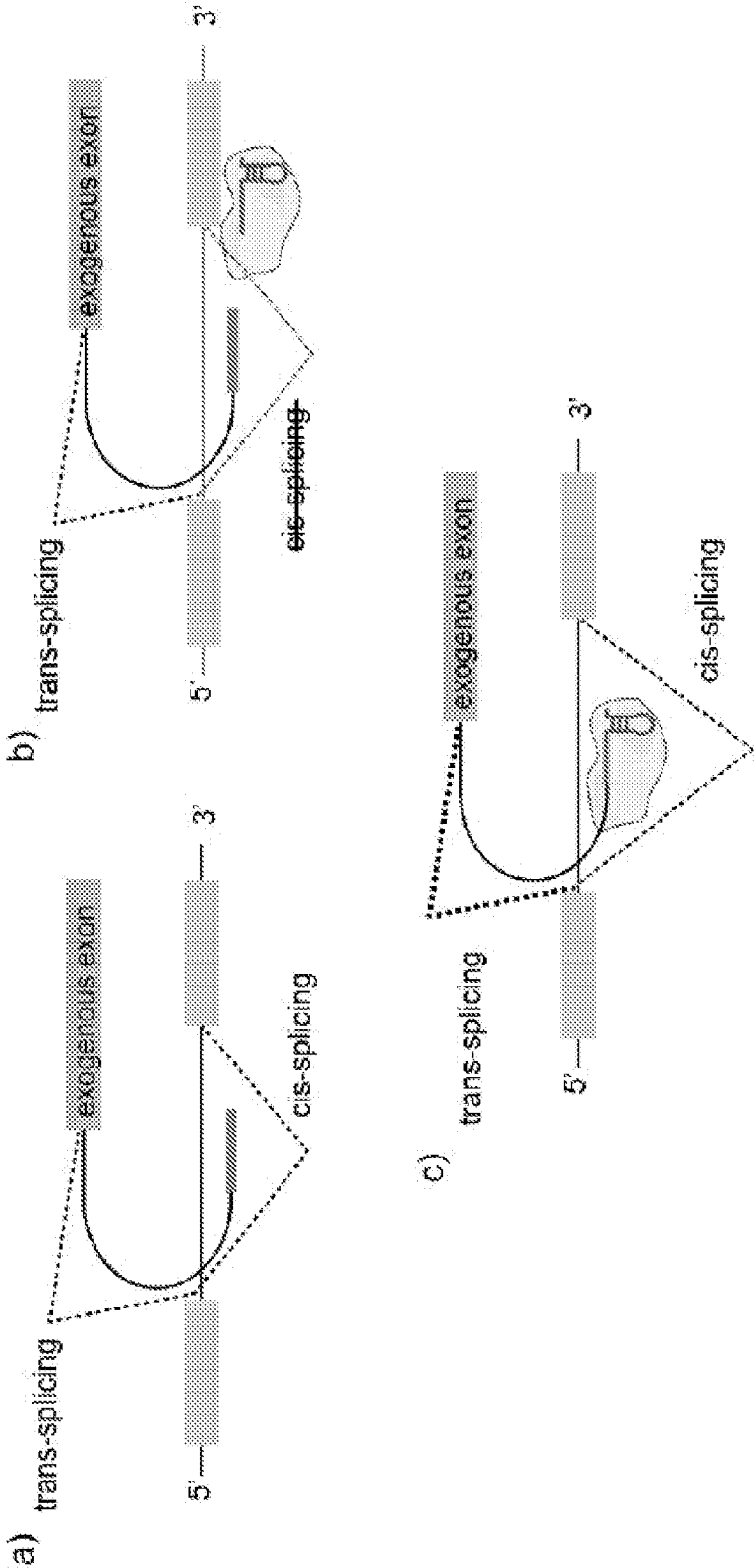


FIG. 2A-2C



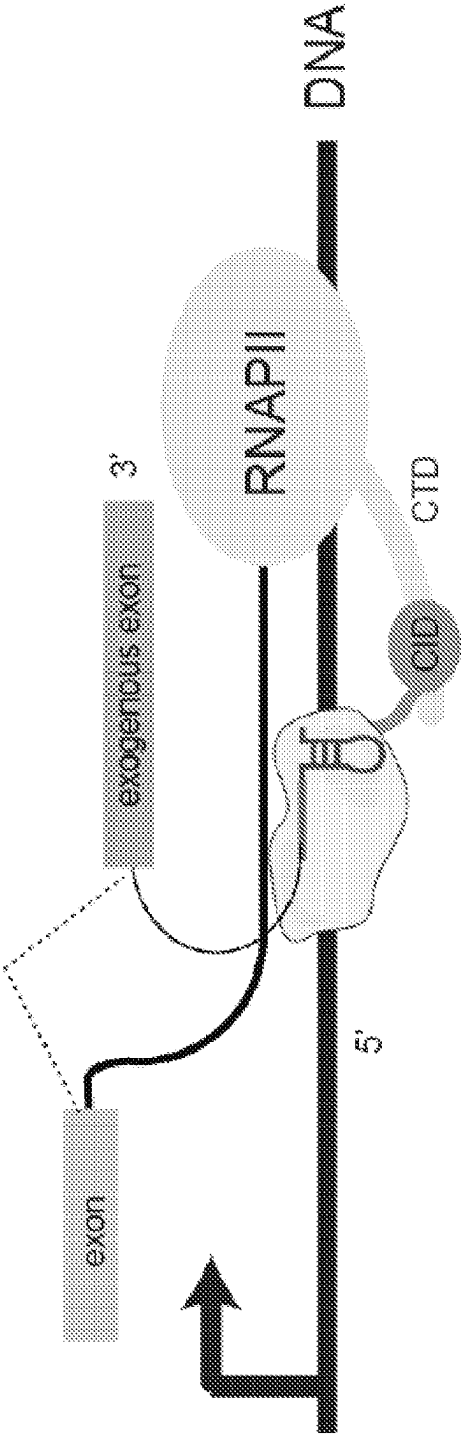


FIG. 4

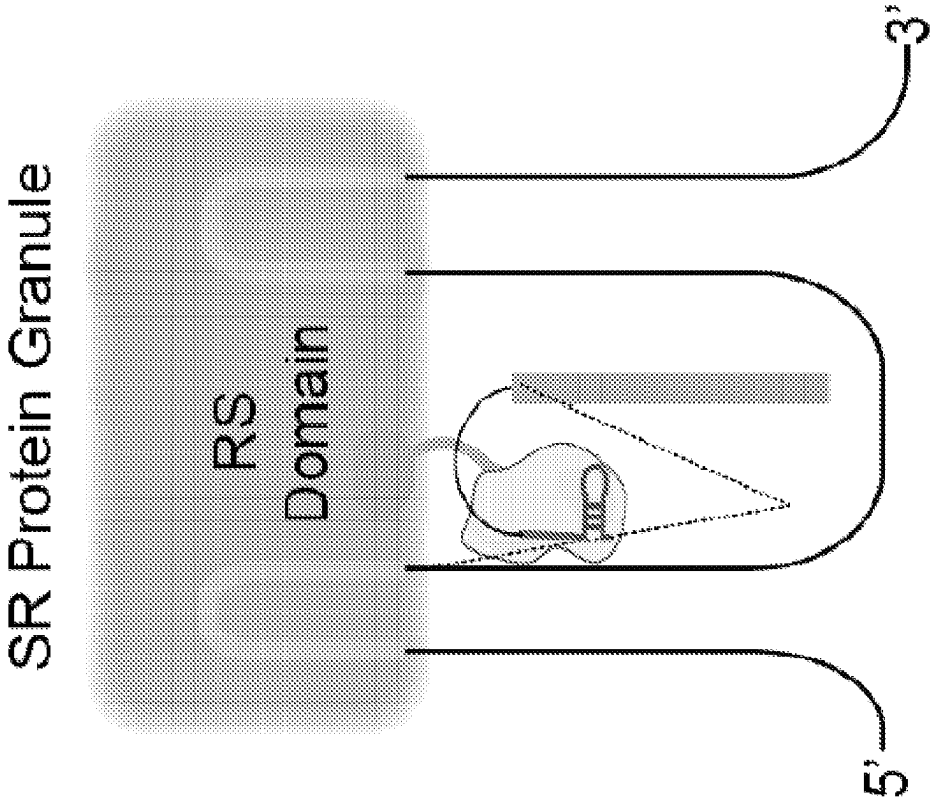


FIG. 5A-5B

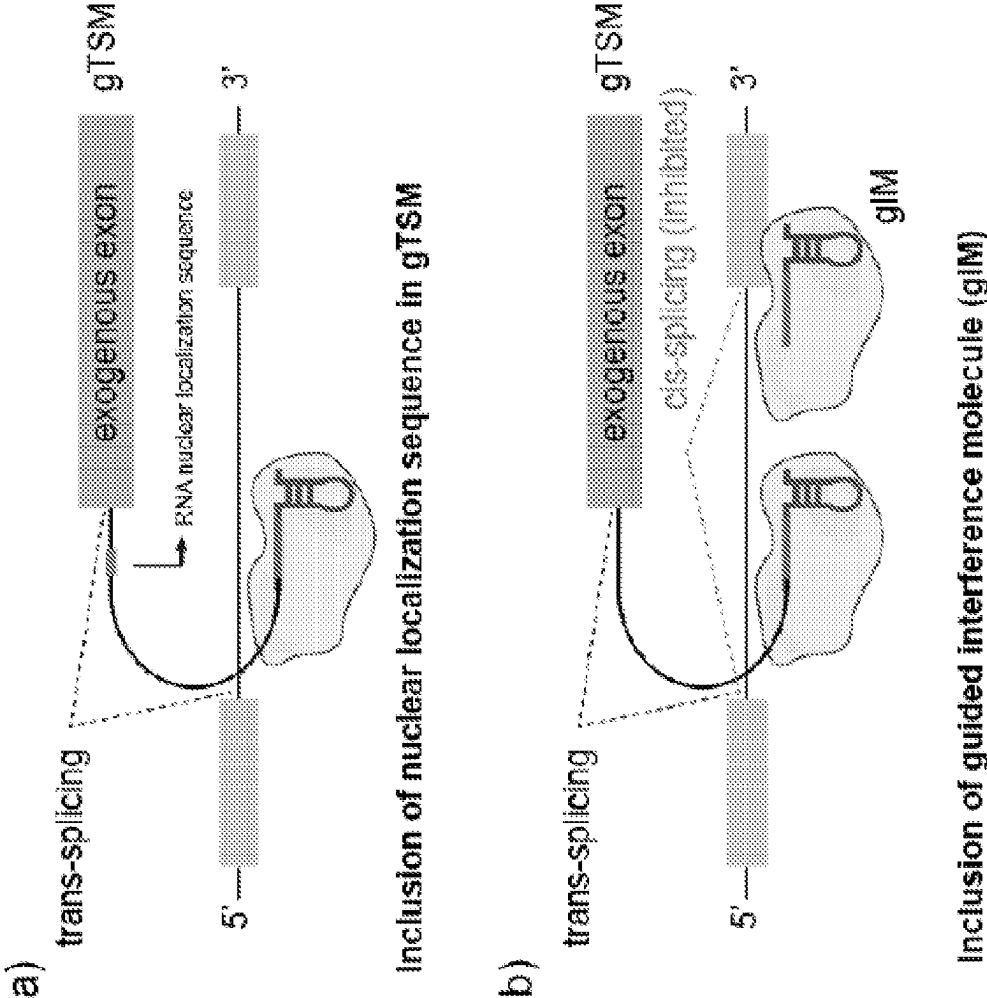


FIG. 5C

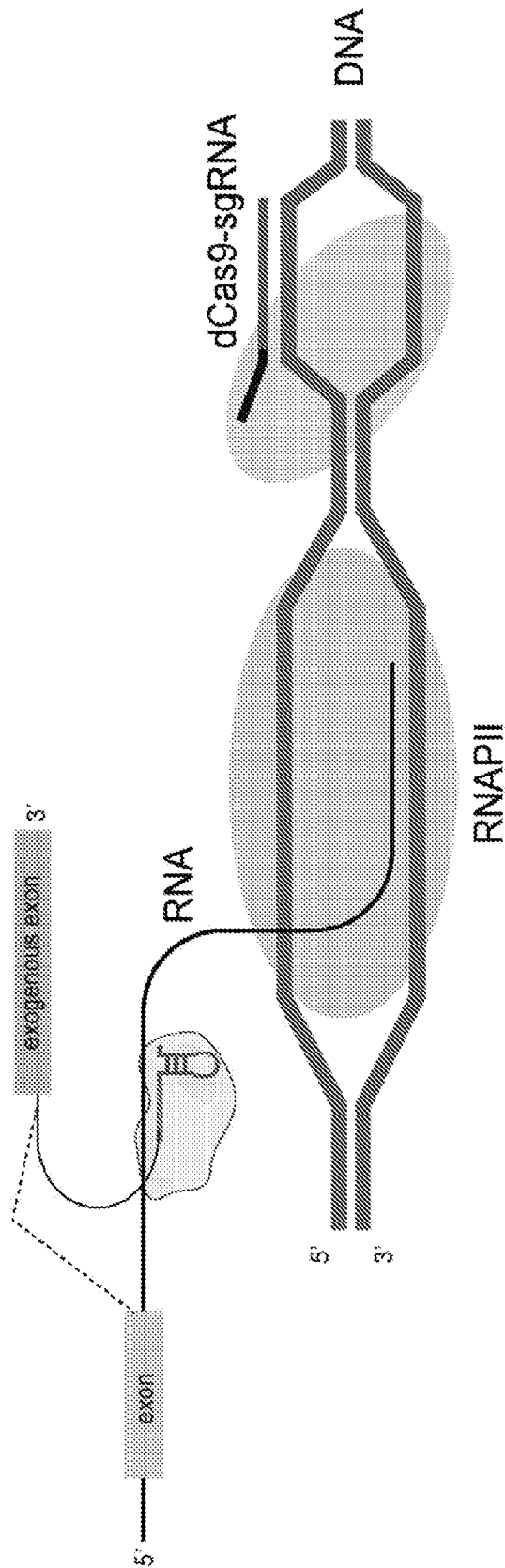


FIG. 6A

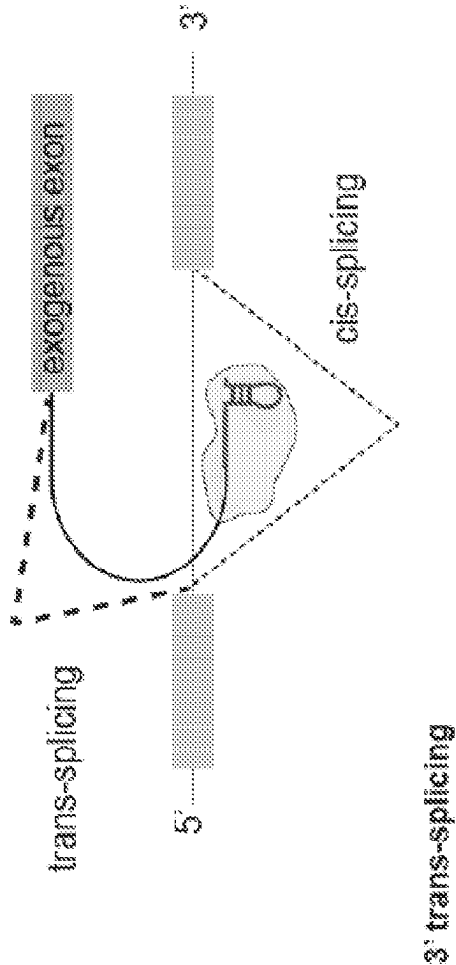


FIG. 6B

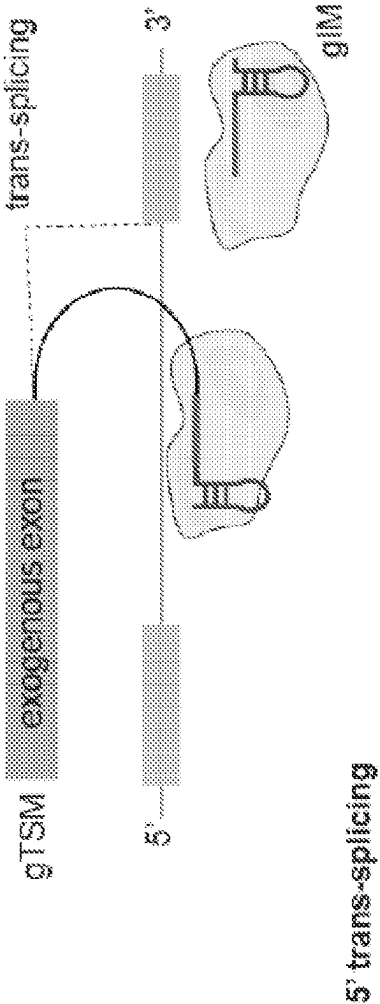


FIG. 6C

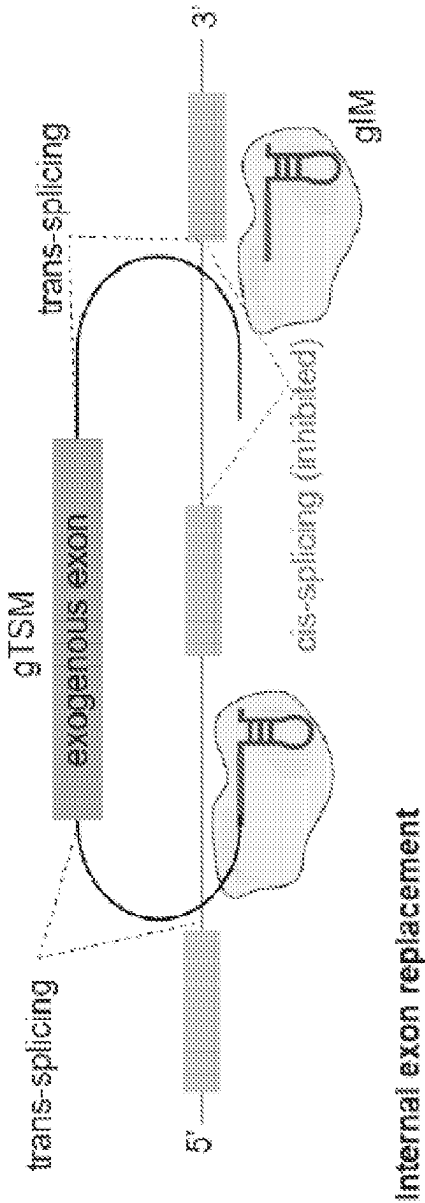


FIG. 7A-7B

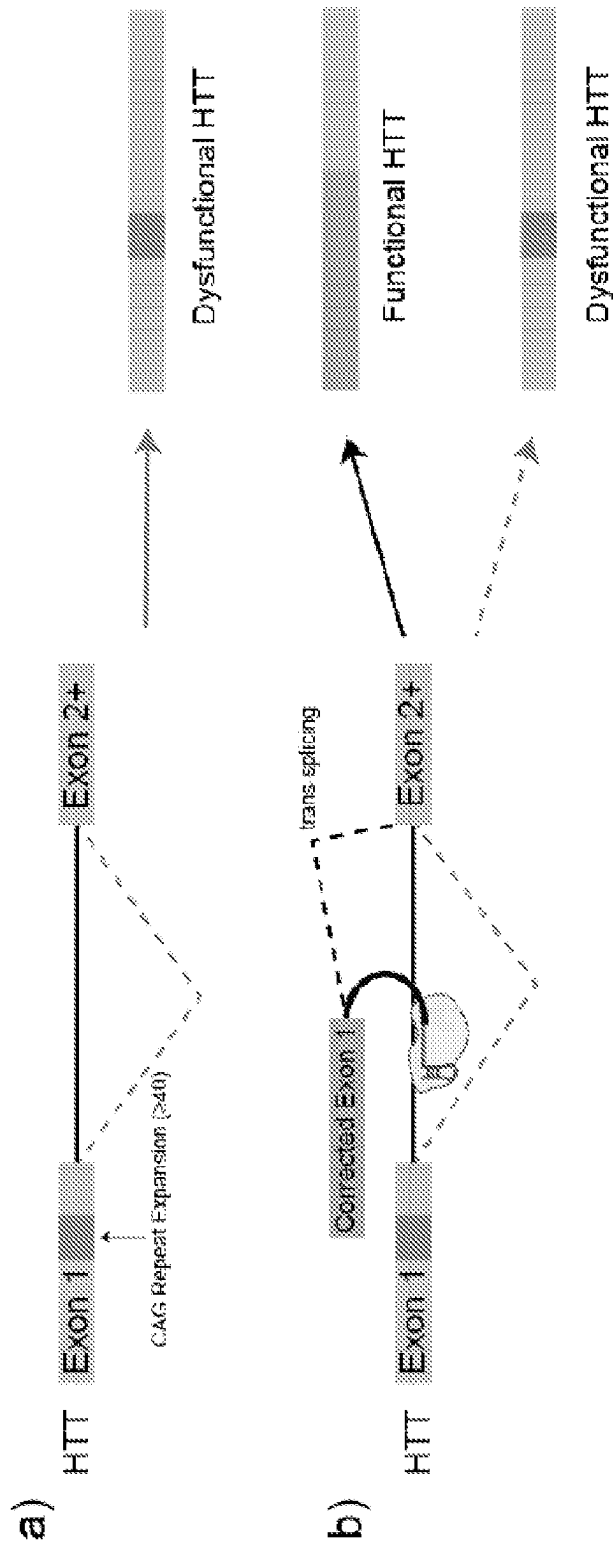


FIG. 8A-8B

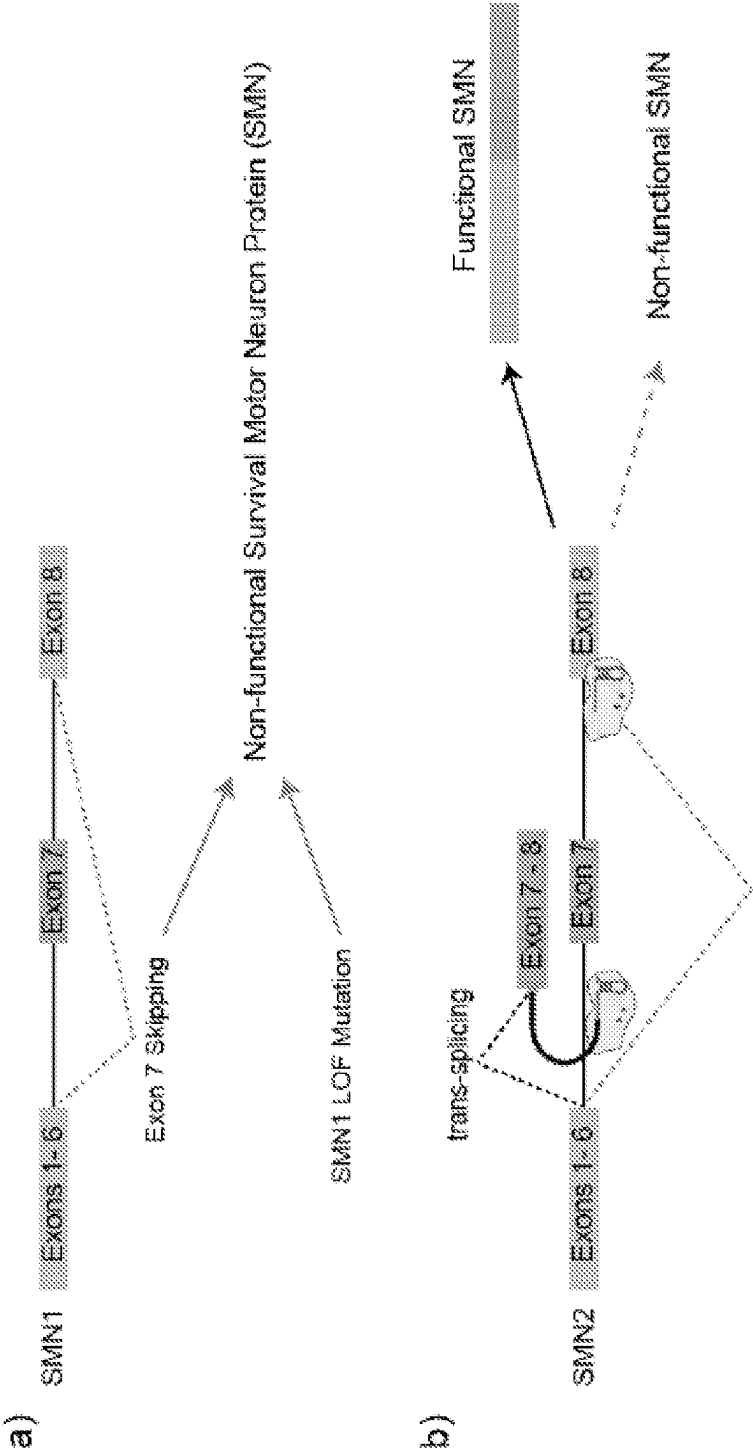


FIG. 9A

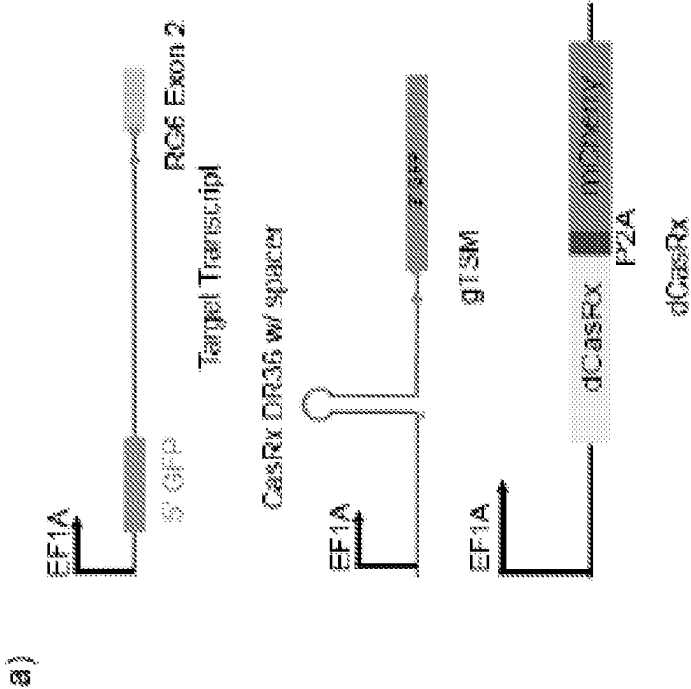
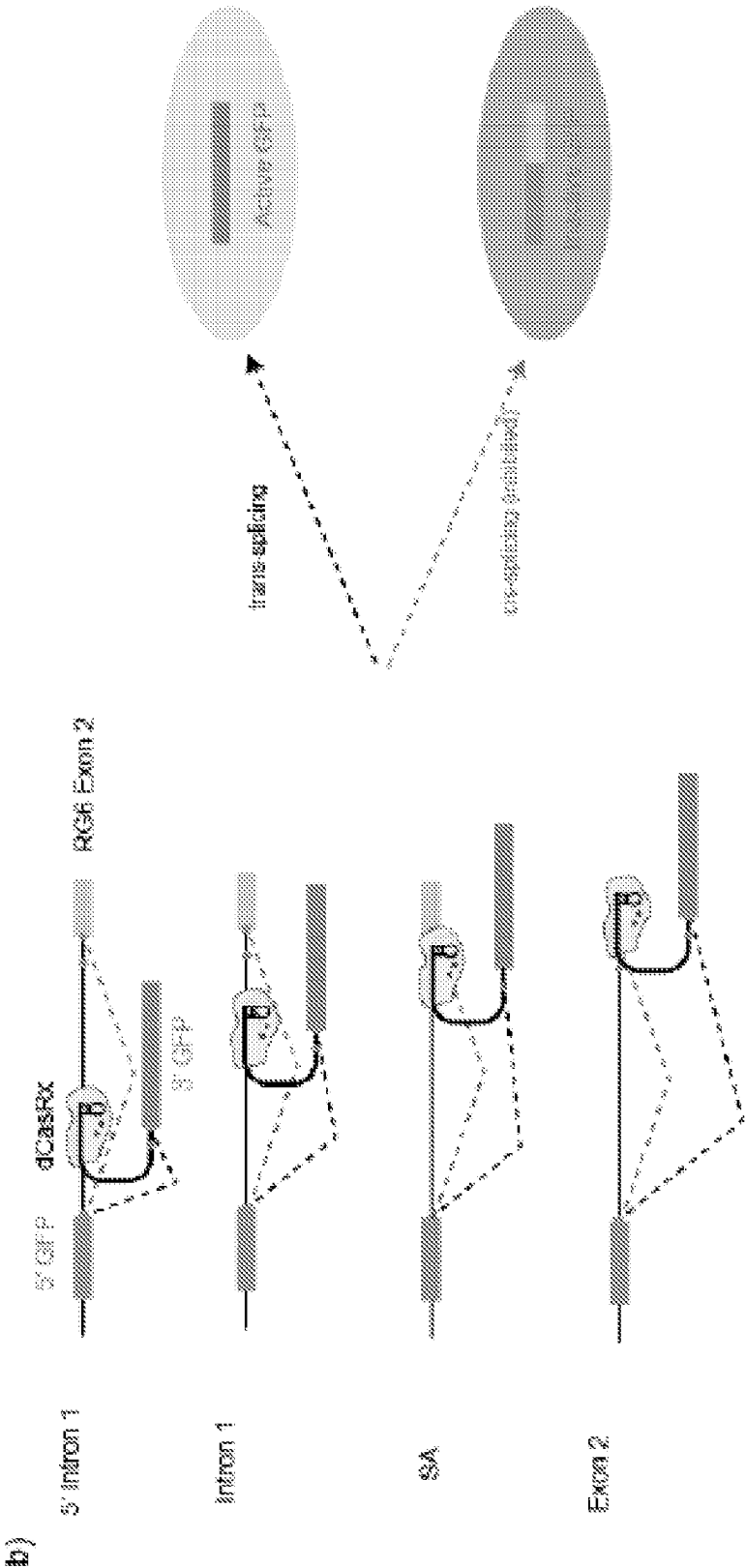


FIG. 9B



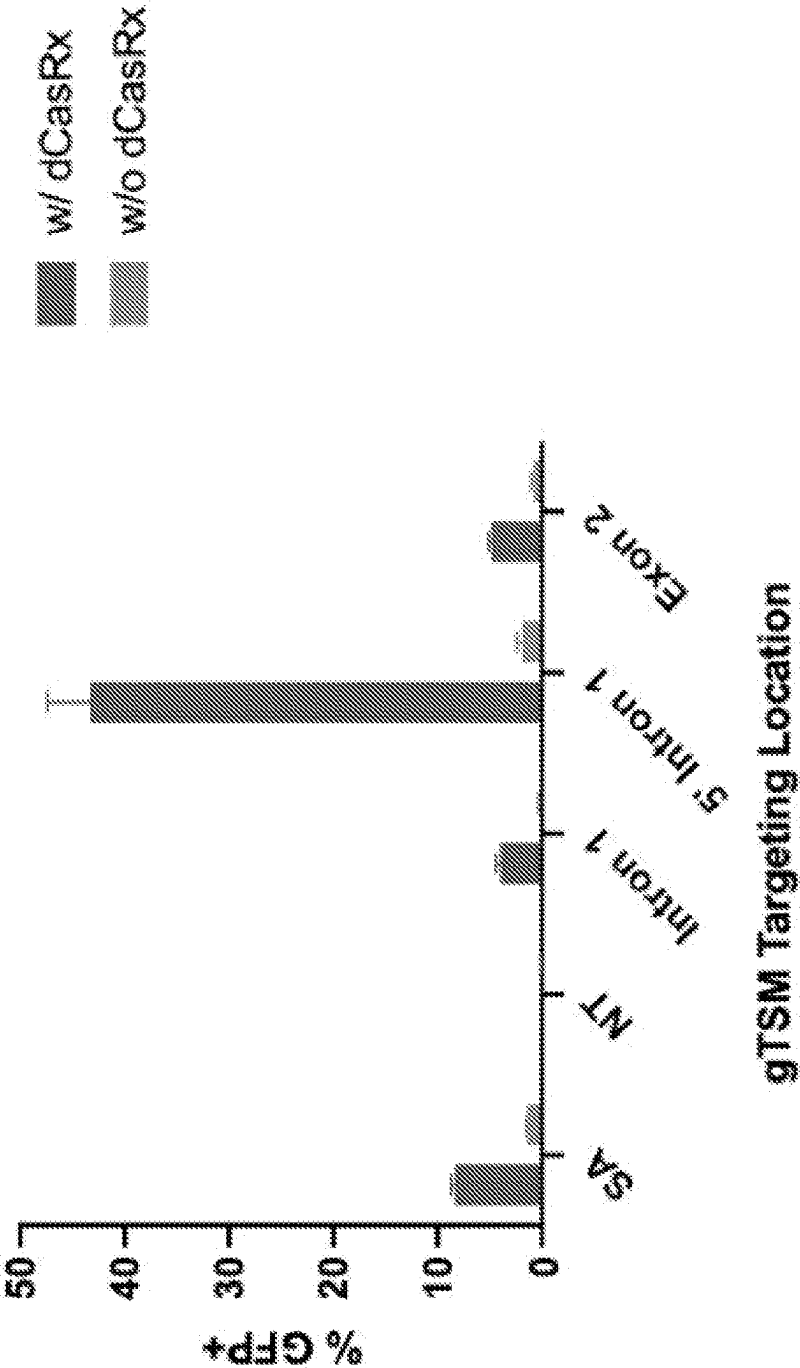


FIG. 10

FIG. 11A

Cas13a

Listeria portmoyi

1 MKITKMRVDG RTIVMEERTSK EGQLVYEGID GNKTTEIIFD KKESFYKSI LNKTVRKLDE
61 KVKDRRQRAI NKAINKIEIAE IMLAVLHQEV PRLKLNLSK LSTESLTKLF KPKFQNMISY
121 PPSKDAEHVQ FCLTDIAVPA IQGLDNIKED WGVFFEKLKP YTDWVESYIH FKQTTIQRSI
181 EQNKIQSPDS PRKVLQKYV AAFNGEPLG LDLVAKKYKL ADLAESEKLV DLNEDKSANY
241 KIKACLOQHQ RNILDELKED PELNQYGIEV KKYIQRYFPI KRAPNRSKHA RADFLKKELI
301 ESIVEQQQFN AVYHYVLEQG KMEAYELTNP KTKDLQDIRS GEAFSEKFIN ACAFASNNLK
361 TILNPECEKD ILKGGEFKS LLNSTTRSDV VKKMIPFFSD EIQNVNFEA IWAIRGSIQQ
421 IRNEVYHCKK HSWERILKIK DFEPNKKMKE DSDMQKLMOK DIAKIPDFIE EKLKSSGVIR
481 FYSHDKLQSI WEMKQGFSL TTNAPFVPSF KRVYAKGHDY QTSKNRYIDL GLTFEDILEY
541 GEEDFRARYF LTKLVYQQF MPWFTADNNA FRDAANFVLR LNKNRQQDAK AFINIREVEE
601 GEMPRDYMGY VQGQIAIHED STEDTPNHFE KFISQVFIKG FDSHMRSADL KFIKNPRNQG
661 LEQSEIEGMS FDIKVEPSFL KNKDDYIAFW TFCMLDARH LSELRNEMIK YDGHLTGEQE
721 IIGLALLGVD SRENDWKQFF SSEWEYEEKIM KGYVGEELYQ CEPYRQSDGK TPILERGVEQ
781 ARKYGTETVI QRLFDASPEF KVSNCNITEW ERQKETIET IRRKELHNE WAKNPEKPQN
841 NAFFKEYKEC CDAIDAYNWH KNKTTLVYVN ELHLLIEIL GRVGYVAIA DRDFQCMANQ
901 YPKHSGITER VEYWGDNRLK SIKKLDTFK KEGLFVSEKN ARNHIAHLNY LSLKSECTLL
961 YLSERLREIF KYDRKLKNV SKSLIDILDR HGMSVVFANL KENKHLVIK SLEPKLRLHL
1021 GEKKIDNGYI ETNQVSEECY GIVKRLEI (SEQ ID NO:1)

FIG. 11B

C2c2 (Cas13a)

>Leptotrichia buccalis (WP_015770004.1)
MKVTKVGGISHKKYTSEGRLVKSESEENRTDERLSALLNMRDMYIKNPSSTETKENQKR
IGKLKFFSNKMVYLKDNLTSLKNGKKENIDREYSETDILESVDVRDKKNFAVLKKIYLNE
NVNSEEELEVFRNDIKKKLNKINSLKYSFEKNKANYQKINENNIEKVEGSKRNIYDYR
ESAKRDAYVSNVKEAFDKLYKEEDIAKLVLIEIENLTKEKYKIREFYHEIIGRKNDKENF
AKIIYEEIQNVNMKELIEKVDPDMSELKKSQVFYKYLDKEELNDKNIKYAFCHFVEIEM
SOLLKNYVYKRLSNISNDKIKRIFEYQNLKKLIENKLLNKLDTYVRNCGKYNYYLQDGEI
ATSDFIARNRQNEAFLRNIIGVSSVAYFSLRNILETENENDITGRMRGKTVKNNKGEKY
VSGEVDKIYNENKKNVKNLKMFYSDFMNDKNIEIDFFANIDEAIISSIRHGIVHFNL
ELECKDIFAFKNIAPSEISKKMFQNEINEKKLKLKIFRQLNSANVFYLEKYKILNYLKR
TRFEFVNKNI PFVPSFTKLYSRIDDLKNSLGIYWKTPKTNDNDNKTEIIDAQIYLLKNIY
YGEFLNYFMSNNGNFFEISKEIIELNKNDRNLKTGFYKLQKFEDIQEKIPKEYLANIQS
LYMINAGNQDEEEKDTYIDFIQKIFLKGEMTYLANNGRLSLIYIGSDEETNTSLAEKKQE
FDKFLKKYEQNNNIKIPYEINEFLREIKLGNILKYTERLNMFYILKLLNHKELTNLKG
LEKYQSANKEEAFSDQLELINLLNDNVRVTEDELEADEIGKFLDENGKVKDNKELKK
FDTNKIYFDGENI IKHRAFYNIKKYGMLNLEK IADKAGYKISIEELKKYSNKKNEIEKN
HKMQENLHRKYARPRKDEKFTDEYESYKQAIENIEEYTHLKNKVEFNELNLLQGLLLRI
LHRLVGYSIWERDLRFRLKGEFPENQYIEEINFENKKNVKYKGGQIVEKYIKFYKELH
QNDEVKINKYSSANIKVLKQEKDLYIRNYIAHFNYIPHAEISLLEVLNLRKLLSYDRK
LKNVAMKSVVDILKEYGFVATFKIGADKKIGIGTLESEKIVHLKNLKKKIMTDRNSEEL
CKLVKIMFEYKMEKKSEN (SEQ ID NO:2)

FIG. 11C

Cas13b

Prevotella intermedia

1 meddkktttes tnmldnkhfw aafnlrlarhn vyitvmhink vlelknkkdq dliicndqdi
61 laikthwekv ngdlnkterl relmtkhhfpl letaiytknk edkkevqgek qakaqsfds1
121 khclflflek lgearnyysh vkyssestkep mlekelllkkm ynifddniql vikdyghnkd
181 inpdedfkh.l drteedinyy ftrnkknit asglffvsl flekkdaism qcklrgfkdn
241 reskkkmthe vfcrsmllp klrlestqtq dwlldmine lircpksiye rlqgddrkkf
301 nvpfdptden ydaeqepfkn tlvrhqdrfp yfalryfdyn evfanlrfaq dlgyhfsiy
361 kkliggqked rhlthklygf eriqlfakqn rpdewkaivk dldtyetsee pyisettphy
421 hlenqkigir frngndiwp sletngenne kskykldkqy qaeafllsvhe ilpmmfyyl
481 lkkekpnnde inasivegti kreiryiyki ydafangein niddlkkyc dkgipkrnlp
541 kmiallesk pkdmveeae kqrkmvkdtk kllatlekqt qgeiedggm irllksgeia
601 rwlvdnmrf qpvqkdhgn plnnskanst eyqmlqrsia lynkeekptr yfrqvnlies
661 snphplnnt ewkecnnils fyrsyltkki eflnelkped weknqyflkl kepktretl
721 vqgwknqfnl prgiftepir ewfkrhqnds keyekvealk rvglvtkvip lffkeeyfke
781 daqkeinncv qpfygfpyv gnihhkpkced flhreeriel wdkkkdkfkq ykakkkkkem
841 tdkeeehrs ylefkswngf erelrivng divtwillcme lidklkvegl nveelqklrl
901 kdiddtdtakq eknnilarvm pmqlpvtvye iddshnivkd rplhtvyiee tktklkqgn
961 fkalvkdrrl nglfsfvds seaelksnpi sklrvveyelg eyqnarieli edmlalegal
1021 iekdknlptd nfsdmlngwl egkdesdkar fqndvkliva vrnafshnqy pmnrriafan
1081 inpslssad iseekgldia nqlkdkthki ikriikieky ietke (SEQ ID NO:3)

FIG. 11D

Cas13b

Prevotella buccae

1 mqkgdklfdv rkknaifafp kytimenke kpepiyvelt dkhfwaafln larnrvytti
61 nhinrrleia elkdggymg ikgswnegak kldkkvrlrd limkhfpfle aaayemtnsk
121 spnnkeqrek egsealsltn lknvlfifle klqvlrnys hykyseespk pifetsllkn
181 mykvfdanv lvkrdymhbe nidmqrdftb lnrrkkqvgrt kniidspnfh yhfackegnm
241 tiagllffvs lfldkkdaiw mqkklkgfkd gnlreqmta evfcrsrl pklklenvqt
301 kdwmgldmln elvrcpsly erlrekdras fkvpdfdifs dynaeepfk ntlvrhqdrf
361 pyfvlyrfdl neifeglrfg idlgyhfsi ynkrigdede vrhlthhlyg fariqdfapq
421 nqpeewrklv kdldhfetsq epyisktaph yhlenekigi kfcsahnnlf psdqtdkton
481 grskfnlgtq ftaeafslsvh eilpmmfyyi lltkdysrke sadkvegilar keisniyaiy
541 dafanneins iadlrrlqn tnllqghlpk qmsilkgrq kdmgkeaerk igemiddtqr
601 rldllckqtn qkirigkna glksgkiad wlvndmmrfq pvqkdqnnip innskanste
661 yrmqlralal fgsefrlka yfnqnnlvgn dnphpflaet qwehqtnils fynyleark
721 kylkgllkpqn wkqyqhflil kvqktnrnti vtgwksfnl prgiftqpir ewfekhnnsk
781 rlydqilsfd rvqfvakaip lyfaeeykdn vqpfydyppn ignrlkpkkr qfldkerve
841 lwqknkeifk nypsekkktd layldflswk kferelriik ngdivtwimf kelfmratve
901 glkigeihlr didtntanee snnilnrimp mklpvktyet dhkgnilker platfyieet
961 etkvlkqgnf kalvkdrrln glfsfaettd lnleehpisk lsvdleliky qttrsisifem
1021 tlgiekkliid kystlptdsf rnmleerwlc kanrpeklky vnsliavna fshnqypmyd
1081 atlfaevckkf tlfpvdtkk ielniaqqli eivgkaikai eksenkn (SEQ ID NO:4)

FIG. 11E

dRfx-Cas13d

IEKKSFAGMGVKSTLVSGSKVYMTTFAEGSDARLEKIVEGDSIRSVNEGEAFSAEMADKNAGYKIGNAKFSPKGYAVVANPLYT
GPVQQDMLGLKETLEKRYFGESADGNDNICIQVIHNILDEKILAEYITNAAYAVNNISGLDKDIIIGFGKFSTVYTYDEFKDPPEHHRAAFN
NNDKLINAIIKAQYDEFDNFLDNPRLGYPFGQAFFSKEGRNYIINYGNECYDILALLSGLRHWVVHNNEESRISRTWLYNLDKNLDNEYIS
TLNYLYDRITNELTNSFSKNSAANVNYYIAETLGINPAEFAEQYPRFSIMKEQKNLGFNITKLREVMLDRKDMSEIRKNHKVFDSTRTKVYT
MMDFVIYRYVYIEEDAKVAAANKSLPDNEKSLSEKDFVNLRGSFNDQKDALYYDEANRIWRKLENIIMHNHNIKEFRGNKTREYKKKDDAP
RLPRILPAGRDVSASFSLMYALTMFLDQKEINDLLTTLINKFDNIQSFLKVMPLIGVNAKFVEEYAFFKDSAKIADDELRLIKSFARMGEPIA
DARRAMYIDAIRILGTNLSYDELKALADTFSLDENGKLLKKGKMGMRNFHNNVISNKRPHYLRJYGDPAHLHEIAKNEAVVKFVLGRIA
DIQKKQQNGKNQIDRYYETCIGKDKGKSVSEKVDALTKIITGMNYDQFDKKRSVIEDTGRENAEREKFKKISLYLTVIYHILKNIVNIN
ARYVIGFHCVERDAQLYKEKGYDINLKKLEEKGFSSVTKLCAGIDETAPDKRKDVEKEMAERAKESIDSLESANPKLYANYIKYSDEKK
AEEFTRQINREKAKTALNAYLRNTKWNVIREDLLRIDNKTCTLFANKAVALEVARYVHAYINDIAEVNSYFQLYHYIMQRIIMNERYEK
SSGKVSSEYFDAVNDEKKYNDRLKLLCVPGYCIPIRFKNLSIEALFDRNEAAKFDKEKKKVSNGNS (SEQ ID NO:5)

FIG. 11F

RfxCas13d

IEKKSFAGMGVKSTLVSGSKVYMTTFAEGSDARLEKIVEGDSIRSVNEGEAFSAEMADKNAGYKIGNAKFSHPKGYAVVANPLYT
GPVQQDMLGLKETLEKRYFGESADGNDNICQVIHNIIDIEKILAEYITNAAYAVNNISGLDKDIIIGFGKFSTVYTYDEFKDPPEHHRAAFN
NNDKLINAIAQYDEFDNFLDNPRUGYFGQAFFSKEGRNYIINYGNECYDILALLSGLRHWVVHNNNEESRISRTWLYNLDKNLDNEYIS
TLNLYLDRITNELTNSFSKNSAANVNYIAETLGINPAEFAEQYFRFSIMKEQKNLGNITKLREVMLDRKDMSEIRKNHKVFDISRTKVYT
MMDFVIYRYYIEEDAKVAAANKSLPDNEKSLSEKDIFVINLRGSFNDDQKDALLYDEANRIWRKLENIIMHNHKEFRGNKTREYKKKDDAP
RLPRILPAGRDVSASFSLMYALTMFLDGGKEINDLLTTLINKFDNIQSFLKVMPLIGVNAKFVEEYAFFKDSAKIADHLHEIAKNEAVVKFVLGRIA
DARRAMYIDAIRILGTNLSYDELKALADTFSLDENGKLLKKGKMGMRNFHINNVISNKRPHYLRJRYGDPAPHLHEIAKNEAVVKFVLGRIA
DIQKKQGQNGKNQIDRYYETCIGKDKGKSVSEKVDALTKIITGMNYDQFDKRRSVIEDTGRENAEREKFKKISLYLTVIYHILKNIVNIN
ARYVIGFHCVERDAQLYKEKGYDINLKKLEEKGFSSVTKLCAGIDETAPDKRKDVEKEMAERAKESIDSLESANPKLYANYIKYSDEKK
AEFTROINREKAKTALNAYLRN'TKWN'VIREDLLRIDNKTCTLFRNKAVHLEVARYVHAYINDIAEVNSYFQLYHYIMQRIIMNERYEK
SSGKVSSEYFDAVNDEKKYNDRLKLLCVPFGYCIPIRFKNLSIEALFDRNEAAKFDKEKKKVS GNS (SEQ ID NO:6)

FIG. 11C

PspCas13b

NQKKYFGTYSVMAMLNAQTVL DHIQKVADIEGEQNNENLWFHPVMSHLYNAKNGYDKQPEKTMFIERLQSYFPFLKIMAEHQREY
SNGKYKQNRVEVNSNDIFEVLKRAFGVLKMYRDLTNHYKTYEELNDGCEFLTSTEQPLSGMINNYTVALRNMNERYGYKTEDLAFI
QDKRFKFKDAYGKKKSQVNTGFFLSLQDYNGDTQKKLHLSGVGIALLICLFELDKQYINIFLSRLPIFSSYNAQSEERRIIRSFSGINSIKLPK
DRIHSEKSNKSVAMDMLNEVKRCPDELFTTILSAEKQSRFRIISDDHNEVLMKRSSDRFVPLLLQYIDYGKLFDRHFHVNMGKRLRYLLKA
DKTCIDGQTRVRVIEQPLNGFGRLEEAETMRKQENGTFGNSGIRIRDFENMKRDDANPANYPIVDYTYHYILENNKVEMFINDKEDSAP
LLPVIEDDRYVVK TIPSCRMSTLEIPAMAFHMF LFGSKKTEKLIJVDVHNRYKRLFQAMQKEEVTAENIASFGIAESDLPQKILDLSGNAH
GKDVDAFIRLTVDDMLTDTERRIKRFKDDRKIRSADNKMGRGEKQISTGKLADFLAKDIVLFQPSVNDGENKITGLNRYRIMQSAIAVY
DSGDDYEAKQQFKLMFEKARLIGKGTTEPHPLYKVFARSIPANA VEFYERYLIERKFYLTGLSNEIKKGNRVDPFIRRDQNKWKTPAM
KTLGRIYSEDLPVLPQMEGIDFNANVTYLAEYMKRVLDDDFQTFYQWNRNRYRYMDMLKGEYDRKGSLOH
CFTSVEEREGLWKERASRTERYRKQASNKIRSNRQMRNASSEIEITLDDKRLSNSRNEYQKSEKVIIRYRVQDALLFLLAKKTLTELADF
DGERFKLKEIMPD AEKGILSEIMPMSFTFEKGKKYTTITSEGMLKNYGDFFVLASDKRIGNLLELVGSDIVSKEDIMEEFNKVDQCRPEI
SSIVFNLEKWAFDTYPELSARVDREKVDFKSILKILLNNKNINKEQSDILRKIRNAFDHNNYPDKGVVEIKALPEIAMSIKKA FGEYAIMK
(SEQ ID NO: 7)

FIG. 11H

dPspCas13b

NQKKYFGTYSVMAMLNAQTVLDHIQKVADIEGEGQNNENLWFHPVMSHLYNAKNGYDKQPEKTMFIERLQSYFPFLKIMAEHQREY
SNGKYKQNRVEVNSNDIFEVLKRAFGVLKMYRDLTNAYKTYEELNDGCEFLTSTEQPLSGMINNYTVALRNMNERYGYKTEDLAFI
QDKRFKVDAYGKKKSQVNTGFFLSLQDYNGDTQKKLHLSGVGIALLICLFELDKQYINIFLSRLPIFSSYNAQSEERRIIRSFGINSIKLPK
DRIHSEKSNKSVAMDMLNEVKRCPDELFTILSAEKQSRFRIISDDHNEVLMKRSSDRFVPLLLQYIDYGKLFDFHIRFHVNMGKRLRYLLKA
DKTICIDGQTRVRVIEQPLNGFGRLEEAEATMRKQENGTFGNSGIRDFENMKRDDANPANYPIYIVDYTYHYILENNKVEMFINDKEDSAP
LLPVIEDDRYVVKTIPTSCRMSTLEIPAMAFHMFELFGSKKTEKLJVDVHNRYKRLFQAMQKEEVTAENIASFGIAESDLPQKILDLSGNAH
GKDVDAFIRLTVDDMLTDTERRIKRFKDDRKIRSADNKMGRGFKQISTGKLADFLAKDIVLFQPSVNDGENKITGLNYRIMQSAIAVY
DSGDDYEAKQQFKLMFEKARLIGKGTTEPHPLYKVFAARSIPANAVEFYERYLIERKFYLTGLSNEIKKGNRVDPFFIRRDQNKWKTPAM
KTLGRIYSEDLPVELPRQMFQDNEIKSHLKSPLQMEGIDFNANVTYLIAEYMKRVLDDDFQTFYQWNRNRYRMDMLKGEYDRKGSQKH
CFTSVEEREGLWKERASRTERYRKQASNKIRSNRQMRNASSEIEETILDKRLSNSRNEYQKSEKVIIRRYRVQDALLFLLAKKTLTELADF
DGERFKLKEIMPDAEKGILSEIMPMSFTFEKGGKKYTTITSEGMKLNKNGDFFVLASDKRIGNLLELVGSDIVSKEDIMEEFNKYDQCRPEI
SSIVFNLEKWAFDTYPELSARVDREKVDFKSILKILLNNKNINKEQSDILRKIRNAFDANNYPDKGVVEIKALPEIAMSIKKAFGEYAIMK
(SEQ ID NO:8)

FIG. 12
C-terminal domain *Homo sapiens* RNA Polymerase II
Amino acids 1593-1960 of NP_000928.1

[illegible]

FIG. 13A

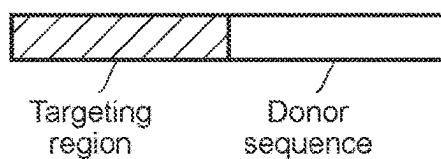


FIG. 13B



FIG. 13C

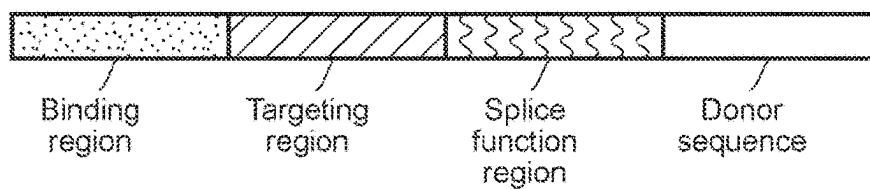


FIG. 13D

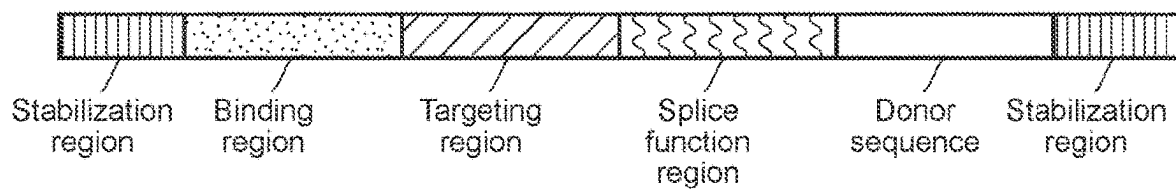


FIG. 14

Examples of Hybrid RNA Molecules

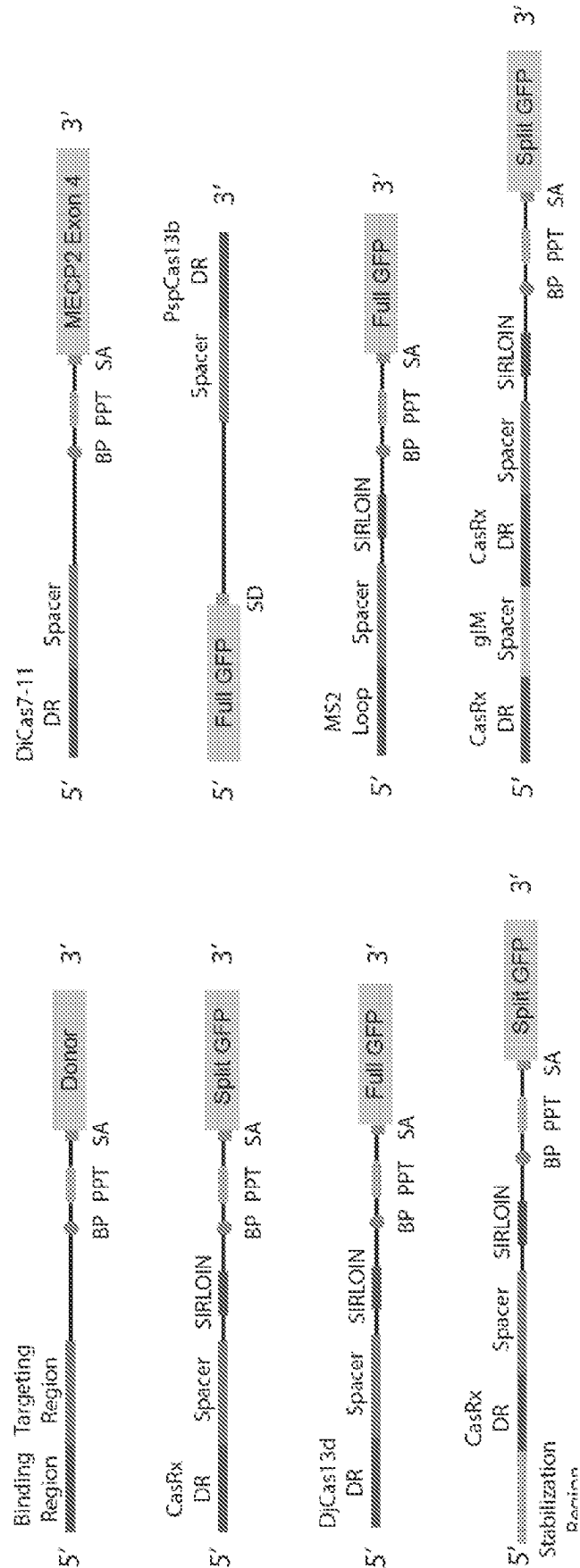


FIG. 15A

EMCV IRES

EMCV (NC_001479; IRES position, 393-881):

GTCTTCTTGACGAGCATTCTAGGGGTCTTTCCCTCTCGCCAAAGGAATGCAAGGTCTGTTGAATGTCG
TGAAGGAAGCAGTTCCTCTGGAAGCTTCTTGAAGACAAACAACGTCTGTAGCGACCCTTGCAGGCAGC
GGAACCCCCACCTGGCGACAGGTGCCTCTGCGGCCAAAAGCCACGTGTATAAGATACACCTGCAAAGG
CGGCACAACCCCAAGTGCCACGTTGTGAGTTGGATAGTTGTGGAAAGAGTCAAATGGCTCTCCTCAAGCG
TATTCAACAAGGGGCTGAAGGATGCCAGAAGGTACCCATTGTATGGGATCTGATCTGGGGCCTCGGT
GCACATGCTTTACATGTGTTAGTCGAGGTTAAAAACGTCTAGGCCCCCGAACCACGGGGACGTGGT
TTTC (SEQ ID NO:10)

FIG. 15B

FMDV IRES

FMDV-a (NC_003992.2, nucleotide 897-1070)

CTTGTTTCTGGAGACAGGCTAAGGATGCCCTTCAGGTACCCGAGGTAACAAGCGACACTCGGGATCT
GAGAAGGGGATTGGGGCTTCTATCCAAGCGCCCAATTTAAAAAGCTTCTATGCCTGAATAGGTGACCGG
AGGCCGGCACCTTTTCCTTTACCCACAACCTACTTT (SEQ ID NO:11)

FIG. 15C

FMDV IRES

FMDV-b (AY593782.1, nucleotide 656-1087)

GTGCAATCTGAAACTCCGCCTGGTCTTTCCAGGTCTAGAGGGGTGACACTTTGTACTGTGCATGACTCCA
CGTTCGATCCACTGGCGAGTGCTAGTAGTAGTACTGCTGTTTCGTAGCGGAACACGACGGCCGTGGGAC
TCCCCCTTGTAACAAGGACCCGCGGGGCCAAAAGCCACGTCTCTGACCCGTCGTGTGTGCAAACCCA
GCACAGCAGCTTTACTGCGAAAACCACTTTAAGGTGACATTGATACTGGTACTCACGCACTGGTGACAG
GCTAAGGATGCCCTTCAGGTACCCGAGGCAACAAGCGACACTCGGGATCTGAGAAGGGGACTGGGGC
TTCTGTAAAAGCGCCCCAGTTTAAAAAGCTTCTATGCCTGAATAGGTGACCGGAGGCCGGCACCTTTCCCT
CTACTACCACTGACTCT (SEQ ID NO:12)

FIG. 15D

CPV IRES

CPV (Cricket paralysis virus, NC_003924.1, nucleotide 357-709)

TTCGATACCAAGAGCTGGTGTCTGAGGGATTACGACTCGCGTGGTCGGACAGCACATGTTAGTCATGTA
TTATGGTGTTCCTAACACGTATCTTTGCTCTAGTATTGAACTTTAGTGATGAGTGTACTTGGCATGTACA
CCTGATTAGTTAGCCCTTGTGGATACGCCACTGCCGCTCAGGTACAGTAGGACATACCCCAAATATTGGG
TTCCACTCGTAGAGAAGAAAATTAGTGCGAAACGTACGGTTTATGTCAAAGGAGAAGGTGATTTAAT
TGCACCTGAAGCAACCCTAGGTCTTCCCTTGATGTGAGCTAATAAGGAGTAACCGATTCTTACAATGTG
ATCA (SEQ ID NO:13)

FIG. 15E

EIF4g3 IRES

EIF4g3 (NM_003760.4):

CCGGGAACCCCTACCCCATCCCCTTATTCAGCACATGAAATAAACAAGGGGCATCCAAATCTTGCGGCAA
CGCCCCGGGACATGCATCGTCCCCTGGACTCTCTCAAACCCCTTATCCCTCTGGACAGAATGCAGGTCC
AACCACGCTGGTATACCCTCAAACCCCTCAGACA (SEQ ID NO:14)

FIG. 15F

BiP IRES

BiP (NM_005347.4, IRES 168-261)

CGGGCCCTGCCGCTGCAAGTCGGAAATTGCGCTGTGCTCCTGTGCTACGGCCTGTGGCTGGACTGCC
TGCTGCTGCCCACTGGCTGGCAAG (SEQ ID NO:15)

FIG. 15G

VEGF IRES

VEGF, (NM_001171630.1, nucleotide 745–1038)

GCTAGCTCGGGCCGGGAGGAGCCGCAGCCGGAGGAGGGGGAGGAGGAAGAAGAGAAGGAAGAGGA
GAGGGGGCCGCAGTGGCGACTCGGCGCTCGGAAGCCGGGCTCATGGACGGGTGAGGCGGCGGTGTG
CGCAGACAGTGCTCCAGCCGCGCGCTCCCCAGGCCCTGGCCCCGGGCCTCGGGCCGGGGAGGAAGA
GTAGCTCGCCGAGGCGCCGAGGAGAGCGGGCCGCCACAGCCCGAGCCGGAGAGGGAGCGCGAGC
CGCGCCGGCCCCGGTCGGGCCTCCGAAACC (SEQ ID NO:16)

FIG. 16
DiCas7-11

MTTMMKISIEFLEPFRMTKWQESTRRNKNKEFVRGQAFARWHRNKKDNTKGRPYITGTLLRSA
VIRSAENLLTSLDGKISEKTCCPGKFDTEKDRLLQLRQRSTLRWTDKNPCPDNAETYCPFCCELL
GRSGNDGKKAEEKDWRFRHFGNLSLPGKPDFDGPKAIGSQRVLNRVDFKSGKAHDFFKAYEV
DHTRFPRFEGETIDNKVSAEARKLLCDSLKFTDRLCGALCVIRFDEYTPAADSGKQTEENVQAEP
NANLAEKTAEQHSILDDNKKTEYTRLLADAIRSLRRSSKLVAGLPKDHGKDDHYLWDIGKKK
KDENSVTIRQILTTSADTKELKNAGKWREFCEKLGEALYLKSKDMSGGLKITRRLGDAEFHGKP
DRLEKSRSVSVIGSVLKETVVCGELVAKTPFFFGAIDEDAKQTDLQVLLTPDNKYRLPRSAVRGILR
RDLQTYFDSPCNAELGGRPCMCKTCRIMRGITYMDARSEYNAPPEIRHRTRINPFTGTVAEGALF
NMEVAPEGIVFPFQLRYRGSEDGLPDALKTVLKWWAEGQAFMSGAASTGKGRFRMENAKYET
LDLSDENQRNDYLKNWGWWRDEKGLEELKKRLNSGLPEPGNYRDPKWHEINVSIEMASPFINGDP
IRAAVDKRGTDVVTFFVKYAEGEAAKPVCAKAESEFRGVIRSAVARIHMEDGVPLTELTHSDCE
CLLCQIFGSEYEAGKIRFEDLVFESDPEPVTFDHVAIDRFTGGAADKKKFDDSPGSPARPLMLK
GSFWIRRDVLEDEEYCKALGKALADVNNGLYPLGGKSAIGYGQVKSGLGKGGDKRISRLMNPAPF
DETDAVAVPEKPKTDAEVRIEAEKVYYPHYFVEPHKKVEREEKPCGHQKFHEGRLTGKIRCKLITK
TPLIVPDTSNDDFFRPADKEARKEKDEYHKSIAFFRLHKQIMIPGSELRGMVSSVYETVTNSCFRI
FDETKRLSWRMDADHQNVLQDFLPGRVTAADGKHIQKFSETARVPFYDKTQKHFDILDEQEIAGE
KPVRMWVKRFIKRLSLVDPKHPQKKQDNKWRRKEGIAFIEQKNGSYFFNVVTNNGCTSFH
LWHKPDNFDQEKLEGIQNGEKLDCWVRDSRYQKAFQEIPENDPDGWECKEGLHVVGPSKVEF
SDKKGDVINNFQGTLPSPNDWKTIRTNDFKNRKRKNPEPVFCCEDDKGNYYTMAKYCETFFFD
LKENEYEIPEKARIKYKELLRVYNNNPQAVPESVFQSRVARENVEKLKSGDLVYFKHNEKYVE
DIVPVRISRTVDDRMIGKRMSADLRPCHGDWVEDGDLSALNAYPEKRLLLRHPKGLCPACRLFG
TGSYKGRVRFGFASLENDPEWLIPGKNPGDPFHGGPVMLSLLERPRPTWSIPGSDNKFVPGRK
YVHHHAWKTIKDGNNHPTTGKAIEQSPNNRTVEALAGGNSFSFEIAFENLKEWELGLLIHSLQLEK
GLAHLKLGMAKSMGFGSVEIDVESVRLRKDWKQWRNGNSEIPNWLKGKFAKLKEWFRDELDFIE
NLKKLLWFPEGDQAPRVCYPMLRKKDDPNGNSGYEELKDGEFKKEDRQKKLTPWTFWA
(SEQ ID NO:17)

FIG. 17

dDiCas7-11

MTTMMKISIEFLEPFRMTKWQESTRRNKNKEFVRGQAFARWHRNKKDNTKGRPYITGTLLRSA
 VIRSAENLLTSDGKISEKTCCPGKFDTEKDRLLQLRQRSTLRWTDKNPCPDNAETYCPFCCELL
 GRSGNDGKKAEEKDWRFRHFGNLSLPGKPDFDGPKAIGSQRVLN RVDFKSGKAHDFFKAYEV
 DHTRFPRFEGEITIDNKVSAEARKLLCDSLKFTDRLCGALCVIRFDEYTPAADSGKQTENVQAEP
 NANLAEKTAEQHISILDDNKKTEYTRLLADAIRSLRRSSKLVAGLPKDHGKDDHYLWDIGKKK
 KDENSVTIRQILTTSADTKELKNAGKWREFCEKLGEALYLKSKDMSGGLKITRRILGDAEFHGKP
 DRLEKSRSVSIGSVLKETVVCGELVAKTPFFFGAIDEDAKQTALQVLLTPDNKYRLPRSAVRGILR
 RDLQTYFDSPCNAELGGRPCMCKTCRIMRGITVMDARSEYNAPPEIRHRTRINPFTGTVAEGALF
 NMEVAPEGIVFPFQLRYRGSEDGLPDALKTVLKWWAEGQAFMSGAASTGKGRFRMENAKYET
 LDLSDENQRNDYLNWGWWRDEKGLEELKKRLNSGLPEPGNYRDPKWHEINVSIMASPFINGDP
 IRAAVDKRGTA VVTFVKYKAEGEEAKPVCAKYAESFRGVIRSAVARIHMEDGVPLTELTHSDCE
 CLLCQIFGSEYEAGKIRFEDLVFESDPEPVTFDHVAIDRFTGGAADKKKFDDSP LPGSPARPLMLK
 GSFWIRRDVLEDEEYCKALGKALADVNNGLYPLGGKSAIGYGQVKSGLGIKGGDKRISRLMNPAF
 DETDVA VPEKPKTDAEVRIEAEKVYYPHYFVEPHKKVEREEKPCGHQKFHEGRLTGKIRCKLITK
 TPLIVPDTSNDDFFRPADKEARKEKDEYHKS YAFFRLHKQIMIPGSELRGMVSSVYETVTNSCFRI
 FDETKRLSWRMDADHQNVLQDFLPGRVTADGKHIQKFSETARVPPYDKTQKHFDILDEQEIAGE
 KPVRMWVKRFIKRLSLVDPKHPQKKQDNKWRRKEGIATFIEQKNGSYYFNVVTNNGCTSFH
 LWHKPDNFDQEKLEGIQNGEKLDCWVRDSRYQKAFQEIPENDPDGWECKEGYLHVVGPSKVEF
 SDKKGDVINNFQGTLPSPNDWKTIRTNDFKNRKRKNEPVFCCEDDKGNYYTMAKYCETFFFD
 LKENEEYEIPEKARIKYKELLRVYNNNPQAVPESVFQSRVARENVEKLKSGDLVYFKHNEKYVE
 DIVPVRISRTVDDRMIGKRMSADLRPCHGDWVEDGDL SALNAYPEKRLLLLRHHPKGLCPACRLFG
 TGSYKGRVRFGFASLENDPEWLIPGKNPGDPFHGGPVMLSLLERPRPTWSIPGSDNKFVKVPGRKF
 YVHHHAWKTIKDG NHPTTGKAIEQSPNNRTVEALAGGNSFSFEIAFENLKEWELGLLIHSLQLEK
 GLAHKLGMAKSMGFGSVEIDVESVRLRKDWKQWRNGNSEIPNWLKGKFAKLKEWFRDELDFIE
 NLKKLLWFPEGDQAPRVCYPMLRKKDDPNGNSGYEELKDGEFKKEDRQKKLTPWTPWA
 (SEQ ID NO:18)

FIG. 18

SEQ ID NO:	TARGETING REGION SEQUENCE	TARGET	
19	TTCTGAATTCTTCCGGATATCGCCTGGATC	RG6 Intron 1 targeting	Example 1
20	AATGGCCTTTTCTGCTCAGTCTGCCCATGG	RG6 Intron 1 targeting	
21	TGCCTTAAAGCTGTGGGGCAGAGCTACGGT	RG6 Intron 1 targeting	
22	GATATCGCCTGGATCCTGAGCCAGGGAGGA	RG6 Intron 1 targeting	
23	GCTCTCCCTGGGCAGGGTGGCTTCGAGGCT	CD81 Intron 4 targeting	Example 1
24	GTGCTCAGCCCCAATGCTGCCTCCTCCATA	CD81 Intron 4 targeting	
25	GGCCCACTAGGTTCACCCGAGGACCAGT	CD81 Intron 5 targeting	
26	CTGCAGAGAGCAGGGGTGCAAGGTCAGCCC	CD81 Intron 5 targeting	
27	TGCCGGGGGGCGGGGGTTCAGGGGCGGGACG	MECP2 Intron 1 Targeting	targeting region for targeting diseased transcripts, for Rett's Syndrome
28	ACAGTATAAAAGTACCCCAAAGACTAAAGA	MECP2 Intron 2 Targeting	
29	CCATGAGGGATCCTTGTCCTGCCCCTCCCT	MECP2 Intron 3 Targeting	
30	CTTAGTTCTACACCTCATTCATTCATTCAG	HTT Intron 1 Targeting	targeting region for targeting diseased transcripts, for Huntington's Disease
31	TACTCCATGTTGCGGGCAGAATGGGGATCT	HTT Intron 1 Targeting	
32	GGGATCTGGACAGGGAAGCACAGGGCACGA	HTT Intron 1 Targeting	
19	TTCTGAATTCTTCCGGATATCGCCTGGATC	RG6 Intron 1 targeting	Example 1

FIG. 19

Target	Gene	Tissue	Protein Size (AAs)	5' or 3' splicing	Minigene?	Location of disease mutation	Genetic Mechanism of Disease	Desired Mechanism of Correction
Spinal muscular atrophy	SMN1/SMN2	CNS	300	3' Splicing	yes	Exon 7	LOF in SMN1 through non-sense mutation or skipping of exon 7	Make functional SMN by including exon 7 in SMN2/SMN1
Huntington's	HTT	CNS	3000	5' Splicing	no	Exon 1	≥40 CAG repeats in exon 1 of HTT	Replace Exon 1 + 5' UTR
Fragile X syndrome	FMR1	CNS	800	5' Splicing	no	5' UTR	Repeat expansion in 5' UTR	Replace Exon1 + 5' UTR
Duchenne Muscular Dystrophy	DMD	Muscle	3664	3' Splicing	no	Exons 43, 51, 53	Variety of deletions, duplications, point mutations resulting in reading frame shift or truncated protein	Replace all exons downstream of faulty exon
Cystic Fibrosis	CFTR	Lung	1480	5' or 3' Splicing	no	Diverse set of mutation types and locations	http://www.350000.56124601.601.028531616165224004.htm	Cargo to replace any diseased exon
Hemophilia A	coagulation factor 8	Liver	2332	3' Splicing	no	Intron 12	Intronic deletion leads to frame shift	replace exons downstream of intron 12
Hemophilia B	coagulation factor 9	Liver	461	5' Splicing	no	Intron 4	Intron 4 point mutation leads to Exon 5 skipping	Replace Exons 1-4 upstream of exon 5
Hemophilia	Factor XIII	Liver	541	3' Splicing	no	any exon	missense or deletion mutation in any exon	Replace unhealthy exons downstream of healthy exons
Rett Syndrome	MECP2	CNS	520	5' or 3' splicing	no	Exon 3 and 4 (Exon 1 and 2 in rare cases)	LOF of MECP2 caused by exonic mutations	Replace 5' or 3' end of transcript to include healthy exons
Endogenous CAR T-Cells	TRB	Spleen	272	5' Splicing	-	-	-	Replace upstream of the constant region of the TCR receptor
PKU	PAH	Liver	452	3' Splicing	no	Across the whole gene	LOF	downstream exon replacement
AATD	SERPINA1/A1AT	Liver	197	3' Splicing	no	Downstream of exon 1	LOF	Exon 1 encodes part of the 5' UTR. Replace with rest of 5' UTR and corrected CDS

FIG. 20

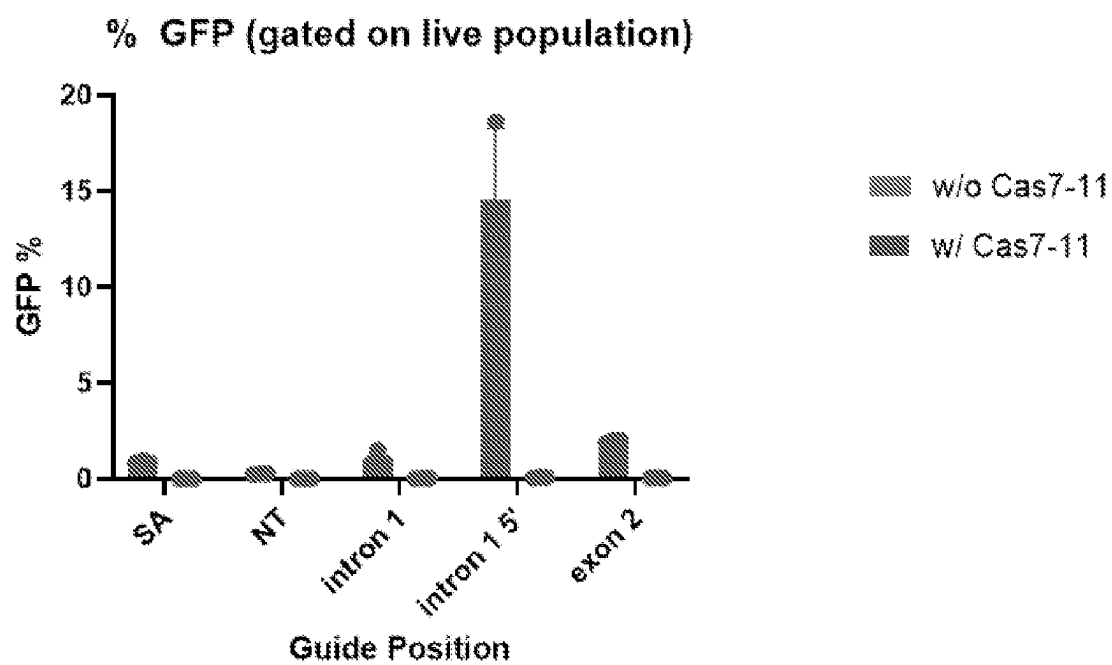


FIG. 21A

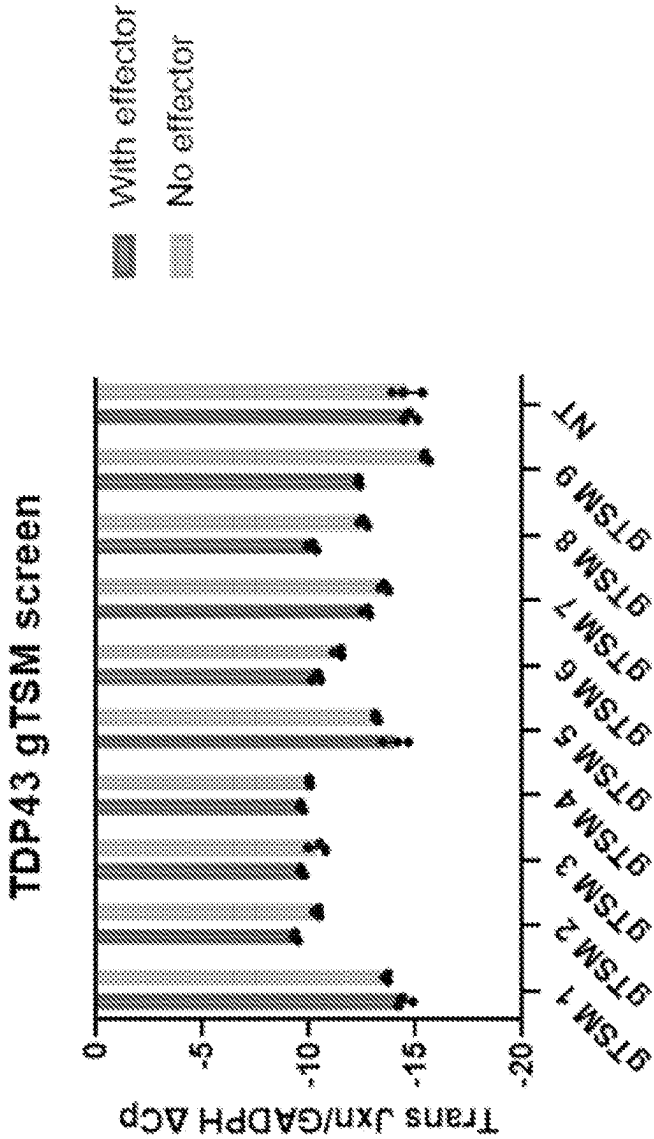


FIG. 21B

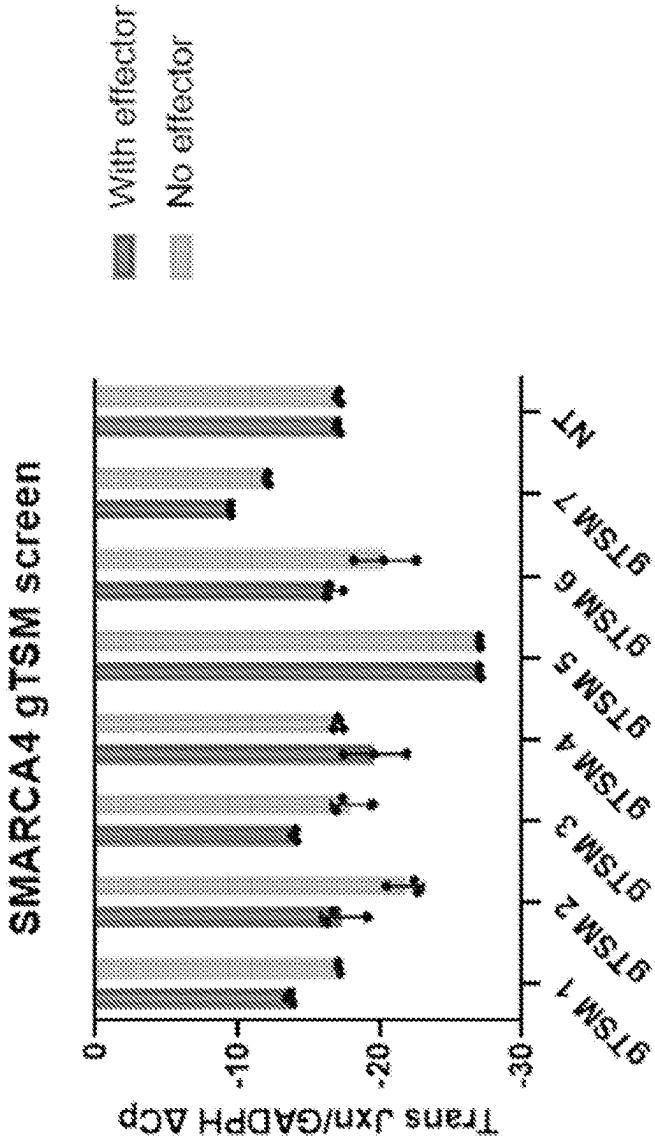


FIG. 21C

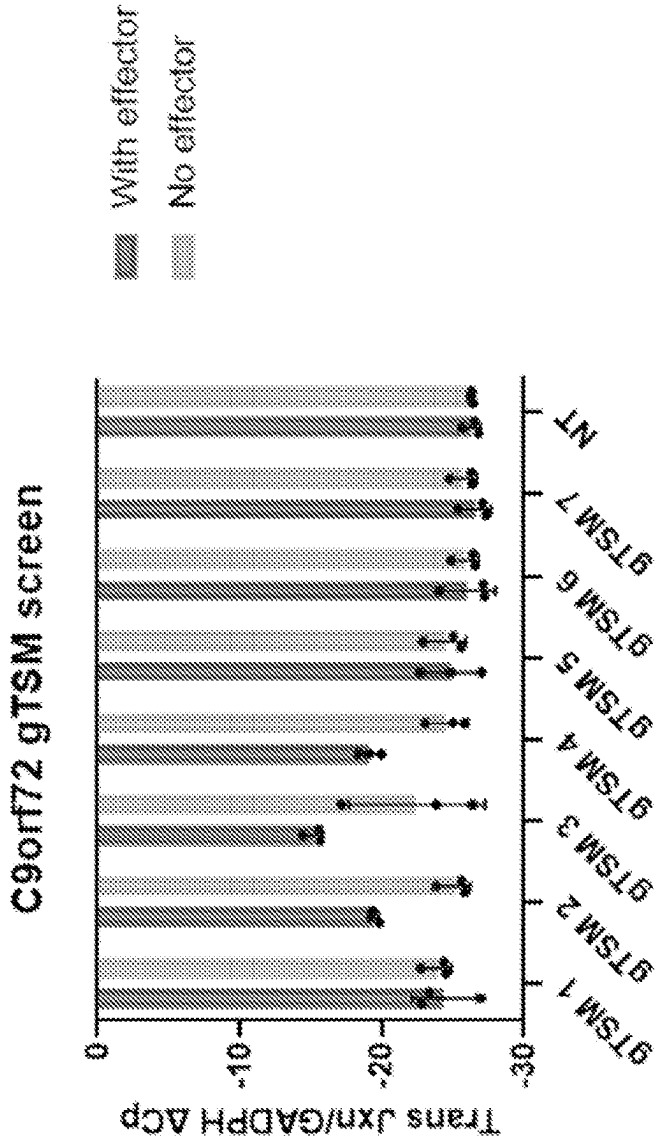


FIG. 21D

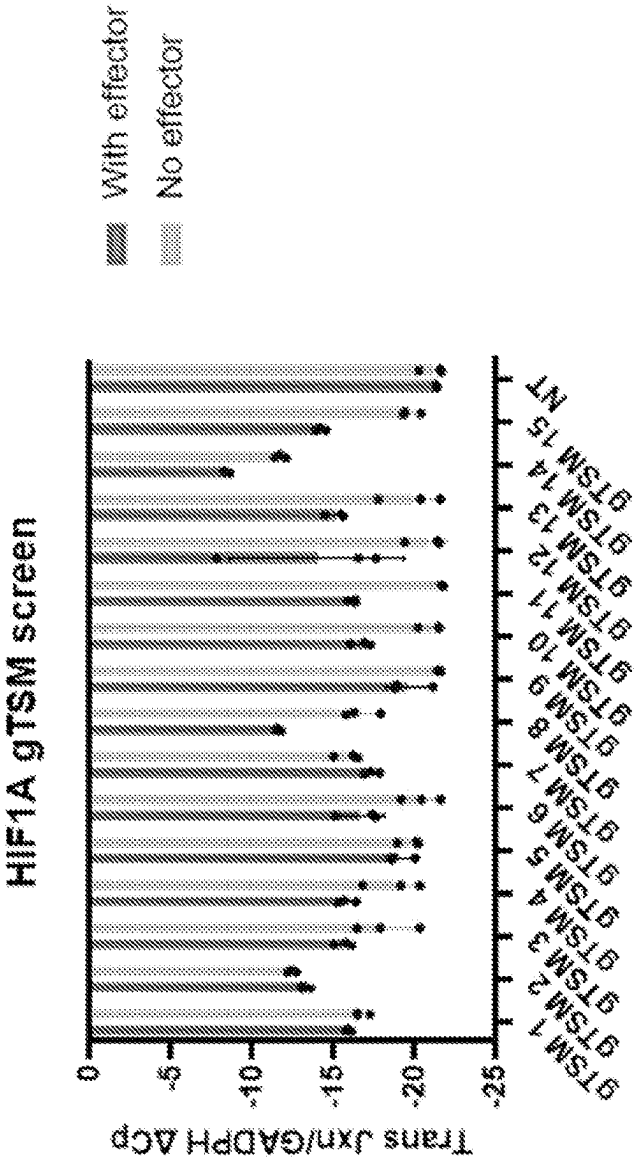


FIG. 21E

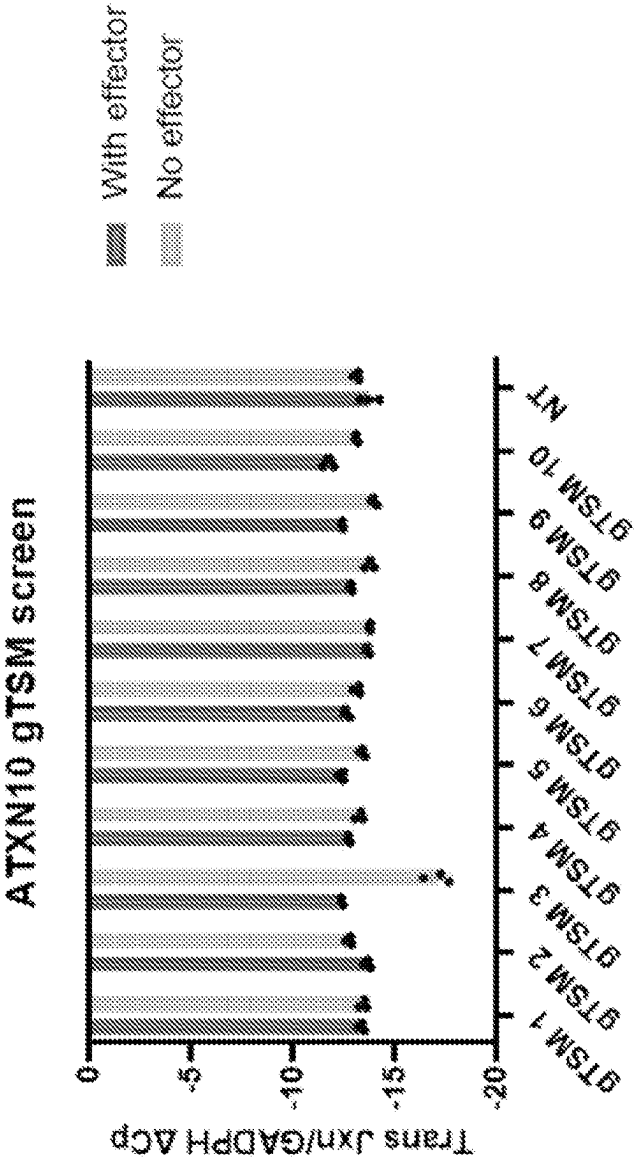


FIG. 22A

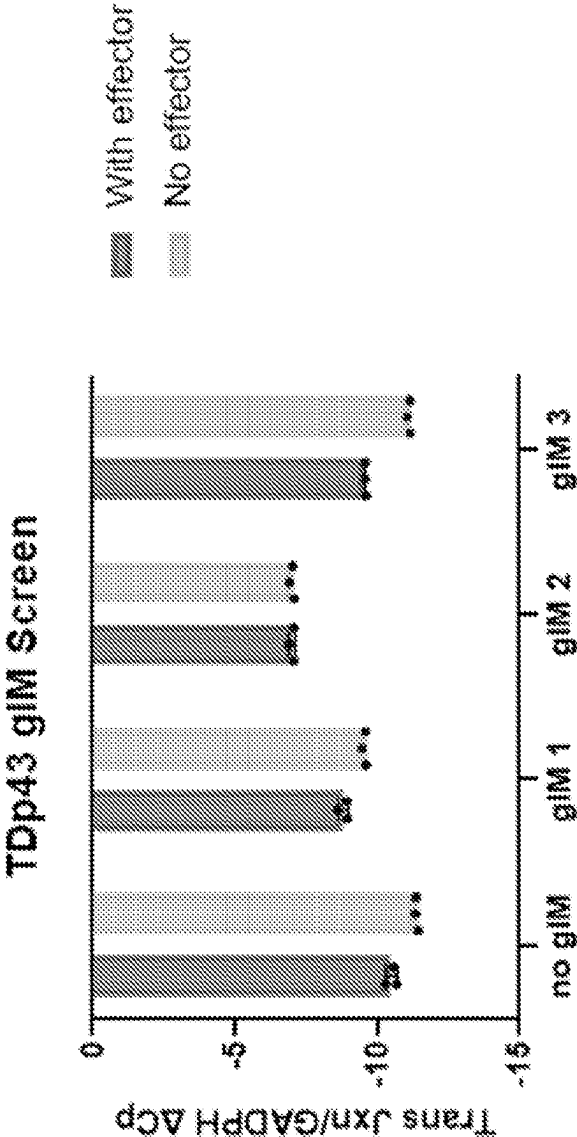


FIG. 22B

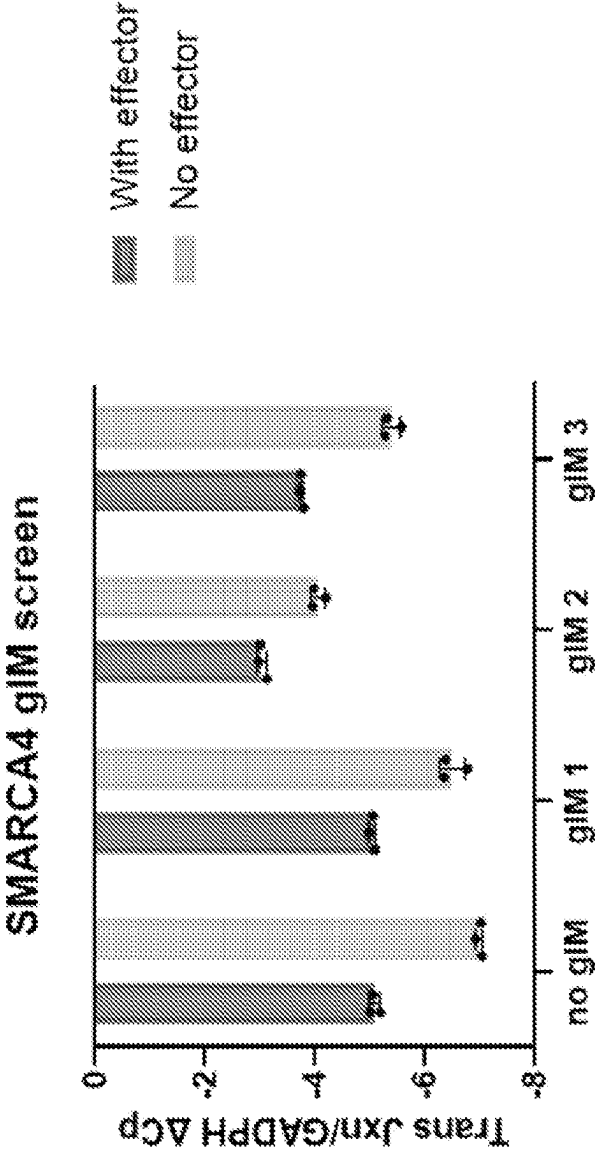


FIG. 23

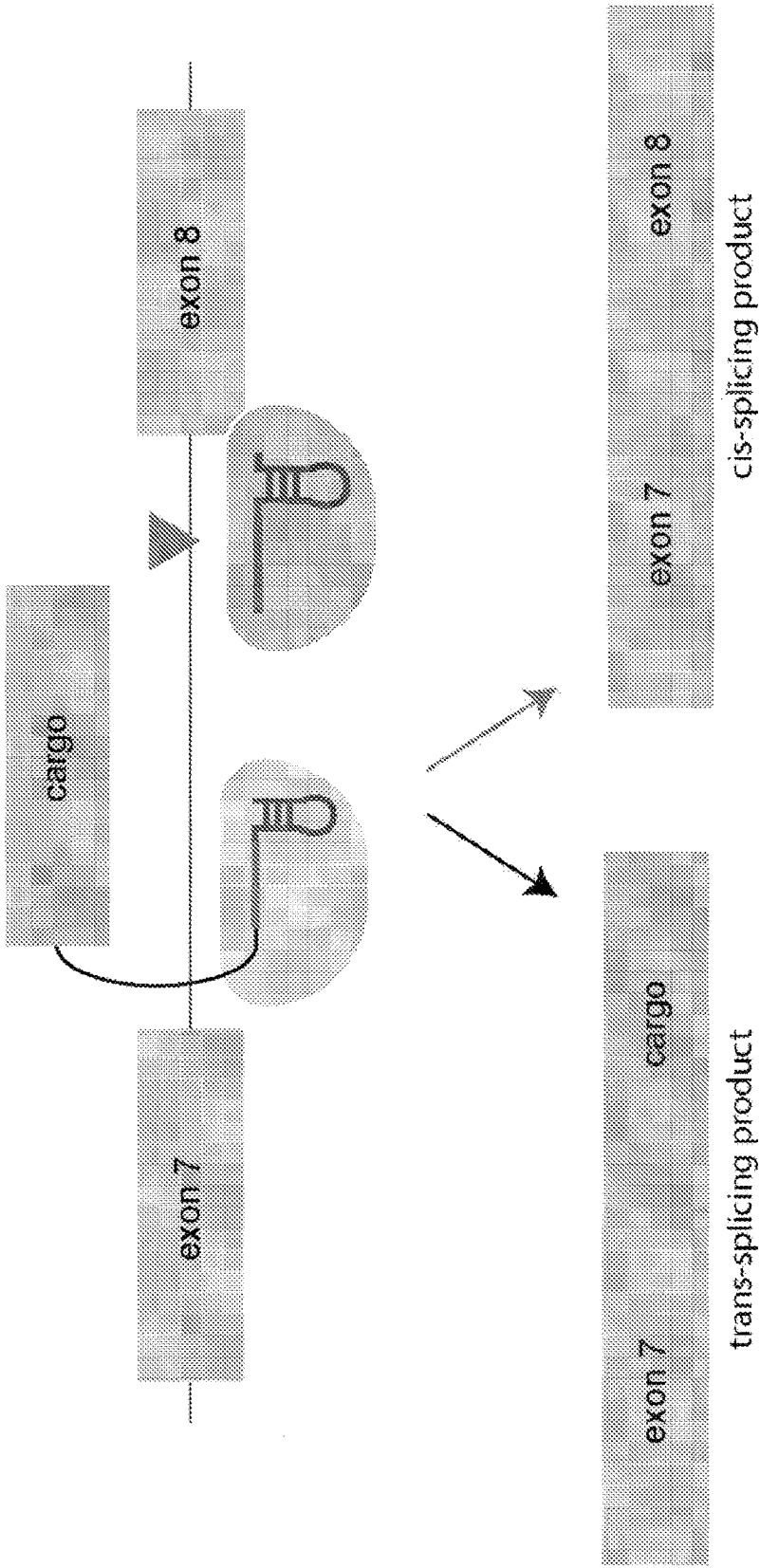


FIG. 24

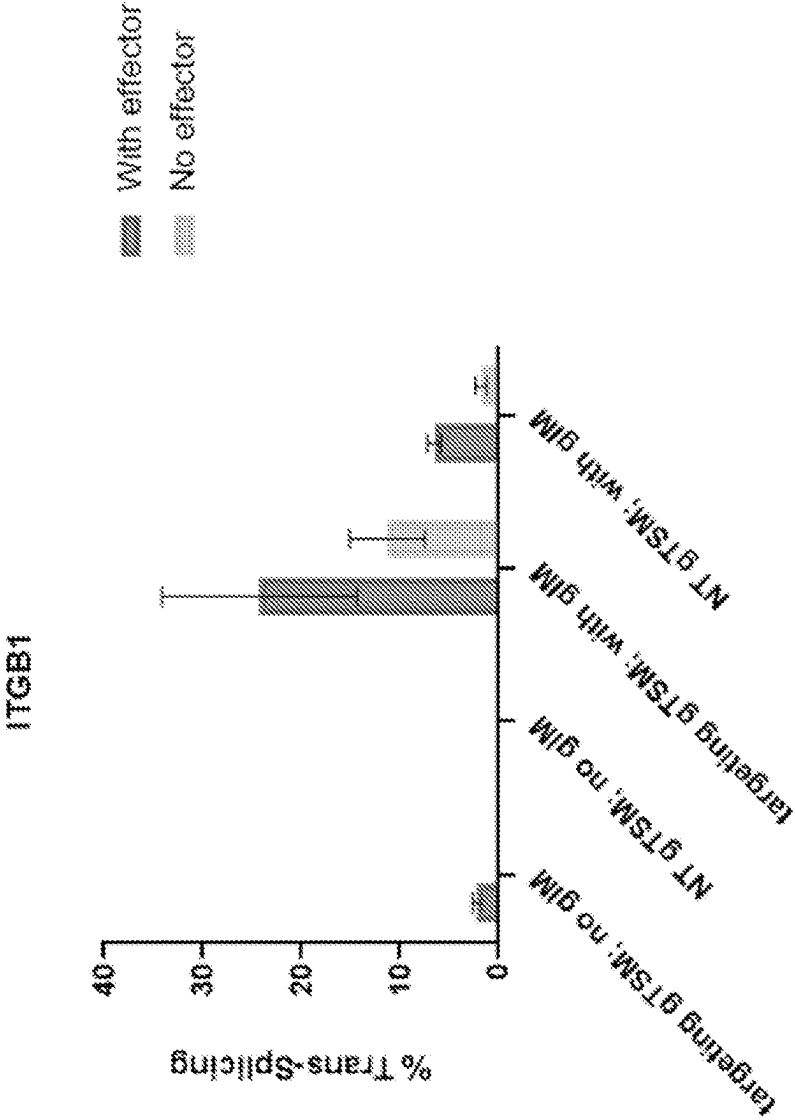
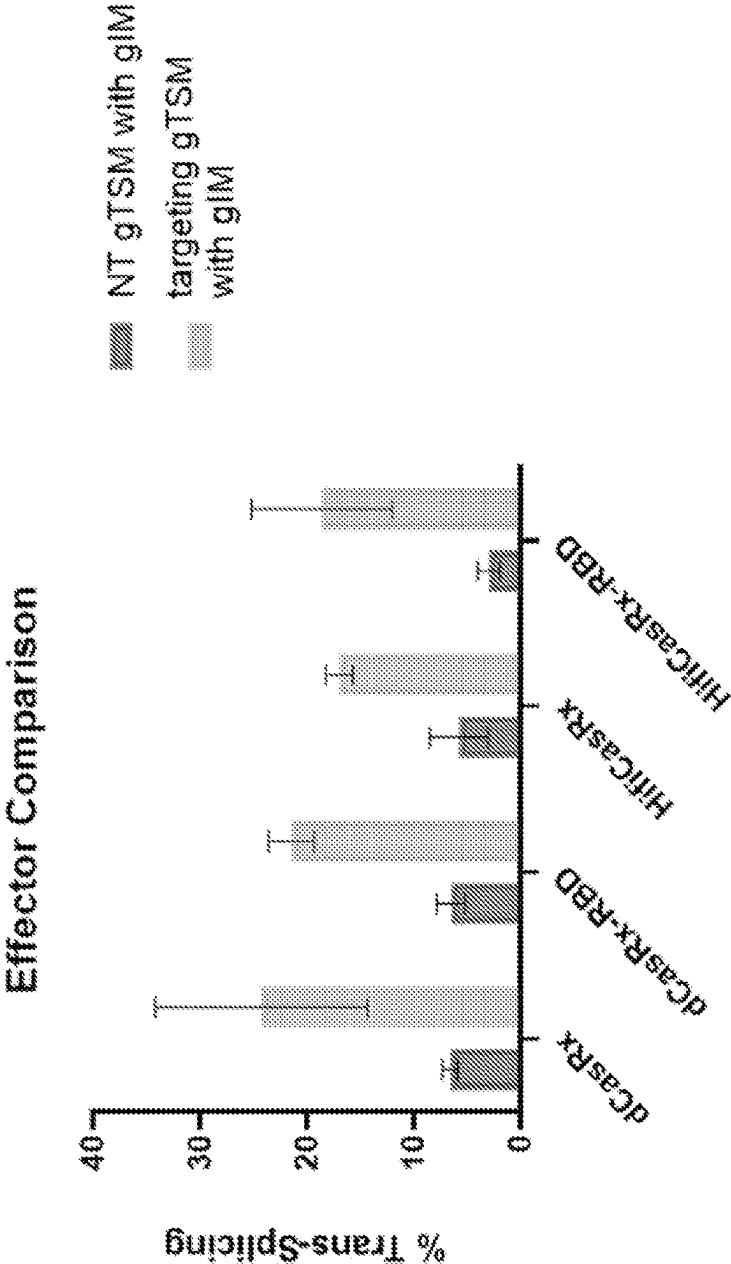


FIG. 25



COMPOSITIONS AND METHODS FOR RNA EDITING

CROSS-REFERENCE

[0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/341,823, filed May 13, 2022, which application is incorporated herein by reference in its entirety.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH

[0002] This invention was made with government support under Grant Nos. GM131073, GM132465, and OD021369, awarded by the National Institutes of Health. The government has certain rights in the invention.

INCORPORATION-BY-REFERENCE OF MATERIAL ELECTRONICALLY SUBMITTED

[0003] A Sequence Listing is provided herewith as a Sequence Listing XML, “BERK-465WO_SEQ_LIST” created on May 10, 2023 and having a size of 70,860 bytes. The contents of the Sequence Listing XML are incorporated by reference herein in their entirety.

INTRODUCTION

[0004] RNA splicing is a fundamental biological process, in which a pre-mRNA transcript is modified by the endogenous spliceosome into a mature mRNA transcript. This standard process involves a single pre-mRNA molecule “in cis.” Some organisms, such as trypanosomes, flatworms, and nematodes, can employ a splicing reaction that joins two distinct RNA molecules “in trans.”

[0005] Methods for editing DNA using editing enzymes have been described and are currently in use for various gene editing applications. There is a need in the art for methods of editing RNA.

SUMMARY

[0006] The present disclosure provides a hybrid RNA molecule comprising a targeting region and a donor RNA, and compositions comprising the hybrid RNA molecule. A hybrid RNA molecule of the present disclosure is useful in methods of modifying a target RNA, which methods are also provided.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1A-1B are a schematic representation of a pathway of cis-splicing (FIG. 1A) and a pathway of trans-splicing (FIG. 1B).

[0008] FIG. 2A-2C schematically depict targeting of a trans-splicing molecule to a transcript of interest (FIG. 2A), interfering with cis-splicing using an RNA targeting protein (FIG. 2B), and guiding a trans-splicing molecule with an RNA targeting protein (FIG. 2C).

[0009] FIG. 3 schematically depicts enhanced localization of a gTSM (a trans splicing molecule (TSM) fused to a guide RNA) to actively transcribed and spliced transcripts using a CTD interacting domain (CID) fused to an RNA targeting protein.

[0010] FIG. 4 schematically depicts enhanced localization of a gTSM to its target through an RS domain fusion to the RNA targeting protein.

[0011] FIG. 5A-5C schematically depict addition of an RNA nuclear localization sequence to the gTSM (FIG. 5A), inclusion of a guided interfering molecule to the gTSM to interrupt cis-splicing (FIG. 5B), and inclusion of a guided interfering molecule targeting DNA to interrupt RNA transcription downstream of the targeted site, thereby reducing the number of transcripts that are able to splice in cis (FIG. 5C).

[0012] FIG. 6A-6C schematically depict 3' trans-splicing (FIG. 6A), 5' trans-splicing (FIG. 6B), and internal trans-splicing (FIG. 6C).

[0013] FIG. 7A-7B schematically depict a mechanism of Huntington's disease (FIG. 7A) and use of trans-splicing to correct Huntington's disease (FIG. 7B).

[0014] FIG. 8A-8B schematically depict a mechanism of spinal muscular atrophy (FIG. 8A) and use of trans-splicing to correct spinal muscular atrophy (FIG. 8B).

[0015] FIG. 9A-9B schematically depict plasmids transfected into HEK293 cells (FIG. 9A) and various gTSM locations (FIG. 9B).

[0016] FIG. 10 depicts the results of transfection of HEK293 cells with the plasmids depicted in FIG. 9A-9B.

[0017] FIG. 11A-11H provide amino acid sequences of various CRISPR-Cas effector polypeptides (SEQ ID NOs: 1-8; respectively).

[0018] FIG. 12 provides the amino acid sequence of a C-terminal domain polypeptide of RNA Polymerase II (SEQ ID NO:9).

[0019] FIG. 13A-13D present schematic depictions of exemplary embodiments of hybrid RNA molecules of the present disclosure.

[0020] FIG. 14 presents schematic depictions of exemplary embodiments of hybrid RNA molecules of the present disclosure.

[0021] FIG. 15A-15G provide examples of suitable internal ribosome entry sites (IRES) (SEQ ID NOs:10-16; respectively).

[0022] FIG. 16 provides the amino acid sequence of a Cas7-11 polypeptide (SEQ ID NOs:17).

[0023] FIG. 17 provides the amino acid sequence of a dCas7-11 polypeptide (SEQ ID NOs:18).

[0024] FIG. 18 provides non-limiting examples of targeting region sequences that can be included in a hybrid RNA molecule of the present disclosure (SEQ ID NOs:19-32 and 19; respectively).

[0025] FIG. 19 provides non-limiting examples of diseases that can be treated using a method of the present disclosure.

[0026] FIG. 20 depicts the effect of catalytically active Cas7-11 on trans-splicing.

[0027] FIG. 21A-21E depict trans-splicing with dCasRx as the effector.

[0028] FIG. 22A-22B depict targeting TDP43 (FIG. 22A) or SMARCA4 (FIG. 22B) with hit gTSM and testing gIMs.

[0029] FIG. 23 schematically depicts use of gTSM and gIM to target the intron between exon 7 and 8 of a target transcript.

[0030] FIG. 24 depicts the percent trans-splicing efficiency for ITGB1, as measured by ddPCR.

[0031] FIG. 25 depicts the percent trans-splicing efficiency for ITGB1, as measured by ddPCR using four different RNA-targeting CRISPR effectors for the gTSM (dCasRx, dCasRx-RBD fusion, HifiCasRx, HifiCasRx-RBD).

DEFINITIONS

[0032] “Heterologous,” as used herein in the context of an RNA, refers to a nucleotide sequence that is not found in the native nucleic acid. For example, a donor RNA is generally heterologous to the targeting region present in a hybrid RNA molecule of the present disclosure. “Heterologous,” as used herein in the context of a polypeptide, refers to an amino acid sequence that is not found in the native polypeptide. For example, a fusion CRISPR-Cas effector polypeptide comprises: a) a CRISPR-Cas effector polypeptide; and b) one or more heterologous polypeptides, where the heterologous polypeptide comprises an amino acid sequence from a protein other than a CRISPR-Cas effector polypeptide.

[0033] “Recombinant,” or “hybrid,” as used herein, means that a particular nucleic acid (DNA or RNA) is the product of various combinations of cloning, restriction, polymerase chain reaction (PCR) and/or ligation steps resulting in a construct having a structural coding or non-coding sequence distinguishable from endogenous nucleic acids found in natural systems. For example, a hybrid RNA molecule of the present disclosure can be generated by either chemical synthesis means, or by the artificial manipulation of isolated segments of nucleic acids, e.g., by genetic engineering techniques.

[0034] The terms “polynucleotide” and “nucleic acid,” used interchangeably herein, refer to a polymeric form of nucleotides of any length, either ribonucleotides or deoxyribonucleotides. Thus, this term includes, but is not limited to, single-, double-, or multi-stranded DNA or RNA, genomic DNA, cDNA, DNA-RNA hybrids, or a polymer comprising purine and pyrimidine bases or other natural, chemically or biochemically modified, non-natural, or derivatized nucleotide bases.

[0035] By “hybridizable” or “complementary” or “substantially complementary” it is meant that a nucleic acid (e.g. RNA, DNA) comprises a sequence of nucleotides that enables it to non-covalently bind, i.e. form Watson-Crick base pairs and/or G/U base pairs, “anneal”, or “hybridize,” to another nucleic acid in a sequence-specific, antiparallel, manner (i.e., a nucleic acid specifically binds to a complementary nucleic acid) under the appropriate in vitro and/or in vivo conditions of temperature and solution ionic strength. Standard Watson-Crick base-pairing includes: adenine (A) pairing with thymidine (T), adenine (A) pairing with uracil (U), and guanine (G) pairing with cytosine (C) [DNA, RNA]. In addition, for hybridization between two RNA molecules (e.g., dsRNA), and for hybridization of a DNA molecule with an RNA molecule (e.g., when a DNA target nucleic acid base pairs with a guide RNA, etc.): guanine (G) can also base pair with uracil (U). For example, G/U base-pairing is at least partially responsible for the degeneracy (i.e., redundancy) of the genetic code in the context of tRNA anti-codon base-pairing with codons in mRNA. Thus, in the context of this disclosure, a guanine (G) (e.g., of dsRNA duplex of a guide RNA molecule; of a guide RNA base pairing with a target nucleic acid, etc.) is considered complementary to both a uracil (U) and to an adenine (A). For example, when a G/U base-pair can be made at a given nucleotide position of a dsRNA duplex of a guide RNA molecule, the position is not considered to be non-complementary, but is instead considered to be complementary.

[0036] It is understood that the sequence of a polynucleotide need not be 100% complementary to that of its target

nucleic acid to be specifically hybridizable or hybridizable. Moreover, a polynucleotide may hybridize over one or more segments such that intervening or adjacent segments are not involved in the hybridization event (e.g., a bulge, a loop structure or hairpin structure, etc.). A polynucleotide can comprise 60% or more, 65% or more, 70% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, 98% or more, 99% or more, 99.5% or more, or 100% sequence complementarity to a target region within the target nucleic acid sequence to which it will hybridize. For example, an antisense nucleic acid in which 18 of 20 nucleotides of the antisense compound are complementary to a target region, and would therefore specifically hybridize, would represent 90 percent complementarity. In this example, the remaining noncomplementary nucleotides may be clustered or interspersed with complementary nucleotides and need not be contiguous to each other or to complementary nucleotides. Percent complementarity between particular stretches of nucleic acid sequences within nucleic acids can be determined using any convenient method. Example methods include BLAST programs (basic local alignment search tools) and PowerBLAST programs (Altschul et al., *J. Mol. Biol.*, 1990, 215, 403-410; Zhang and Madden, *Genome Res.*, 1997, 7, 649-656), the Cap program (Wisconsin Sequence Analysis Package, Version 8 for Unix, Genetics Computer Group, University Research Park, Madison Wis.), e.g., using default settings, which uses the algorithm of Smith and Waterman (*Adv. Appl. Math.*, 1981, 2, 482-489), and the like.

[0037] The terms “regulatory sequences,” “control elements,” and “regulatory elements,” used interchangeably herein, refer to transcriptional and translational control sequences, such as promoters, enhancers, polyadenylation signals, terminators, protein degradation signals, and the like, that provide for and/or regulate transcription.

[0038] As used herein, a “promoter” or a “promoter sequence” is a DNA regulatory region capable of binding RNA polymerase and initiating transcription of a downstream (3' direction) coding or non-coding sequence. For purposes of the present disclosure, the promoter sequence is bounded at its 3' terminus by the transcription initiation site and extends upstream (5' direction) to include the minimum number of bases or elements necessary to initiate transcription at levels detectable above background. Within the promoter sequence will be found a transcription initiation site, as well as protein binding domains responsible for the binding of RNA polymerase. Eukaryotic promoters will often, but not always, contain “TATA” boxes and “CAT” boxes. Various promoters, including inducible promoters, may be used to drive expression of the various nucleic acids of the present disclosure.

[0039] By “embryonic germ stem cell” (EGSC) or “embryonic germ cell” or “EG cell” is meant a PSC that is derived from germ cells and/or germ cell progenitors, e.g. primordial germ cells, i.e. those that would become sperm and eggs. Embryonic germ cells (EG cells) are thought to have properties similar to embryonic stem cells as described above. Examples of methods of generating and characterizing EG cells may be found in, for example, U.S. Pat. No. 7,153,684; Matsui, Y., et al., (1992) *Cell* 70:841; Shambloott, M., et al. (2001) *Proc. Natl. Acad. Sci. USA* 98: 113; Shambloott, M., et al. (1998) *Proc. Natl. Acad. Sci. USA*,

95:13726; and Koshimizu, U., et al. (1996) Development, 122:1235, the disclosures of which are incorporated herein by reference.

[0040] The term “induced pluripotent stem cell” or “iPSC” refers to a PSC that is derived from a cell that is not a PSC (i.e., from a cell this is differentiated relative to a PSC). iPSCs can be derived from multiple different cell types, including terminally differentiated cells. iPSCs have an ES cell-like morphology, growing as flat colonies with large nucleocytoplasmic ratios, defined borders and prominent nuclei. In addition, iPSCs express one or more key pluripotency markers known by one of ordinary skill in the art, including but not limited to Alkaline Phosphatase, SSEA3, SSEA4, Sox2, Oct3/4, Nanog, TRA160, TRA181, TDGF1, Dnmt3b, FoxD3, GDF3, Cyp26a1, TERT, and zfp42. Examples of methods of generating and characterizing iPSCs may be found in, for example, U.S. Patent Publication Nos. US20090047263, US20090068742, US20090191159, US20090227032, US20090246875, and US20090304646, the disclosures of which are incorporated herein by reference. Generally, to generate iPSCs, somatic cells are provided with reprogramming factors (e.g. Oct4, SOX2, KLF4, MYC, Nanog, Lin28, etc.) known in the art to reprogram the somatic cells to become pluripotent stem cells.

[0041] The term “somatic cell” refers to any cell in an organism that, in the absence of experimental manipulation, does not ordinarily give rise to all types of cells in an organism. In other words, somatic cells are cells that have differentiated sufficiently that they will not naturally generate cells of all three germ layers of the body, i.e. ectoderm, mesoderm and endoderm. For example, somatic cells would include both neurons and neural progenitors, the latter of which may be able to naturally give rise to all or some cell types of the central nervous system but cannot give rise to cells of the mesoderm or endoderm lineages.

[0042] The term “mitotic cell” refers to a cell undergoing mitosis. Mitosis is the process by which a eukaryotic cell separates the chromosomes in its nucleus into two identical sets in two separate nuclei. It is generally followed immediately by cytokinesis, which divides the nuclei, cytoplasm, organelles and cell membrane into two cells containing roughly equal shares of these cellular components.

[0043] The term “post-mitotic cell” refers to a cell that has exited from mitosis, i.e., it is “quiescent”, i.e. it is no longer undergoing divisions. This quiescent state may be temporary, i.e. reversible, or it may be permanent.

[0044] The term “meiotic cell” refers to a cell that is undergoing meiosis. Meiosis is the process by which a cell divides its nuclear material for the purpose of producing gametes or spores. Unlike mitosis, in meiosis, the chromosomes undergo a recombination step which shuffles genetic material between chromosomes. Additionally, the outcome of meiosis is four (genetically unique) haploid cells, as compared with the two (genetically identical) diploid cells produced from mitosis.

[0045] As used herein, the terms “treatment,” “treating,” and the like, refer to obtaining a desired pharmacologic and/or physiologic effect. The effect may be prophylactic in terms of completely or partially preventing a disease or symptom thereof and/or may be therapeutic in terms of a partial or complete cure for a disease and/or adverse effect attributable to the disease. “Treatment,” as used herein, covers any treatment of a disease in a mammal, e.g., in a human, and includes: (a) preventing the disease from occur-

ring in a subject which may be predisposed to the disease but has not yet been diagnosed as having it; (b) inhibiting the disease, i.e., arresting its development; and (c) relieving the disease, i.e., causing regression of the disease.

[0046] The terms “individual,” “subject,” “host,” and “patient,” used interchangeably herein, refer to an individual organism, e.g., a mammal, including, but not limited to, murines, simians, humans, non-human primates, ungulates, felines, canines, bovines, ovines, mammalian farm animals, mammalian sport animals, and mammalian pets.

[0047] Before the present invention is further described, it is to be understood that this invention is not limited to particular embodiments described, as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the present invention will be limited only by the appended claims.

[0048] Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the invention. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges, and are also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the invention.

[0049] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although any methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the present invention, the preferred methods and materials are now described. All publications mentioned herein are incorporated herein by reference to disclose and describe the methods and/or materials in connection with which the publications are cited.

[0050] It must be noted that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a hybrid RNA molecule” includes a plurality of such molecules and reference to “the secondary structure” includes reference to one or more secondary structures and equivalents thereof known to those skilled in the art, and so forth. It is further noted that the claims may be drafted to exclude any optional element. As such, this statement is intended to serve as antecedent basis for use of such exclusive terminology as “solely,” “only” and the like in connection with the recitation of claim elements, or use of a “negative” limitation.

[0051] It is appreciated that certain features of the invention, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the invention, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable sub-combination. All combinations of the embodiments pertaining to the invention are specifically embraced by the present invention and are disclosed herein just as if each and every combination was individually and explicitly disclosed. In addition, all sub-combinations of the

various embodiments and elements thereof are also specifically embraced by the present invention and are disclosed herein just as if each and every such sub-combination was individually and explicitly disclosed herein.

[0052] The publications discussed herein are provided solely for their disclosure prior to the filing date of the present application. Nothing herein is to be construed as an admission that the present invention is not entitled to antedate such publication by virtue of prior invention. Further, the dates of publication provided may be different from the actual publication dates which may need to be independently confirmed.

DETAILED DESCRIPTION

[0053] The present disclosure provides a hybrid RNA molecule comprising a targeting region and a donor RNA, and compositions comprising the hybrid RNA molecule. A hybrid RNA molecule of the present disclosure is useful in methods of modifying a target RNA, which methods are also provided.

Hybrid RNA Molecules

[0054] The present disclosure provides a hybrid RNA molecule comprising: a) a targeting region that comprises a nucleotide sequence that is complementary to a target sequence of a target RNA (“targeting region”); and b) a donor RNA (“donor region” or “donor sequence”).

[0055] A hybrid RNA molecule will in some cases comprise, in addition to a targeting region and a donor RNA, one or more of: (i) a binding region that binds to an RNA binding or modifying polypeptide (“binding region”); (ii) a protein-recruiting region; (iii) a splice function region; and (iv) a stabilization region. FIG. 13A-13D provide schematic depictions of various exemplary, non-limiting embodiments of a hybrid RNA molecule of the present disclosure. For example, in some cases, a hybrid RNA molecule comprises: (a) a targeting region; and (b) a donor sequence, as depicted schematically in FIG. 13A. As another example, in some cases, a hybrid RNA molecule comprises: (a) a targeting region; (b) a donor sequence; and (c) a binding region, as depicted schematically in FIG. 13B. As another example, in some cases, a hybrid RNA molecule comprises: (a) a targeting region; (b) a donor sequence; (c) a binding region; and (d) a splice function region, as depicted schematically in FIG. 13C. As another example, in some cases, a hybrid RNA molecule comprises: (a) a targeting region; (b) a donor sequence; (c) a binding region; (d) a splice function region; and (e) one or more stabilization regions, as depicted schematically in FIG. 13D. Any of these embodiments may further include a protein recruiting region.

[0056] In some cases, the targeting region and the donor region are in the same RNA molecule; e.g., some cases, the targeting region and the donor region are linked via a phosphodiester bond, or another type of covalent bonds, such as a phosphorothioate linkage, in the same RNA molecule. Alternatively, the targeting region and the donor region can be in two separate RNA molecules that are non-covalently associated with one another, where the non-covalent association can be via: i) hybridization of a stretch of contiguous nucleotides in a first RNA molecule comprising the targeting region with a stretch of contiguous nucleotides in a second RNA molecule comprising the donor region; ii) binding of an avidin moiety present in a first RNA

molecule comprising the targeting region with a biotin moiety present in a second RNA molecule comprising the donor region. As another alternative, a first RNA molecule comprising the targeting region and a second RNA molecule comprising the donor region can be covalently linked to one another via a linkage other than a phosphodiester bond, e.g., where the linkage is via a click chemistry reaction (see, e.g., Fantoni et al. (2021) Chem. Rev. 12:7122).

Targeting Region

[0057] As noted above, a hybrid RNA molecule of the present disclosure comprises a targeting region that comprises a nucleotide sequence that is complementary to a target sequence of a target RNA. The nucleotide sequence that is complementary to a target sequence of a target RNA can have a length of from about 10 nucleotides (nt) to about 200 nt (e.g., from about 10 nt to about 15 nt, from about 15 nt to about 20 nt, from about 20 nt to about 25 nt, from about 25 nt to about 30 nt, from about 30 nt to about 40 nt, from about 40 nt to about 50 nt, from about 50 nt to about 60 nt, from about 60 nt to about 70 nt, from about 70 nt to about 80 nt, from about 80 nt to about 90 nt, from about 90 nt to about 100 nt, from about 100 nt to about 125 nt, from about 125 nt to about 150 nt, from about 150 nt to about 175 nt, or from about 175 nt to about 200). In some cases, the nucleotide sequence that is complementary to a target sequence of a target RNA has a length of from about 10 nt to about 20 nt. In some cases, the nucleotide sequence that is complementary to a target sequence of a target RNA has a length of from about 10 nt to about 15 nt. In some cases, the nucleotide sequence that is complementary to a target sequence of a target RNA has a length of from about 15 nt to about 20 nt. In some cases, the nucleotide sequence that is complementary to a target sequence of a target RNA has a length of from about 10 nt to about 50 nt. In some cases, the nucleotide sequence that is complementary to a target sequence of a target RNA has a length of from about 25 nt to about 50 nt. In some cases, the nucleotide sequence that is complementary to a target sequence of a target RNA has a length of from about 50 nt to about 100 nt. In some cases, the nucleotide sequence that is complementary to a target sequence of a target RNA has a length of from about 100 nt to about 200 nt.

[0058] The nucleotide sequence that is complementary to a target sequence of a target RNA will in some cases have 100% complementarity to the target sequence, e.g., 100% complementarity over the same number of contiguous nucleotides. For example, if the nucleotide sequence that is complementary to a target sequence of a target RNA has a length of 20 nt, in some cases, the target sequence in the target RNA will have a length of 20 nt and in some cases, the nucleotide sequence that is complementary to the target sequence will have 100% complementarity to the target sequence.

[0059] In other cases, the nucleotide sequence that is complementary to the target sequence in a target RNA will have less than 100% complementarity to a target sequence of a target RNA, e.g., the nucleotide sequence that is complementary to the target sequence will have 1, 2, 3, 4, or 5 mismatches compared to the target sequence. In some cases, the nucleotide sequence that is complementary to the target sequence in a target RNA will have a single mismatch to the target sequence. In some cases, the nucleotide sequence that is complementary to the target sequence in a

target RNA will have 2 mismatches to the target sequence. In some cases, the nucleotide sequence that is complementary to the target sequence in a target RNA will have 3 mismatches to the target sequence. In some cases, the nucleotide sequence that is complementary to the target sequence in a target RNA will have 4 mismatches to the target sequence. In some cases, the nucleotide sequence that is complementary to the target sequence in a target RNA will have 5 mismatches to the target sequence.

[0060] The target sequence in the target RNA can be any sequence of interest. For example, the target sequence in the target RNA can be: (i) a nucleotide sequence in an exon (e.g., a nucleotide sequence encoding all or a portion of a polypeptide); (ii) a nucleotide sequence in a 5' splice sequence; (iii) a nucleotide sequence in a 3' splice sequence; (iv) a nucleotide sequence in an intron; (v) a nucleotide sequence in a regulatory region of a target RNA; (vi) a nucleotide sequence forming a secondary structure in a target RNA; (vii) a nucleotide sequence in a protein-binding site of an RNA; and the like.

Donor RNA

[0061] As noted above, a hybrid RNA molecule of the present disclosure comprises a donor RNA. The donor RNA comprises a nucleotide sequence that will be inserted into a target RNA or that will replace a portion of a target RNA.

[0062] In some cases, a donor RNA has a length of from 1 nucleotide (nt) to 9000 nt; e.g., the donor RNA can have a length of from 1 nt to 10 nt, from 10 nt to 15 nt, from 15 nt to 20 nt, from 20 nt to 25 nt, from 25 nt to 30 nt, from 30 nt to 40 nt, from 40 nt to 50 nt, from 50 nt to 60 nt, from 60 nt to 70 nt, from 70 nt to 80 nt, from 80 nt to 90 nt, from 90 nt to 100 nt, from 100 nt to 250 nt, from 250 nt to 500 nt, from 500 nt to 1000 nt, from 1000 nt to 2000 nt, from 2000 nt to 3000 nt, from 3000 nt to 4000 nt, from 4000 nt to 5000 nt, from 5000 nt to 6000 nt, from 6000 nt to 7000 nt, from 7000 nt to 8000 nt, or from 8000 nt to 9000 nt. In some cases, a donor RNA has a length greater than 8000 nt. In some cases, a donor RNA has a length of 1 nt, 2 nt, 3 nt, 4 nt, 5 nt, 6 nt, 7 nt, 8 nt, 9 nt, or 10 nt. In some cases, a donor RNA has a length of from 10 nt to 20 nt (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 nt). In some cases, a donor RNA has a length of from 20 nt to 30 nt (e.g., 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nt). In some cases, a donor RNA has a length of from 20 nt to 100 nt. In some cases, a donor RNA has a length of from 100 nt to 200 nt. In some cases, a donor RNA has a length of from 100 nt to 250 nt. In some cases, a donor RNA has a length of from 100 nt to 500 nt. In some cases, a donor RNA has a length of from 100 nt to 1000 nt. In some cases, a donor RNA has a length of from 1000 nt to 1500 nt. In some cases, a donor RNA has a length of from 1500 nt to 2000 nt. In some cases, a donor RNA has a length of from 1500 nt to 2500 nt. In some cases, a donor RNA has a length of from 2000 nt to 8000 nt. In some cases, a donor RNA has a length of from 2000 nt to 7000 nt. In some cases, a donor RNA has a length of from 2000 nt to 6000 nt. In some cases, a donor RNA has a length of from 2000 nt to 5000 nt.

[0063] The donor RNA can include one or more of: a) a coding region comprising a nucleotide sequence encoding all or a portion of a polypeptide; b) an untranslated region (UTR); c) a bar code (i.e., a nucleotide sequence that is unique to the donor RNA that distinguishes it from other nucleic acids); d) an internal ribosomal entry site (IRES); e)

a poly(pyrimidine) tract (e.g., a poly(A) tract or a poly(U) tract); f) an RNA response element; g) an intron; h) an aptamer (e.g., a fluorescent aptamer).

Coding Region

[0064] In some cases, the donor RNA comprises a nucleotide sequence encoding all or a portion of a polypeptide. For example, in some cases, the donor RNA comprises a nucleotide sequence encoding from 1 to 3000 contiguous amino acids (e.g., from 1 to 5, from 5 to 10, from 10 to 15, from 15 to 20, from 20 to 25, from 25 to 30, from 30 to 40, from 40 to 50, from 50 to 60, from 60 to 70, from 70 to 80, from 80 to 90, from 90 to 100, from 100 to 150, from 150 to 200, from 200 to 250, from 250 to 300, from 300 to 400, from 400 to 500, from 500 to 1000, from 1000 to 1250, from 1250 to 1500, from 1500 to 2000, from 2000 to 2500, or from 2500 to 3000, contiguous amino acids). In some cases, the polypeptide, or portion thereof, encoded by a donor RNA is a non-diseased form of a diseased protein (or portion thereof). In some cases, the polypeptide, or portion thereof, encoded by a donor RNA, is a functional form of a non-functional protein (or portion thereof). In some cases, the polypeptide, or portion thereof, encoded by a donor RNA is a fluorescent protein, thus adding a fluorescent tag to the protein encoded by the target RNA. Suitable fluorescent proteins include, but are not limited to, a green fluorescent protein, a yellow fluorescent protein, a blue fluorescent protein, and a red fluorescent protein. In some cases, the polypeptide, or portion thereof, encoded by a donor RNA, is an epitope tag, thus enabling affinity purification of the protein encoded by the target RNA. Suitable epitope tags include, but are not limited to, a FLAG tag, a hemagglutinin (HA), a SNAP tag, a c-Myc tag, and a HALO tag. In some cases, the polypeptide, or portion thereof, encoded by a donor RNA, is an effector polypeptide, thus fusing an effector polypeptide to the protein encoded in the target RNA. Suitable effector polypeptides include, but are not limited to, polypeptides that can modify a nucleotide or ribonucleotide (e.g., a cytidine deaminase, a pentatricopeptide repeat (PPR) protein, an adenosine deaminase, an adenosine deaminase acting on RNA (ADAR) family protein, or an APOBEC family protein); polypeptides that can enable proximity-based protein labeling and identification (e.g., a biotin ligase (such as BirA) or a peroxidase (such as APEX2) in order to biotinylate proteins that interact with the protein encoded by the target RNA); RNA editing enzymes (e.g., RNA deaminases, e.g., ADAR polypeptides, including A to I and/or C to U editing enzymes); helicases; RNA-binding proteins; and the like.

IRES

[0065] As noted above, in some cases, the donor RNA present in a hybrid RNA molecule of the present disclosure includes one or more (e.g., 1, 2, 3, 4, or 5, or more) internal ribosome entry sites (IRES). IRES elements are able to bypass the ribosome scanning model of 5' methylated cap-dependent translation and begin translation at internal sites. Thus, an IRES provides a structure to which a ribosome can bind that does not need to be at the 5' end of the mRNA. An IRES can therefore allow a ribosome to initiate translation at a second (or additional) initiation codon within a mRNA, allowing more than one polypeptide to be produced from a single mRNA.

[0066] Suitable IRESs include, but are not limited to, a picornavirus IRES, an Aphthovirus IRES, a Kaposi's sarcoma-associated herpesvirus IRES, a hepatitis A virus IRES, a hepatitis C virus IRES, a pestivirus IRES, a crippavirus IRES, a Friend murine leukemia virus IRES, a Moloney murine leukemia virus IRES, a Rouse sarcoma virus IRES, a human immunodeficiency virus IRES, a mammalian IRES, and the like. In some cases, the IRES is an IRES from a picornavirus (e.g., a pest viruses (CFPV), a polio virus (PV), an encephalomyocarditis virus (ECMV), a foot-and-mouth disease virus (FMDV), a hepatitis C virus (HCV), a classical swine fever virus (CSFV), a murine leukemia virus (MLV), a simian immune deficiency virus (SIV), or a cricket paralysis viruses (CPV). An IRES can be derived from a eukaryotic non-translated region, e.g., from BiP, fibroblast growth factor-1 (FGF-1), FGF-2, and the like. An IRES can include a naturally-occurring sequence. An IRES can be a synthetic IRES (e.g., comprising a nucleotide sequence not found in nature). Non-limiting examples of IRES sequences are provided in FIG. 15A-15G.

[0067] In some cases, the donor sequence comprises an eToehold. See, e.g., Zhao et al. (2021) *Nature Biotechnol.* 40:539.

[0068] In some cases, the donor RNA comprises a nucleotide sequence that provides a functional 3' splice site. In some cases, the donor RNA comprises a nucleotide sequence that provides a functional 5' splice site. In some cases, the donor RNA comprises a nucleotide sequence that is all or a portion of an intron. Any of these components can comprises a naturally-occurring nucleotide sequence, or can be a synthetic (non-naturally-occurring) sequence.

Splice Function Region

[0069] In some cases, a hybrid RNA molecule of the present disclosure comprises a splice function region. A splice function region can include one or more of: a) a splice modifying sequence; b) a splice exclusion sequence; c) an intron; d) a branch point; e) an intronic splice enhancer; f) an intronic splice silencer; g) a poly(pyrimidine tract); and h) a SIRLOIN. Any of these components can comprises a naturally-occurring nucleotide sequence, or can be a synthetic (non-naturally-occurring) sequence.

[0070] Branch points map within short motifs upstream of acceptor splice sites and are essential for splicing of pre-mature mRNA. Branch points are known in the art; see, e.g., Leman et al. (2020) *BMC Genomics* 21:86. As one non-limiting example, a branch point can comprise the sequence: TGCTAAC. As another non-limiting example, a branch point can comprise the sequence: CTGAT.

[0071] Suitable poly(pyrimidine) tracts include, e.g., TCTCTTTTTTTCTC (SEQ ID NO:35); and TTCCCTCTCTTCCC (SEQ ID NO:36).

[0072] In some cases, a hybrid RNA molecule comprises a "stuffer" sequence that separates the targeting region from the splice function region and/or that separates the splice function region and the donor sequence. A suitable "stuffer" sequence is as follows:

(SEQ ID NO: 37)
GGAGCAGGCGCCGATCCGGCGCAGGAGCCGGCGCCACC.

Binding Region

[0073] As noted above, in some cases, a hybrid RNA molecule includes a binding region, e.g., a stretch of nucleotides that binds to an RNA targeting polypeptide such as an RNA-targeting Clustered Regularly Interspaced Short Palindromic Repeat (CRISPR)-associated (CRISPR-Cas) polypeptide (also referred to herein as a "CRISPR-Cas effector polypeptide"). Suitable RNA-targeting CRISPR-Cas effector polypeptides include Type II CRISPR-Cas effector polypeptides, Type III CRISPR Cas effector polypeptides, and Type VI CRISPR-Cas effector polypeptides. Examples of Type TT CRISPR-Cas polypeptides include Cas9 polypeptides, e.g., *Staphylococcus aureus* Cas9, *Streptococcus pyogenes* Cas9, etc. Examples of Type VI CRISPR-Cas effector polypeptides include a Cas13a polypeptide, a Cas13b polypeptide, a Cas13c polypeptide, a Cas13d polypeptide, a Cas13e polypeptide, a Cas13f polypeptide, a Cas13X polypeptide, and a Cas13Y polypeptide. A suitable Type III CRISPR-Cas effector polypeptide is a Cas7-11 polypeptide (see, e.g., Ozcan et al. (2021) *Nature* 597:710). A suitable CRISPR-Cas effector polypeptide is an RNA-binding CRISPR-Cas effector polypeptide.

[0074] For example, in some cases, the CRISPR-Cas effector polypeptide is an Rfx Cas13d polypeptide (see FIG. 11F). As another example, in some cases, the CRISPR-Cas effector polypeptide is a dRfxCas13d polypeptide (see FIG. 11E). As another example, in some cases, the CRISPR-Cas effector polypeptide is an enhanced RfxCas13d that cleaves target RNA with minimal collateral degradation of off-target RNAs (see, eg. Tong et al. (2022) *Nature Biotechnology* 41.1 (2023): 108-119). As another example, in some cases, the CRISPR-Cas effector polypeptide is an RfxCas13d fused to a dsRNA binding domain to increase specificity (see, e.g. Han et al. (2020) *PNAS* 117.36 (2020): 22068-22079).

[0075] As another example, in some cases, the CRISPR-Cas effector polypeptide is a DjCas13d polypeptide. As another example, in some cases, the CRISPR-Cas effector polypeptide is a PspCas13b polypeptide, e.g., *Prevotella* sp. P5-125 Cas13b (see FIG. 11G). As another example, in some cases, the CRISPR-Cas effector polypeptide is a dPspCas13b polypeptide (see FIG. 11H). In some cases, a suitable CRISPR-Cas effector polypeptide comprises an amino acid sequence having at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to any of the amino acid sequences depicted in FIG. 11A-11H. In some cases, a suitable CRISPR-Cas effector polypeptide comprises an amino acid sequence having at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the amino acid sequence depicted in FIG. 11F. In some cases, a suitable CRISPR-Cas effector polypeptide comprises an amino acid sequence having at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the amino acid sequence depicted in FIG. 11E. In some cases, a suitable CRISPR-Cas effector polypeptide comprises an amino acid sequence having at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the amino acid sequence depicted in FIG. 11G. In some cases, a suitable CRISPR-Cas effector polypeptide comprises an amino acid

sequence having at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the amino acid sequence depicted in FIG. 11H.

[0076] In some cases, a Cas13 polypeptide (e.g., a Cas13a polypeptide, a Cas13b polypeptide, a Cas13c polypeptide, a Cas13d polypeptide, a Cas13e polypeptide, a Cas13f polypeptide, a Cas13X polypeptide, or a Cas13Y polypeptide) comprises two Higher Eukaryotes and Prokaryotes Nucleotide-binding (HEPN) domains, each comprising a HEPN motif, where each HEPN motif is RXXXXH, RXXXXXH, or RXXXXXXH, where X is any amino acid.

[0077] In some cases, the CRISPR-Cas effector polypeptide is a variant that exhibits reduced catalytic activity compared to a wild-type CRISPR-Cas effector polypeptide. For example, where the CRISPR-Cas effector polypeptide (e.g., a Cas13 polypeptide) comprises a HEPN domain, the CRISPR-Cas effector polypeptide can comprise a substitution of the Arg and/or the His in the HEPN motif, where such a mutation reduces the catalytic activity of the CRISPR-Cas effector polypeptide. In some cases, the RNA-binding CRISPR-Cas effector polypeptide binds, but does not cleave, a target RNA.

[0078] As another example, in some cases, the CRISPR-Cas effector polypeptide is a Cas7-11 polypeptide (e.g., a DiCas7-11 polypeptide). See, e.g., Ozcan et al. (2021) *Nature* 597:720. In some cases, a suitable CRISPR-Cas effector polypeptide comprises an amino acid sequence having at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the amino acid sequence depicted in FIG. 16 (DiCas7-11). In some cases, a suitable CRISPR-Cas effector polypeptide comprises an amino acid sequence having at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the amino acid sequence depicted in FIG. 17 (dDiCas7-11). In some cases, the Cas7-11 polypeptide is a variant that exhibits reduced catalytic activity compared to a wild-type Cas7-11 effector polypeptide. For example, in some cases, the variant Cas7-11 polypeptide comprises a substitution of one or more of D177, D429, D654, D758, E959, and D998 (where the amino acid numbering is as set forth in FIG. 16). D177, D429, D654, D758, E959, and D998 are in bold in FIG. 16. For example, as shown in FIG. 17, a variant Cas7-11 can have an Asp at position 177, an Ala at position 429, an Ala at position 654, an Asp at position 758, a Glu at position E959, and an Asp at position 998. In some cases, a variant Cas7-11 comprises a D429A substitution.

Fusion CRISPR-Cas Effector Polypeptides

[0079] In some cases, a suitable CRISPR-Cas effector polypeptide is a fusion polypeptide comprising: (i) a CRISPR-Cas effector polypeptide; and (ii) one or more heterologous polypeptides (also referred to herein as “fusion partners”). Such polypeptides may be referred to herein as “fusion CRISPR-Cas effector polypeptides.” The fusion partner can be at the N-terminus of the CRISPR-Cas effector polypeptide. The fusion partner can be at the C-terminus of the CRISPR-Cas effector polypeptide.

[0080] As one example, in some cases, a fusion CRISPR-Cas effector polypeptide comprises: (i) a CRISPR-Cas effector polypeptide; (ii) a C-terminal interacting domain (CID)

polypeptide. As another example, in some cases, a fusion CRISPR-Cas effector polypeptide comprises: (i) a CRISPR-Cas effector polypeptide; (ii) a C-terminal interacting domain (CID) polypeptide; and (iii) a double-stranded RNA Binding Domain (ds-RBD). As another example, in some cases, a fusion CRISPR-Cas effector polypeptide comprises: (i) a CRISPR-Cas effector polypeptide; (ii) a CID polypeptide; and (iii) a protein that is associated with the C-terminal domain (CTD) of RNA polymerase II. A CID domain interacts with the C-terminal domain (CTD) of RNA polymerase II (RNAPII), and allows the CRISPR-Cas effector polypeptide to associate with RNAPII as it transcribes, leading to faster localization of the donor RNA to the target RNA. FIG. 3 provides a schematic illustration of use of such a fusion CRISPR-Cas effector polypeptide. In some cases, the CTD polypeptide includes up to 52 repeats of the amino acid sequence YSPTSPS (SEQ ID NO:63). See, e.g., Barron-Casella and Corden (1992) *J. Mol. Evol.* 35:405; and Winterzerith et al. (1992) *Nucl. Acids Res.* 20:910; and NP_000928. As an example, a CTD can comprise an amino acid sequence having at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100%, amino acid sequence identity to the CTD amino acid sequence depicted in FIG. 12; and can have a length of from about 300 amino acids to about 375 amino acids (e.g., from 300 amino acids to 350 amino acids, from 350 amino acids to 370 amino acids, or from 370 amino acids to 375 amino acids; e.g., about 368 amino acids in length).

[0081] As another example, a fusion CRISPR-Cas effector polypeptide comprises: (i) a CRISPR-Cas effector polypeptide; and (ii) an Arg/Ser-rich (RS) domain polypeptide from a Serine-arginine (SG) polypeptide. Such a fusion partner can increase localization to SR protein granules, which are located proximal to the splicing reaction. See, e.g., Graveley and Maniatis (1998) *Molecular Cell* 1:765. FIG. 4 provides a schematic illustration of use of such a fusion CRISPR-Cas effector polypeptide.

[0082] As another example, at least one of the one or more heterologous polypeptides can comprise a nuclear localization sequence (NLS). In some cases, the fusion partner can modulate transcription (e.g., inhibit transcription, increase transcription) of a target DNA. For example, in some cases the fusion partner is a protein (or a domain from a protein) that inhibits transcription (e.g., a transcriptional repressor, a protein that functions via recruitment of transcription inhibitor proteins, modification of target DNA such as methylation, recruitment of a DNA modifier, modulation of histones associated with target DNA, recruitment of a histone modifier such as those that modify acetylation and/or methylation of histones, and the like). In some cases, the fusion partner is a protein (or a domain from a protein) that increases transcription (e.g., a transcription activator, a protein that acts via recruitment of transcription activator proteins, modification of target DNA such as demethylation, recruitment of a DNA modifier, modulation of histones associated with target DNA, recruitment of a histone modifier such as those that modify acetylation and/or methylation of histones, and the like). In some cases, the fusion partner is a reverse transcriptase. In some cases, the fusion partner is a base editor. In some cases, the fusion partner is a deaminase.

[0083] Other suitable fusion partners include, e.g., polypeptides that can cleave RNA (e.g., a PIN endonuclease, an NYN domain, an SMR domain from SOT1, or an RNase

domain from a Staphylococcal nuclease); polypeptides that can affect RNA stability (e.g., tristetraprolin (TTP) or domains from UPF1, EXOSC5, and STAU1); polypeptides that can modify a nucleotide or ribonucleotide (e.g., a cytidine deaminase, PPR protein, adenosine deaminase, an adenosine deaminase acting on RNA (ADAR) family protein, or an APOBEC family protein); polypeptides that can activate translation (e.g., eIF4E and other translation initiation factors, a domain of the yeast poly(A)-binding protein or GLD2), those that can repress translation (e.g., *Pumilio* or FBF PUF proteins, deadenylases, CAF1, Argonaute proteins); polypeptides that can methylate RNA (e.g., domains from m6A methyltransferase factors such as METTL14, METTL3, or WTAP); polypeptides that can demethylate RNA (e.g., human alkylation repair homolog 5 or Alkbh5); polypeptides that can affect splicing (e.g., the RS-rich domain of SRSF1, the Gly-rich domain of hnRNP A1, the alanine-rich motif of RBM4, or the proline-rich motif of DAZAP1); polypeptides that can enable affinity purification or immunoprecipitation (e.g., FLAG, hemagglutinin (HA), biotin, or HALO tags); and polypeptides that can enable proximity-based protein labeling and identification (e.g., a biotin ligase (such as BirA) or a peroxidase (such as APEX2) in order to biotinylate proteins that interact with the target RNA). Suitable heterologous polypeptides include splicing factors (e.g., RS domains); protein translation components (e.g., translation initiation, elongation, and/or release factors; e.g., eIF4G); RNA methylases; RNA editing enzymes (e.g., RNA deaminases, e.g., ADAR polypeptides, including A to I and/or C to U editing enzymes); helicases; RNA-binding proteins; and the like. In some cases, a heterologous polypeptide (a fusion partner) is a transglycosylase. In some cases, a heterologous polypeptide (a fusion partner) is capable of catalyzing aminoacylation (attachment of an amino acid to a tRNA). In some cases, a heterologous polypeptide (a fusion partner) is capable of catalyzing pseudouridylation (conversion of a uridine to a pseudouridine in an RNA). In some cases, a heterologous polypeptide (a fusion partner) provides for subcellular localization, i.e., the heterologous polypeptide contains a subcellular localization sequence (e.g., a nuclear localization signal (NLS) for targeting to the nucleus; a sequence to keep the fusion protein out of the nucleus, e.g., a nuclear export sequence (NES); a sequence to keep the fusion protein retained in the cytoplasm; a mitochondrial localization signal for targeting to the mitochondria; a chloroplast localization signal for targeting to a chloroplast; an endoplasmic reticulum (ER) retention signal; and the like).

Hybrid RNA Comprising a CRISPR-Cas Guide RNA and a Donor RNA

[0084] In some cases, a hybrid RNA molecule of the present disclosure comprises, in a single RNA molecule: a) a CRISPR-Cas guide RNA; and b) a donor RNA. A CRISPR-Cas guide RNA (referred to herein as a “guide RNA” or “gRNA”) comprises: (i) a targeting region comprising a nucleotide sequence that hybridizes to a target sequence in a target RNA; and (ii) a binding region comprising a nucleotide sequence that binds to (and may also activate) a CRISPR-Cas effector polypeptide. In some cases, the binding region comprises a direct repeat.

[0085] In some cases, the binding region (also referred to herein as the “protein-binding region” or “constant region”) is 5' of the targeting region. The targeting region comprises

a guide sequence (also referred to as a “spacer” sequence) that hybridizes to a target sequence in a target RNA.

[0086] In some cases, the guide sequence is 16-30 nucleotides (nt) in length (e.g., 16-26, 16-24, 16-22, 16-20, 16-18, 17-26, 17-24, 17-22, 17-20, 17-18, 18-26, 18-24, 18-22, or 18-30 nt in length). In some cases, the guide sequence is 18-24 nucleotides (nt) in length. In some cases, the guide sequence is at least 16 nt long (e.g., at least 18, 20, or 22 nt long). In some cases, the guide sequence is at least 17 nt long. In some cases, the guide sequence is at least 18 nt long. In some cases, the guide sequence is at least 20 nt long. In some cases, the guide sequence is 16-28 nt long.

[0087] In some cases, the guide sequence has 80% or more (e.g., 85% or more, 90% or more, 95% or more, or 100% complementarity) with the target sequence of the single stranded target RNA. In some cases, the guide sequence is 100% complementary to the target sequence of the single stranded target RNA. In some cases, the guide sequence has 1, 2, 3, or 4 mismatches with a corresponding stretch of nucleotides in the target sequence. For example, where the guide sequence has a length of 20 nt, the guide sequence can have 1, 2, 3, or 4 mismatches with a corresponding stretch of 20 nt in the target sequence.

[0088] The constant region of the guide RNA binds to a CRISPR-Cas effector polypeptide. Such constant region sequences are known in the art. For example, the constant region suitable for use with a *Listeria seeligeri* Cas13a polypeptide can comprise the following nucleotide sequence:

GACUACCUCUAUAUGAAAGAGGAC-UAAAAC (SEQ ID NO:38).

As another example, the constant region suitable for use with a *Leptotrichia buccalis* Cas13A polypeptide can comprise the following nucleotide sequence:

GACCACCCCAAAAUGAAGGGGAC-UAAAACA (SEQ ID NO:39).

As another example, a guide RNA can comprise a direct repeat (CRISPR-Cas binding region) comprising the sequence: CAAGTAAACCCCTAC-CAACTGGTCGGGGTTTGAAAC (SEQ ID NO:40), e.g., where the CRISPR-Cas effector polypeptide is an RfxCas13d polypeptide (see, e.g., FIG. 11E and FIG. 11F, respectively, for dead RfxCas13d (dRfxCas13d) and RfxCas13d amino acid sequences). As another example, a guide RNA can comprise a direct repeat (CRISPR-Cas binding region) comprising the sequence: GTTGTC-GAAGGTCCAGTTTGTAGGGGCTATTACAAC (SEQ ID NO:41), e.g., where the CRISPR-Cas effector polypeptide is a PspCas13b polypeptide or a dPspCas13b polypeptide (see, e.g., FIG. 11G and FIG. 11H, respectively, for PspCas13b and dead PspCas13b (dPspCas13b) amino acid sequences).

As another example, a guide RNA can comprise a direct repeat (CRISPR-Cas binding region) comprising the sequence: GTTGATGTCACGGAAC (SEQ ID NO:42), e.g., where the CRISPR-Cas effector polypeptide is a Cas7-11 polypeptide or a dCas7-11 polypeptide (see, e.g., FIG. 16 for the amino acid sequence of a Cas7-11 polypeptide; and FIG. 17 for the amino acid sequence of a dead Cas7-11 (dCas7-11) polypeptide). As another example, a guide RNA can comprise a direct repeat (CRISPR-Cas binding region) comprising the sequence: CAACTA-CAACCCCGTAAAAATACGGGGTTCTGAAAC (SEQ ID NO:43), where the CRISPR-Cas effector polypeptide is a DjCas13d polypeptide. As another example, a guide RNA can comprise a direct repeat (CRISPR-Cas binding region) comprising the sequence: GAC-CAACACCTCTGCAAACTGCAGGGGTCTAAAC

(SEQ ID NO:44), where the CRISPR-Cas effector polypeptide is an AdmCas13D polypeptide.

[0089] The following description relates to Cas13a guide RNAs. However, the description can be generally applied to guide RNAs.

[0090] In some cases, the Cas13a guide RNA includes a double stranded RNA duplex (dsRNA duplex).

[0091] In some cases, a Cas13a guide RNA includes a dsRNA duplex that is 4 base pairs (bp) in length. In some cases, a Cas13a guide RNA includes a dsRNA duplex with a length of from 2 to 12 bp (e.g., from 2 to 10 bp, 2 to 8 bp, 2 to 6 bp, 2 to 5 bp, 2 to 4 bp, 3 to 12 bp, 3 to 10 bp, 3 to 8 bp, 3 to 6 bp, 3 to 5 bp, 3 to 4 bp, 4 to 12 bp, 4 to 10 bp, 4 to 8 bp, 4 to 6 bp, or 4 to 5 bp). In some cases, a Cas13a guide RNA includes a dsRNA duplex that is 2 or more bp in length (e.g., 3 or more, 4 or more, 5 or more, 6 or more, or 7 or more bp in length). In some cases, a Cas13a guide RNA includes a dsRNA duplex that is longer than the dsRNA duplex of a corresponding wild type Cas13a guide RNA.

[0092] In some cases, the region of a Cas13a guide RNA that is 5' of the guide sequence is 15 or more nucleotides (nt) in length (e.g., 18 or more, 20 or more, 21 or more, 22 or more, 23 or more, 24 or more, 25 or more, 26 or more, 27 or more, 28 or more, 29 or more, 30 or more, 31 or more nt, 32 or more, 33 or more, 34 or more, or 35 or more nt in length). In some cases, the region of a Cas13a guide RNA that is 5' of the guide sequence is 29 or more nt in length.

[0093] In some cases, the region of a Cas13a guide RNA that is 5' of the guide sequence has a length in a range of from 12 to 100 nt (e.g., from 12 to 90, 12 to 80, 12 to 70, 12 to 60, 12 to 50, 12 to 40, 15 to 100, 15 to 90, 15 to 80, 15 to 70, 15 to 60, 15 to 50, 15 to 40, 20 to 100, 20 to 90, 20 to 80, 20 to 70, 20 to 60, 20 to 50, 20 to 40, 25 to 100, 25 to 90, 25 to 80, 25 to 70, 25 to 60, 25 to 50, 25 to 40, 28 to 100, 28 to 90, 28 to 80, 28 to 70, 28 to 60, 28 to 50, 28 to 40, 29 to 100, 29 to 90, 29 to 80, 29 to 70, 29 to 60, 29 to 50, or 29 to 40 nt). In some cases, the region of a Cas13a guide RNA that is 5' of the guide sequence has a length in a range of from 28 to 100 nt. In some cases, the region of a Cas13a guide RNA that is 5' of the guide sequence has a length in a range of from 28 to 40 nt.

[0094] In some cases, the region of the Cas13a guide RNA that is 5' of the guide sequence is truncated relative to (shorter than) the corresponding region of a corresponding wild type Cas13a guide RNA. For example, a mature wild-type Cas13a guide RNA can include a region 5' of the guide sequence that is 30 nucleotides (nt) in length, and a truncated Cas13a guide RNA (relative to the mature wild-type Cas13a guide RNA) can therefore have a region 5' of the guide sequence that is less than 30 nt in length (e.g., less than 29, 28, 27, 26, 25, 22, or 20 nt in length). In some cases, a truncated Cas13a guide RNA includes a region 5' of the guide sequence that has a length in a range of from 12 to 29 nt (e.g., from 12 to 28, 12 to 27, 12 to 26, 12 to 25, 12 to 22, 12 to 20, 12 to 18 nt). In some cases, the truncated Cas13a guide RNA is truncated by one or more nt (e.g., 2 or more, 3 or more, 4 or more, 5 or more, or 10 or more nt), e.g., relative to a corresponding wild type Cas13a guide).

[0095] In some cases, the region of the Cas13a guide RNA that is 5' of the guide sequence is extended relative to (longer than) the corresponding region of a corresponding wild type Cas13a guide RNA. For example, a mature wild-type Cas13a guide RNA can include a region 5' of the guide sequence that is 30 nucleotides (nt) in length, and an

extended Cas13a guide RNA (relative to the mature wild-type Cas13a guide RNA) can therefore have a region 5' of the guide sequence that is longer than 30 nt (e.g., longer than 31, longer than 32, longer than 33, longer than 34, or longer than 35 nt). In some cases, an extended Cas13a guide RNA includes a region 5' of the guide sequence that has a length in a range of from 30 to 100 nt (e.g., from 30 to 90, 30 to 80, 30 to 70, 30 to 60, 30 to 50, or 30 to 40 nt). In some cases, the extended Cas13a guide RNA includes a region 5' of the guide sequence that is extended (e.g., relative to the corresponding region of a corresponding wild type Cas13a guide RNA) by one or more nt (e.g., 2 or more, 3 or more, 4 or more, 5 or more, or 10 or more nt).

[0096] In some cases, a Cas13a guide RNA is 30 or more nucleotides (nt) in length (e.g., 34 or more, 40 or more, 45 or more, 50 or more, 55 or more, 60 or more, 65 or more, 70 or more, or 80 or more nt in length). In some cases, the Cas13a guide RNA is 35 or more nt in length.

[0097] In some cases, a subject Cas13a guide RNA has a length in a range of from 30 to 120 nt (e.g., from 30 to 110, 30 to 100, 30 to 90, 30 to 80, 30 to 70, 30 to 60, 35 to 120, 35 to 110, 35 to 100, 35 to 90, 35 to 80, 35 to 70, 35 to 60, 40 to 120, 40 to 110, 40 to 100, 40 to 90, 40 to 80, 40 to 70, 40 to 60, 50 to 120, 50 to 110, 50 to 100, 50 to 90, 50 to 80, or 50 to 70 nt). In some cases, the Cas13a guide RNA has a length in a range of from 33 to 80 nt. In some cases, the Cas13a guide RNA has a length in a range of from 35 to 60 nt.

[0098] In some cases, a subject Cas13a guide RNA is truncated relative to (shorter than) a corresponding wild type Cas13a guide RNA. For example, a mature wild-type Cas13a guide RNA can be 50 nucleotides (nt) in length, and a truncated Cas13a guide RNA (relative to the Lse Cas13a guide RNA) can therefore in some cases be less than 50 nt in length (e.g., less than 49, 48, 47, 46, 45, 42, or 40 nt in length). In some cases, a truncated Cas13a guide RNA has a length in a range of from 30 to 49 nt (e.g., from 30 to 48, 30 to 47, 30 to 46, 30 to 45, 30 to 42, 30 to 40, 35 to 49, 35 to 48, 35 to 47, 35 to 46, 35 to 45, 35 to 42, or 35 to 40 nt). In some cases, the truncated Cas13a guide RNA is truncated by one or more nt (e.g., 2 or more, 3 or more, 4 or more, 5 or more, or 10 or more nt), e.g., relative to a corresponding wild type Cas13a guide).

[0099] In some cases, a subject Cas13a guide RNA is extended relative to (longer than) a corresponding wild type Cas13a guide RNA. For example, a mature wild-type Cas13a guide RNA can be 50 nucleotides (nt) in length, and an extended Cas13a guide RNA (relative to the mature wild-type Cas13a guide RNA) can therefore in some cases be longer than 50 nt (e.g., longer than 51, longer than 52, longer than 53, longer than 54, or longer than 55 nt). In some cases, an extended Cas13a guide RNA has a length in a range of from 51 to 100 nt (e.g., from 51 to 90, 51 to 80, 51 to 70, 51 to 60, 53 to 100, 53 to 90, 53 to 80, 53 to 70, 53 to 60, 55 to 100, 55 to 90, 55 to 80, 55 to 70, or 55 to 60 nt). In some cases, the extended Cas13a guide RNA is extended (e.g., relative to a corresponding wild type Cas13a guide RNA) by one or more nt (e.g., 2 or more, 3 or more, 4 or more, 5 or more, or 10 or more nt).

Protein-Recruiting Region

[0100] As noted above, in some cases, a hybrid RNA molecule of the present disclosure will comprise, in addition to a targeting region and a donor RNA, a protein-recruiting

region. In some cases, the protein-recruiting region comprises a nucleotide sequence that binds one or more spliceosome polypeptides. Spliceosome polypeptides include, e.g., U1, U2, U4, U5, and U6.

Stabilization Region

[0101] As noted above, in some cases, a hybrid RNA molecule of the present disclosure will comprise, in addition to a targeting region and a donor RNA, a stabilization region. In some cases, the stabilization region is capable of forming a secondary structure. Examples of such secondary structures include, e.g., pseudoknots, stem-loops, and tetraloops. See, e.g., Shalybkova et al. (2021) RNA 27:907. In some cases, stabilization region comprises a toehold switch. A stabilization region can include a structured RNA motif such as AsCpf1BB, BoxB, a pseudoknot (decoy), a pseudoknot (tEvoPreQ1), fmpknot, mpknot, MS2, PP7, SLBP, TAR, or ThermoPh. In some cases, the stabilization region comprises a pseudoknot, where the pseudoknot is a naturally occurring pseudoknot or a synthetic pseudoknot. The term “pseudoknot,” as used herein, includes, but is not limited to, hairpins, multiloops, kissing loops, coaxial stacking, triplexes, pseudoknot-like structures, a pseudoknotted hairpins and/or a decoy pseudoknotted hairpins or other RNA structural motifs. A hybrid RNA molecule of the present disclosure can include a stabilization region at the 5' end of the hybrid RNA molecule, at the 3' end of the hybrid RNA molecule, or at both the 5' end and the 3' end of the hybrid RNA molecule.

[0102] As one non-limiting example, a stabilization region can comprise the following nucleotide sequence: TTGACGCGGTTCTATCTAGTTACGCGTTAAAC-CAACTAGAAA (SEQ ID NO:45) (or where the Ts are replaced with Us); where such a sequence is a modified prequosine riboswitch aptamer (evopreQ1).

[0103] As another non-limiting example, a stabilization region can comprise the following nucleotide sequence: GGGTCAGGAGCCCCCCCCCTGAACCCAGGA-TAACCTCAAAGTCGGGGGGCAACCC (SEQ ID NO:46); e.g., a Moloney murine leukemia virus (MMLV) (mpknot-1 trimmed).

[0104] As another non-limiting example, a stabilization region can comprise an mpknot. See, e.g., Nelson et al. (2022) Nat. Biotechnol. 40:402. For example, a stabilization region can comprise the following nucleotide sequence: GTCAGGGTCAGGAGCCCCCCCCCTGAACCCAGGA-TAACCTCAAAGTCGGGGGGCAACCC (SEQ ID NO:47). As another example, a stabilization region can comprise the following nucleotide sequence: GGGTCAG-GAGCCCCCCCCCTGAACCCAGGAAAACCTCAAAGTCGGGGGGCAACCC (SEQ ID NO:48). As another example, a stabilization region can comprise the following nucleotide sequence:

(SEQ ID NO: 49)

GGGTCAGGAGCCCCCCCCCTGCACCCAGGAAAACCTCA
AAGTCGGGGGGCAACCC.

[0105] In some cases, a pseudoknot suitable for including in a stabilization region is a tEvoPreQ1 Pseudoknot comprising the following nucleotide sequence: UAAUUUCUA-CUAAGUGUAGAU (SEQ ID NO:50). In some cases, a pseudoknot suitable for including in a stabilization region is

a pseudoknot EvoPreQ1 comprising the following nucleotide sequence: TTGACGCGGTTCTATCTAGTTACGCGT-TAAACCAACUAGAAA (SEQ ID NO:51). In some cases, a pseudoknot suitable for including in a stabilization region is a pseudoknot comprising the following nucleotide sequence: TAAGTCTCCATAGAATGGAGG (SEQ ID NO:52). In some cases, a pseudoknot suitable for including in a stabilization region is a pseudoknot comprising the following nucleotide sequence: UAAGUCUCCAUA-GAAUGGAGG (SEQ ID NO:52).

[0106] In some cases, a stabilization region comprises a structured RNA motif, where suitable structured RNA motifs include, but are not limited to, AsCpf1BB, comprising the following nucleotide sequence: TAATTCT-TACTCTGTAGAT (SEQ ID NO:53); BoxB, comprising the following nucleotide sequence: GGGCCCT-GAAGAAGGGCCC (SEQ ID NO:54); pseudoknot (decoy), comprising the following nucleotide sequence: TAAGTCTCCATAGAATGGAGG (SEQ ID NO:52); pseudoknot (tEvoPreQ1), comprising the following nucleotide sequence: TTGACGCGGTTCTATCTAGTTACGCGT-TAAACCAACUAGAAA (SEQ ID NO:51); fmpknot, comprising the following nucleotide sequence: GGAGGTCAGGGTCAGGAGCCCCCCCCCT-GAACCCAGGATAACCTCAAAGTCG GGGGGCAACCC (SEQ ID NO:55); mpknot, comprising the following nucleotide sequence: GGGTCAG-GAGCCCCCCCCCTGAACCCAGGATAACCTCAAAGTCGGGGGGCAACCC (SEQ ID NO:56); MS2, comprising the following nucleotide sequence: GGCCAA-CATGAGGATCACCCATGTCTGCAGGGCC (SEQ ID NO:57); PP7, comprising the following nucleotide sequence: CCGGAGCAGACGATATGGCGTCGCTCCGG (SEQ ID NO:58); SLBP, comprising the following nucleotide sequence: CCAAAGGCTCTCTCAGAGCCACCCA (SEQ ID NO:59); TAR, comprising the following nucleotide sequence: GGCCAGATCTGAGCCTGG-GAGCTCTCTGGCC (SEQ ID NO:60); and ThermoPh, comprising the following nucleotide sequence: ATATAACCTTCACCATTAGGTTCAAATAATGGTAAT (SEQ ID NO:61).

Nuclear Localization Motif

[0107] As noted above, in some cases, a hybrid RNA molecule of the present disclosure will comprise, in addition to a targeting region and a donor RNA, a SINE-derived nuclear RNA Localization (SIRLOIN) motif. See, e.g., Lubelsky and Ulitsky (2018) Nature 555:107. A SIRLOIN motif is a 42-nucleotide motif. In some cases, a SIRLOIN motif comprises the following nucleotide sequence:

(SEQ ID NO: 62)

CGCCTCCCGGGTTCAAGCGATTCTCTGCTCAGCCTCCCGA

RNA Modifications

[0108] A hybrid RNA molecule of the present disclosure can include one or more of: a) a modified nucleoside comprising a modified nucleoside base; b) a modified nucleoside comprising a modified sugar; c) a modified internucleoside linkage; and d) a modified backbone linking two or more nucleosides.

[0109] A nucleoside is a base-sugar combination. The base portion of the nucleoside is normally a heterocyclic base. The two most common classes of such heterocyclic bases are the purines and the pyrimidines. Nucleotides are nucleosides that further include a phosphate group covalently linked to the sugar portion of the nucleoside. For those nucleosides that include a pentofuranosyl sugar, the phosphate group can be linked to the 2', the 3', or the 5' hydroxyl moiety of the sugar. In forming oligonucleotides, the phosphate groups covalently link adjacent nucleosides to one another to form a linear polymeric compound. In turn, the respective ends of this linear polymeric compound can be further joined to form a circular compound, however, linear compounds are suitable. In addition, linear compounds may have internal nucleotide base complementarity and may therefore fold in a manner as to produce a fully or partially double-stranded compound. Within oligonucleotides, the phosphate groups are commonly referred to as forming the internucleoside backbone of the oligonucleotide. The normal linkage or backbone of RNA and DNA is a 3' to 5' phosphodiester linkage.

[0110] Suitable nucleic acid modifications include, but are not limited to: 2'-O-methyl modified nucleotides, 2' Fluoro modified nucleotides, locked nucleic acid (LNA) modified nucleotides, peptide nucleic acid (PNA) modified nucleotides, nucleotides with phosphorothioate linkages, and a 5' cap (e.g., a 7-methylguanylate cap (m7G)). Additional details and additional modifications are described below.

[0111] A 2'-O-Methyl modified nucleoside (also referred to as 2'-O-Methyl RNA) is a naturally occurring modification of RNA found in tRNA and other small RNAs that arises as a post-transcriptional modification. RNA can be directly synthesized so that it contains one or more 2'-O-Methyl nucleosides. Such an RNA is stable with respect to attack by single-stranded ribonucleases.

[0112] 2' Fluoro modified nucleosides (e.g., 2' Fluoro bases) have a fluorine modified ribose which increases binding affinity (T_m) and also confers some relative nuclease resistance when compared to native RNA.

[0113] LNA bases have a modification to the ribose backbone that locks the base in the C3'-endo position, which favors RNA A-type helix duplex geometry. This modification significantly increases T_m and is also very nuclease resistant. Multiple LNA insertions can be placed in an RNA at any position except the 3'-end.

[0114] The phosphorothioate (PS) bond (i.e., a phosphorothioate linkage) substitutes a sulfur atom for a non-bridging oxygen in the phosphate backbone of a nucleic acid (e.g., an oligo). This modification renders the internucleoside linkage resistant to nuclease degradation. Phosphorothioate bonds can be introduced between the last 3-5 nucleosides at the 5'- or 3'-end of an RNA molecule to inhibit exonuclease degradation. Including phosphorothioate bonds within an RNA molecule can help reduce attack by endonucleases as well.

[0115] In some cases, a subject hybrid RNA molecule has one or more nucleotides that are 2'-O-Methyl modified nucleosides. In some cases, a subject hybrid RNA molecule has one or more 2' Fluoro modified nucleosides. In some cases, a subject hybrid RNA molecule has one or more LNA bases. In some cases, a subject hybrid RNA molecule has one or more nucleosides that are linked by a phosphorothioate bond (i.e., the subject nucleic acid has one or more phosphorothioate linkages).

Modified Backbones and Modified Internucleoside Linkages

[0116] In some cases, a hybrid RNA molecule of the present disclosure comprises a modified backbone or one or more non-natural internucleoside linkage. Nucleic acids having modified backbones include those that retain a phosphorus atom in the backbone and those that do not have a phosphorus atom in the backbone.

[0117] Suitable modified oligonucleotide backbones containing a phosphorus atom therein include, for example, phosphorothioates, chiral phosphorothioates, phosphorodithioates, phosphotriesters, aminoalkylphosphotriesters, methyl and other alkyl phosphonates including 3'-alkylene phosphonates, 5'-alkylene phosphonates and chiral phosphonates, phosphinates, phosphoramidates including 3'-amino phosphoramidate and aminoalkylphosphoramidates, phosphorodiamidates, thionophosphoramidates, thionoalkylphosphonates, thionoalkylphosphotriesters, selenophosphates and boranophosphates having normal 3'-5' linkages, 2'-5' linked analogs of these, and those having inverted polarity wherein one or more internucleoside linkages is a 3' to 3', 5' to 5' or 2' to 2' linkage. Suitable RNA molecules having inverted polarity comprise a single 3' to 3' linkage at the 3'-most internucleoside linkage i.e. a single inverted nucleoside residue which may be a basic (the nucleobase is missing or has a hydroxyl group in place thereof). Various salts (such as, for example, potassium or sodium), mixed salts and free acid forms are also included.

[0118] In some cases, a subject hybrid RNA molecule comprises one or more phosphorothioate and/or heteroatom internucleoside linkages, in particular $-\text{CH}_2-\text{NH}-\text{O}-\text{CH}_2-$, $-\text{CH}_2-\text{N}(\text{CH}_3)-\text{O}-\text{CH}_2-$ (known as a methylene (methylimino) or MMI backbone), $-\text{CH}_2-\text{O}-\text{N}(\text{CH}_3)-\text{CH}_2-$, $-\text{CH}_2-\text{N}(\text{CH}_3)-\text{N}(\text{CH}_3)-\text{CH}_2-$ and $-\text{O}-\text{N}(\text{CH}_3)-\text{CH}_2-\text{CH}_2-$ (wherein the native phosphodiester internucleoside linkage is represented as $-\text{O}-\text{P}(=\text{O})(\text{OH})-\text{O}-\text{CH}_2-$). MMI type internucleoside linkages are disclosed in U.S. Pat. No. 5,489,677, the disclosure of which is incorporated herein by reference in its entirety. Suitable amide internucleoside linkages are disclosed in U.S. Pat. No. 5,602,240, the disclosure of which is incorporated herein by reference in its entirety.

[0119] Also suitable are hybrid RNA molecule having morpholino backbone structures as described in, e.g., U.S. Pat. No. 5,034,506. For example, in some cases, a subject hybrid RNA molecule comprises a 6-membered morpholino ring in place of a ribose ring. In some of these instances, a phosphorodiamidate or other non-phosphodiester internucleoside linkage replaces a phosphodiester linkage.

[0120] Suitable modified polynucleoside backbones that do not include a phosphorus atom therein have backbones that are formed by short chain alkyl or cycloalkyl internucleoside linkages, mixed heteroatom and alkyl or cycloalkyl internucleoside linkages, or one or more short chain heteroatomic or heterocyclic internucleoside linkages. These include those having morpholino linkages (formed in part from the sugar portion of a nucleoside); siloxane backbones; sulfide, sulfoxide and sulfone backbones; formacetyl and thioformacetyl backbones; methylene formacetyl and thioformacetyl backbones; riboacetyl backbones; alkene containing backbones; sulfamate backbones; methyleneimino and methylenehydrazino backbones; sulfonate and sulfonamide backbones; amide backbones; and others having mixed N, O, S and CH_2 component parts.

Modified Sugar Moieties

[0121] A subject hybrid RNA molecule can also include one or more substituted sugar moieties.

[0122] Suitable polynucleotides comprise a sugar substituent group selected from: OH; F; O—, S—, or N-alkyl; O—, S—, or N-alkenyl; O-, S- or N-alkynyl; or O-alkyl-O-alkyl, wherein the alkyl, alkenyl and alkynyl may be substituted or unsubstituted C₁ to C₁₀ alkyl or C₂ to C₁₀ alkenyl and alkynyl. Particularly suitable are O((CH₂)_nO)_mCH, O(CH₂)_nOCH₃, O(CH₂)_nNH₂, O(CH₂)_nCH₃, O(CH₂)_nONH₂, and O(CH₂)_nON((CH₂)_n—CH₃)₂, where n and m are from 1 to about 10. Other suitable polynucleotides comprise a sugar substituent group selected from: C₁ to C₁₀ lower alkyl, substituted lower alkyl, alkenyl, alkynyl, alkaryl, aralkyl, O-alkaryl or O-aralkyl, SH, SCH₃, OCN, Cl, Br, CN, CF₃, OCF₃, SOCH₃, SO₂CH₃, ONO₂, NO₂, N₃, NH₂, heterocycloalkyl, heterocycloalkaryl, aminoalkylamino, polyalkylamino, substituted silyl, an RNA cleaving group, a reporter group, an intercalator, a group for improving the pharmacokinetic properties of an RNA, or a group for improving the pharmacodynamic properties of an RNA, and other substituents having similar properties. A suitable modification includes 2'-methoxyethoxy (2'-O—CH₂—CH₂—OCH₃, also known as 2'-O-(2-methoxyethyl) or 2'-MOE) (Martin et al., *Helv. Chim. Acta*, 1995, 78, 486-504, the disclosure of which is incorporated herein by reference in its entirety) i.e., an alkoxyalkoxy group. A further suitable modification includes 2'-dimethylaminooxyethoxy, i.e., a O(CH₂)₂₀N(CH₃)₂ group, also known as 2'-DMAOE, as described in examples hereinbelow, and 2'-dimethylaminoethoxyethoxy (also known in the art as 2'-O-dimethyl-amino-ethoxy-ethyl or 2'-DMAEOE), i.e., 2'-O—CH₂—O—CH₂—N(CH₃)₂.

[0123] Other suitable sugar substituent groups include methoxy (—O—CH₃), aminopropoxy (—O—CH₂—CH₂—CH₂—NH₂), allyl (—CH₂—CH=CH₂), —O-allyl (—O—CH₂—CH=CH₂) and fluoro (F). 2'-sugar substituent groups may be in the arabino (up) position or ribo (down) position. A suitable 2'-arabino modification is 2'-F. Similar modifications may also be made at other positions on the RNA, particularly the 3' position of the sugar on the 3' terminal nucleoside or in 2'-5' linked oligonucleosides and the 5' position of 5' terminal nucleoside.

Base Modifications and Substitutions

[0124] A subject hybrid RNA molecule may also include nucleobase (often referred to in the art simply as "base") modifications or substitutions. As used herein, "unmodified" or "natural" nucleobases include the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C) and uracil (U). Modified nucleobases include other synthetic and natural nucleobases such as 5-methylcytosine (5-me-C), 5-hydroxymethyl cytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-methyl and other alkyl derivatives of adenine and guanine, 2-propyl and other alkyl derivatives of adenine and guanine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-halouracil and cytosine, 5-propynyl (—C≡C—CH₃) uracil and cytosine and other alkynyl derivatives of pyrimidine bases, 6-azo uracil, cytosine and thymine, 5-uracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl and other 8-substituted adenines and guanines, 5-halo particularly 5-bromo, 5-trifluoromethyl and other 5-substituted uracils and cytosines, 7-methylguanine and 7-methyladenine, 2-F-adenine,

2-amino-adenine, 8-azaguanine and 8-azaadenine, 7-deazaguanine and 7-deazaadenine and 3-deazaguanine and 3-deazaadenine. Further modified nucleobases include tricyclic pyrimidines such as phenoxazine cytidine(1H-pyrimido(5,4-b)(1,4)benzoxazin-2(3H)-one), phenothiazine cytidine (1H-pyrimido(5,4-b)(1,4)benzothiazin-2(3H)-one), G-clamps such as a substituted phenoxazine cytidine (e.g. 9-(2-aminoethoxy)-H-pyrimido(5,4-b)(1,4)benzoxazin-2(3H)-one), carbazole cytidine (2H-pyrimido(4,5-b)indol-2-one), pyridoindole cytidine (H-pyrido(3',2':4,5)pyrrolo(2,3-d)pyrimidin-2-one).

[0125] Heterocyclic base moieties may also include those in which the purine or pyrimidine base is replaced with other heterocycles, for example 7-deaza-adenine, 7-deazaguanosine, 2-aminopyridine and 2-pyridone. Further nucleobases include those disclosed in U.S. Pat. No. 3,687,808, those disclosed in *The Concise Encyclopedia Of Polymer Science And Engineering*, pages 858-859, Kroschwitz, J. I., ed. John Wiley & Sons, 1990, those disclosed by Englisch et al., *Angewandte Chemie, International Edition*, 1991, 30, 613, and those disclosed by Sanghvi, Y. S., Chapter 15, *Antisense Research and Applications*, pages 289-302, Crooke, S. T. and Lebleu, B., ed., CRC Press, 1993; the disclosures of which are incorporated herein by reference in their entirety. Certain of these nucleobases are useful for increasing the binding affinity of an oligomeric compound. These include 5-substituted pyrimidines, 6-azapyrimidines and N-2, N-6 and 0-6 substituted purines, including 2-aminopropyladenine, 5-propynyluracil and 5-propynylcytosine. 5-methylcytosine substitutions have been shown to increase nucleic acid duplex stability by 0.6-1.2° C. (Sanghvi et al., eds., *Antisense Research and Applications*, CRC Press, Boca Raton, 1993, pp. 276-278; the disclosure of which is incorporated herein by reference in its entirety) and are suitable base substitutions, e.g., when combined with 2'-O-methoxyethyl sugar modifications.

Nucleic Acids; DNA Molecules; Recombinant Expression Vectors)

[0126] The present disclosure provides a DNA molecule comprising a nucleotide sequence encoding a hybrid RNA molecule of the present disclosure. In some cases, the nucleotide sequence is operably linked to transcriptional control element, such as a promoter, e.g., a promoter functional in a eukaryotic cell. In some cases, a DNA molecule of the present disclosure comprises a nucleotide sequence encoding a hybrid RNA molecule and a CRISPR-Cas effector polypeptide or a CRISPR-Cas fusion polypeptide. In some cases, a hybrid RNA molecule and a CRISPR-Cas effector polypeptide (or CRISPR-Cas fusion polypeptide) are encoded on separate DNA molecules. In some cases, a nucleotide sequence encoding a hybrid RNA molecule is codon optimized, e.g., for production in a particular organism.

[0127] The present disclosure provides a recombinant expression vector comprising a DNA molecule of the present disclosure, e.g., comprising a nucleotide sequence encoding a hybrid RNA molecule of the present disclosure. The present disclosure provides a recombinant expression vector comprising: a) a first nucleotide sequence encoding a hybrid RNA molecule of the present disclosure; and b) a second nucleotide sequence encoding a CRISPR-Cas effector polypeptide or a CRISPR-Cas fusion polypeptide. The present disclosure provides a first recombinant expression

vector comprising a nucleotide sequence encoding a hybrid RNA molecule of the present disclosure and a second recombinant expression vector comprising a nucleotide sequence encoding a CRISPR-Cas effector polypeptide or a CRISPR-Cas fusion polypeptide.

[0128] Suitable expression vectors include viral expression vectors (e.g., viral vectors based on vaccinia virus; poliovirus; adenovirus; adeno-associated virus; lentivirus, e.g., a human immunodeficiency virus; a retrovirus; etc.). In some cases, a recombinant expression vector of the present disclosure is a recombinant adeno-associated virus (AAV) vector. In some cases, a recombinant expression vector of the present disclosure is a recombinant lentivirus vector. In some cases, a recombinant expression vector of the present disclosure is a recombinant retroviral vector.

[0129] As noted above, in some cases, a nucleotide sequence encoding a hybrid RNA molecule of the present disclosure is operably linked to a transcriptional control element. Similarly, a nucleotide sequence encoding a CRISPR-Cas effector polypeptide or a CRISPR-Cas fusion polypeptide can be operably linked to a transcriptional control element. The transcriptional control element can be a promoter. In some cases, the promoter is a constitutively active promoter. In some cases, the promoter is a regulatable promoter. In some cases, the promoter is an inducible promoter. In some cases, the promoter is a tissue-specific promoter. In some cases, the promoter is a cell type-specific promoter. In some cases, the transcriptional control element (e.g., the promoter) is functional in a targeted cell type or targeted cell population. In some cases, the transcriptional control element is one that is functional in eukaryotic cells.

[0130] A promoter can be a constitutively active promoter (i.e., a promoter that is constitutively in an active/“ON” state), it may be an inducible promoter (i.e., a promoter whose state, active/“ON” or inactive/“OFF”, is controlled by an external stimulus, e.g., the presence of a particular temperature, compound, or protein.), it may be a spatially restricted promoter (i.e., transcriptional control element, enhancer, etc.) (e.g., tissue specific promoter, cell type specific promoter, etc.), and it may be a temporally restricted promoter (i.e., the promoter is in the “ON” state or “OFF” state during specific stages of embryonic development or during specific stages of a biological process, e.g., hair follicle cycle in mice).

[0131] RNA polymerase III (Pol III) promoters can be used to drive the expression of an RNA. In some cases, a suitable promoter is a Pol III promoter. Non-limiting examples of Pol III promoters include a U6 promoter, an H1 promoter, a 5S promoter, an Adenovirus 2 (Ad2) VAI promoter, a tRNA promoter, and a 7SK promoter. See, for example, Schramm and Hernandez (2002) *Genes & Development* 16:2593-2620. In some cases, a Pol III promoter is selected from the group consisting of a U6 promoter, an H1 promoter, a 5S promoter, an Adenovirus 2 (Ad2) VAI promoter, a tRNA promoter, and a 7SK promoter. In some cases, a hybrid RNA molecule-encoding nucleotide sequence is operably linked to a promoter selected from the group consisting of a U6 promoter, an H1 promoter, a 5S promoter, an Adenovirus 2 (Ad2) VAI promoter, a tRNA promoter, and a 7SK promoter.

Compositions

[0132] The present disclosure provides a composition comprising one or more hybrid RNA molecules of the present disclosure. The present disclosure provides a composition comprising a DNA molecule of the present disclosure. The present disclosure provides a composition comprising a recombinant expression vector of the present disclosure. A composition of the present disclosure can comprise, in addition to a hybrid RNA molecule of the present disclosure, one or more of: a salt, e.g., NaCl, MgCl₂, KCl, MgSO₄, etc.; a buffering agent, e.g., a Tris buffer, N-(2-Hydroxyethyl)piperazine-N'-(2-ethanesulfonic acid) (HEPES), 2-(N-Morpholino)ethanesulfonic acid (MES), 2-(N-Morpholino)ethanesulfonic acid sodium salt (MES), 3-(N-Morpholino)propanesulfonic acid (MOPS), N-tris[Hydroxymethyl]methyl-3-aminopropanesulfonic acid (TAPS), etc.; a solubilizing agent; a detergent, e.g., a non-ionic detergent such as Tween-20, etc.; a nuclease inhibitor; glycerol; and the like.

[0133] A composition of the present disclosure can include: a) a hybrid RNA molecule of the present disclosure, or a nucleic acid or recombinant expression vector comprising a nucleotide sequence encoding the hybrid RNA molecule; and b) one or more of: a buffer, a surfactant, an antioxidant, a hydrophilic polymer, a dextrin, a chelating agent, a suspending agent, a solubilizer, a thickening agent, a stabilizer, a bacteriostatic agent, a wetting agent, a nuclease inhibitor, and a preservative. Suitable buffers include, but are not limited to, (such as N,N-bis(2-hydroxyethyl)-2-aminoethanesulfonic acid (BES), bis(2-hydroxyethyl)amino-tris(hydroxymethyl)methane (BIS-Tris), N-(2-hydroxyethyl)piperazine-N'-3-propanesulfonic acid (EPPS or HEPES), glycylglycine, N-2-hydroxyethylpiperazine-N'-2-ethanesulfonic acid (HEPES), 3-(N-morpholino)propane sulfonic acid (MOPS), piperazine-N,N'-bis(2-ethanesulfonic acid) (PIPES), sodium bicarbonate, 3-(N-tris(hydroxymethyl)-methyl-amino)-2-hydroxy-propanesulfonic acid) TAPSO, (N-tris(hydroxymethyl)methyl-2-aminoethanesulfonic acid (TES), N-tris(hydroxymethyl)methyl-glycine (Tricine), tris(hydroxymethyl)-aminomethane (Tris), etc.). Suitable salts include, e.g., NaCl, MgCl₂, KCl, MgSO₄, etc.

[0134] The composition may comprise a pharmaceutically acceptable excipient, a variety of which are known in the art and need not be discussed in detail herein. Pharmaceutically acceptable excipients have been amply described in a variety of publications, including, for example, “Remington: The Science and Practice of Pharmacy”, 19th Ed. (1995), or latest edition, Mack Publishing Co; A. Gennaro (2000) “Remington: The Science and Practice of Pharmacy”, 20th edition, Lippincott, Williams, & Wilkins; *Pharmaceutical Dosage Forms and Drug Delivery Systems* (1999) H. C. Ansel et al., eds 7th ed., Lippincott, Williams, & Wilkins; and *Handbook of Pharmaceutical Excipients* (2000) A. H. Kibbe et al., eds., 3rd ed. Amer. Pharmaceutical Assoc.

[0135] A pharmaceutical composition can comprise a hybrid RNA molecule of the present disclosure, and a pharmaceutically acceptable excipient. In some cases, a subject pharmaceutical composition will be suitable for administration to a subject, e.g., will be sterile. For example, in some cases, a subject pharmaceutical composition will be suitable for administration to a human subject, e.g., where the composition is sterile and is free of detectable pyrogens and/or other toxins.

[0136] The protein compositions may comprise other components, such as pharmaceutical grades of mannitol, lactose, starch, magnesium stearate, sodium saccharin, talcum, cellulose, glucose, sucrose, magnesium, carbonate, and the like. The compositions may contain pharmaceutically acceptable auxiliary substances as required to approximate physiological conditions such as pH adjusting and buffering agents, toxicity adjusting agents and the like, for example, sodium acetate, sodium chloride, potassium chloride, calcium chloride, sodium lactate, hydrochloride, sulfate salts, solvates (e.g., mixed ionic salts, water, organics), hydrates (e.g., water), and the like.

[0137] Formulation suitable for parenteral administration can include isotonic sterile injection solutions, anti-oxidants, bacteriostats, and solutes that render the formulation isotonic with the blood of the intended recipient, suspending agents, solubilizers, thickening agents, stabilizers, and preservatives. For example, a subject pharmaceutical composition can be present in a container, e.g., a sterile container, such as a syringe. The formulations can be presented in unit-dose or multi-dose sealed containers, such as ampules and vials, and can be stored in a freeze-dried (lyophilized) condition requiring only the addition of the sterile liquid excipient, for example, water, for injections, immediately prior to use. Extemporaneous injection solutions and suspensions can be prepared from sterile powders, granules, and tablets.

[0138] In some cases, a hybrid RNA molecule is formulated with a lipid, e.g., a lipid such as 1,2-dioleoyl-3-trimethylammonium-propane (DOTAP) or 1,2-ditetradecanoyl-sn-glycero-3-phosphocholine (DMPC). In some cases, a hybrid RNA molecule is present in a liposome.

[0139] In some cases, a hybrid RNA molecule is in a particle or associated with a particle. A variety of particles (e.g., nanoparticles) can be used, e.g., sugar-based particles, lipid-based particles, and the like. For example, a lipid nanoparticle can be used, where the lipid nanoparticle comprises one or more lipids such as 1,2-dilinoeoyl-3-dimethylammonium-propane (DLinDAP), 1,2-dilinoeoyloxy-3-N,N-dimethylaminopropane (DLinDMA), 1,2-dilinoeoyloxy-keto-N,N-dimethyl-3-aminopropane (DLinKDMA), and 1,2-dilinoeoyl-4-(2-dimethylaminoethyl)-[1,3]-dioxolane (DLinKC2-DMA).

[0140] In some cases, a hybrid RNA molecule is formulated with a stable nucleic-acid-lipid particle (SNALP). A SNALP may contain the lipids 3-N-[(methoxypoly(ethylene glycol) 2000) carbamoyl]-1,2-dimyristyloxy-propylamine (PEG-C-DMA), 1,2-dilinoeoyloxy-N,N-dimethyl-3-aminopropane (DLinDMA), 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC) and cholesterol, in a 2:40:10:48 molar percent ratio. The SNALP liposomes may be prepared by formulating D-Lin-DMA and PEG-C-DMA with distearoylphosphatidylcholine (DSPC), Cholesterol and siRNA using a 25:1 lipid/siRNA ratio and a 48/40/10/2 molar ratio of Cholesterol/D-Lin-DMA/DSPC/PEG-C-DMA. The resulting SNALP liposomes can be about 80-100 nm in size. A SNALP may comprise synthetic cholesterol (Sigma-Aldrich, St Louis, Mo., USA), dipalmitoylphosphatidylcholine (Avanti Polar Lipids, Alabaster, Ala., USA), 3-N-[(w-methoxy poly(ethylene glycol)₂₀₀₀) carbamoyl]-1,2-dimyristyloxypropylamine, and cationic 1,2-dilinoeoyloxy-3-N,N-dimethylaminopropane. A SNALP may comprise synthetic cholesterol (Sigma-Aldrich), 1,2-distearoyl-sn-glycero-3-

phosphocholine (DSPC; Avanti Polar Lipids Inc.), PEG-cDMA, and 1,2-dilinoeoyloxy-3-(N,N-dimethyl)aminopropane (DLinDMA).

[0141] In some cases, a hybrid RNA molecule is present in a ribonucleoprotein (RNP) complex with a CRISPR-Cas effector polypeptide or a CRISPR-Cas fusion polypeptide. The present disclosure thus provides an RNP comprising: a) a hybrid RNA molecule of the present disclosure; and b) a CRISPR-Cas effector polypeptide or a CRISPR-Cas fusion polypeptide. In these cases, the hybrid RNA molecule comprises, in addition to a targeting region and a donor RNA, a binding region that binds to the CRISPR-Cas effector polypeptide or CRISPR-Cas fusion polypeptide.

Methods for Modifying a Target RNA

[0142] The present disclosure provides methods for modifying a target RNA, such that a modified RNA is produced. The methods comprise contacting a target RNA with a hybrid RNA molecule of the present disclosure.

[0143] FIG. 1B depicts trans splicing of a pre-mRNA as one non-limiting embodiment. As depicted in FIG. 2A-2C, a method of the present disclosure can include one or more of: a) initiation of trans-splicing via the binding of a trans-splicing RNA molecule (e.g., a hybrid RNA molecule of the present disclosure) to a target RNA; b) inhibition of the competing cis-splicing reaction; and c) enhancement of trans-splicing efficiency. In FIG. 2C, a CRISPR-Cas effector polypeptide binds to the hybrid RNA molecule to increase efficiency of the trans-splicing region and to inhibit cis-splicing.

[0144] In some cases, the target RNA is present in a eukaryotic cell. In some cases, the target RNA is present in a eukaryotic cell in vitro. In some cases, the target RNA is present in a eukaryotic cell in vivo. In some cases, the cell is a mammalian cell (e.g., a human cell, a non-human primate cell, a bovine cell, an ovine cell, a porcine cell, an equine cell, a feline cell, a canine cell, a rodent cell, etc.). In some cases, the cell is a stem cell, e.g., a pluripotent cell, a hematopoietic stem cell, a stem cell that gives rise to myocytes, etc. In some cases, the cell is an embryonic germ cell, an induced pluripotent stem cell, a mitotic cell, a meiotic cell, or a somatic cell.

[0145] Target RNA that can be modified using a method of the present disclosure include pre-mRNA, microRNA (miRNA), transfer RNA, ribosomal RNA, a circular RNA, a partially spliced RNA, a non-coding RNA, a non-host cell RNA, a regulatory RNA, a coding RNA, a transfer RNA (tRNA), a pre-ribosomal RNA, a ribosomal RNA, a mature RNA with cryptic splice sites, and a long non-coding RNA (lncRNA). In some cases, the target RNA is a pre-mRNA, e.g., an unspliced RNA. In some cases, the target RNA comprises a coding region that encodes an abnormal polypeptide, e.g., a polypeptide that is associated with a disease state. In some cases, the target RNA comprises a defective splice donor and/or a defective splice acceptor.

[0146] A hybrid RNA molecule is introduced into a cell (a target cell) comprising a target RNA. A hybrid RNA molecule can be introduced into a target cell using any of a variety of methods. Suitable methods include e.g., viral infection, transfection, lipofection, electroporation, calcium phosphate precipitation, polyethyleneimine (PEI)-mediated transfection, DEAE-dextran mediated transfection, liposome-mediated transfection, particle gun technology, cal-

cium phosphate precipitation, direct microinjection, nanoparticle-mediated nucleic acid delivery, and the like.

[0147] In some cases, a method of the present disclosure comprises administering to an individual in need thereof an effective amount of a hybrid RNA molecule of the present disclosure, or a DNA molecule (or a recombinant expression vector comprising the DNA molecule) comprising a nucleotide sequence encoding a hybrid RNA molecule of the present disclosure. In some cases, a hybrid RNA molecule of the present disclosure is administered to an individual in need thereof, where the hybrid RNA molecule is formulated in a lipid nanoparticle. In some cases, a method of the present disclosure comprises administering to an individual in need thereof an effective amount of a DNA molecule comprising a nucleotide sequence encoding a hybrid RNA molecule of the present disclosure.

[0148] In some cases, the DNA molecule is integrated into the genome of a cell in the individual. In some cases, a method of the present disclosure comprises administering to an individual in need thereof an effective amount of a recombinant expression vector comprising a DNA molecule comprising a nucleotide sequence encoding a hybrid RNA molecule of the present disclosure. In some cases, the DNA molecule is integrated into the genome of a cell in the individual. In some cases, the recombinant expression vector, or the nucleotide sequence encoding the hybrid RNA molecule, is integrated into the genome of a cell in the individual. In some cases, the recombinant expression vector is not integrated into the genome of a cell in the individual.

[0149] In some cases, a method of the present disclosure for modifying a target RNA comprises contacting the target RNA with a composition comprising: a) a hybrid RNA molecule; and b) a CRISPR-Cas effector polypeptide or a CRISPR-Cas fusion polypeptide. In some cases, the method comprises introducing into a target cell comprising a target RNA: a) a hybrid RNA molecule; and b) a CRISPR-Cas effector polypeptide or a CRISPR-Cas fusion polypeptide. In some cases, the method comprises introducing into a target cell comprising a target RNA: a) a DNA molecule (e.g. a recombinant expression vector) comprising a nucleotide sequence encoding a hybrid RNA molecule; and b) a CRISPR-Cas effector polypeptide or a CRISPR-Cas fusion polypeptide. In some cases, the method comprises introducing into a target cell comprising a target RNA: a) a hybrid RNA molecule; and b) a DNA molecule (e.g. a recombinant expression vector) comprising a nucleotide sequence encoding a CRISPR-Cas effector polypeptide or a CRISPR-Cas fusion polypeptide.

[0150] A method of the present disclosure increases trans-splicing, thereby generating a modified RNA that comprises a nucleotide sequence present in the donor RNA of a hybrid RNA molecule of the present disclosure. In some cases, the method further comprises inhibiting cis-splicing. For example, in some cases, the method further comprises contacting a target RNA (e.g., introducing into a target cell comprising a target RNA), with a RNA-targeting CRISPR-Cas guide RNA comprising a targeting region that comprises a nucleotide sequence that is complementary to a second target sequence of the target RNA, wherein the RNA-targeting CRISPR-Cas guide RNA sterically inhibits cis-splicing of the target RNA. Such a molecule is referred to as a “guided interference molecule” (gIM). This is illustrated schematically in FIG. 5A-5B. In some cases, the method further comprises contacting a target DNA (e.g., introducing

into a target cell comprising a target DNA) with a DNA-targeting CRISPR-Cas guide RNA comprising a targeting region that comprises a nucleotide sequence that is complementary to a target sequence of the target DNA, wherein the DNA-targeting CRISPR-Cas guide RNA interrupts RNA transcription downstream of the targeted site thereby reducing the number transcripts that are able to splice in cis. This is illustrated schematically in FIG. 5C.

[0151] A method for modifying a target RNA according to the present disclosure can be used to insert one or more nucleotides into a target RNA and/or to replace one or more nucleotides in a target RNA. FIG. 6A-6C schematically illustrate various embodiments of modification of a target RNA using trans-splicing.

[0152] In some cases, a method of the present disclosure provides for replacement of from 1 nucleotide (nt) to 9000 nt of a target RNA; e.g., a method of the present disclosure provides for replacement of from 1 nt to 10 nt, from 10 nt to 15 nt, from 15 nt to 20 nt, from 20 nt to 25 nt, from 25 nt to 30 nt, from 30 nt to 40 nt, from 40 nt to 50 nt, from 50 nt to 60 nt, from 60 nt to 70 nt, from 70 nt to 80 nt, from 80 nt to 90 nt, from 90 nt to 100 nt, from 100 nt to 250 nt, from 250 nt to 500 nt, from 500 nt to 1000 nt, from 1000 nt to 2000 nt, from 2000 nt to 3000 nt, from 3000 nt to 4000 nt, from 4000 nt to 5000 nt, from 5000 nt to 6000 nt, from 6000 nt to 7000 nt, from 7000 nt to 8000 nt, or from 8000 nt to 9000 nt, of a target RNA. In some cases, a method of the present disclosure provides for replacement of 1 nt, 2 nt, 3 nt, 4 nt 5 nt, 6 nt, 7 nt, 8 nt, 9 nt, or 10 nt, of a target RNA. In some cases, a method of the present disclosure provides for replacement of from 10 nt to 20 nt (e.g., 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 nt). In some cases, a method of the present disclosure provides for replacement of from 20 nt to 30 nt (e.g., 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nt). In some cases, a method of the present disclosure provides for replacement of from 20 nt to 100 nt. In some cases, a method of the present disclosure provides for replacement of from 100 nt to 200 nt. In some cases, a method of the present disclosure provides for replacement of from 100 nt to 250 nt. In some cases, a method of the present disclosure provides for replacement of from 100 nt to 500 nt. In some cases, a method of the present disclosure provides for replacement of from 100 nt to 1000 nt. In some cases, a method of the present disclosure provides for replacement of from 1000 nt to 1500 nt. In some cases, a method of the present disclosure provides for replacement of from 1500 nt to 2000 nt. In some cases, a method of the present disclosure provides for replacement of from 1500 nt to 2500 nt. In some cases, a method of the present disclosure provides for replacement of from 2000 nt to 8000 nt. In some cases, a method of the present disclosure provides for replacement of from 2000 nt to 7000 nt. In some cases, a method of the present disclosure provides for replacement of from 2000 nt to 6000 nt. In some cases, a method of the present disclosure provides for replacement of from 2000 nt to 5000 nt.

[0153] The following are non-limiting examples of use of a method of the present disclosure. These non-limiting examples are depicted schematically in FIG. 7A-7B and in FIG. 8A-8B.

[0154] In some cases, a method of the present disclosure modifies a target RNA, where the target RNA is a pre-mRNA encoding a protein comprising a Gln repeat, where the Gln repeat comprises (Gln)*n*, where *n* is greater than 30. For example, the pre-mRNA comprises a CAG trinucleotide

repeat expansion such that the encoded protein comprises a Gln repeat. Such proteins can give rise to a disease state. For example, a huntingtin protein comprising a poly(Gln) expansion of more than about 30 Gln residues can result in Huntington's disease. As another example, an ataxin polypeptide comprising a poly(Gln) expansion of more than about 30 Gln residues can result in a spinocerebellar ataxia (SCA). An ataxin-1 polypeptide comprising a poly(Gln) expansion of more than about 30 Gln residues can result in SCA type 1 (SCA1). An ataxin-2 polypeptide comprising a poly(Gln) expansion of more than about 30 Gln residues can result in SCA type 2 (SCA2). An ataxin-3 polypeptide comprising a poly(Gln) expansion of more than about 30 Gln residues can result in SCA type 2 (SCA3). As another example, an atrophin-1 polypeptide comprising a poly(Gln) expansion of more than about 30 Gln residues can result in dentatorubral-pallidoluysian atrophy (DRPLA).

[0155] In some cases, a method of the present disclosure modifies a target RNA, where the target RNA is a pre-mRNA comprising a GGGGCC hexanucleotide expansion repeat, e.g., where the GGGGCC expansion repeat comprises more than about 25 repeats of the hexanucleotide GGGGCC. Diseases associated with such a hexanucleotide expansion repeat include amyotrophic lateral sclerosis (ALS) and frontotemporal dementia (FTD). Such a pre-mRNA can encode a pathological form of C9orf72, leading to ALS or FTD.

[0156] In some cases, a method of the present disclosure modifies a target RNA, where the target RNA is a pre-mRNA comprising a GAA trinucleotide repeat expansion, e.g., where the GAA trinucleotide expansion repeat comprises more than about 25-30 repeats of the trinucleotide GAA. Diseases associated with such trinucleotide repeat expansions include Friedreich's ataxia.

[0157] In some cases, a method of the present disclosure provides for reduced production of a disease-associated expansion repeat protein in a cell in an individual. For example, a method of the present disclosure can provide for a reduction of at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 50%, or more than 50%, of a disease-associated expansion repeat protein in a cell in an individual. In some cases, a method of the present disclosure provides for increased production of a normal form (e.g., a form comprising an expansion repeat that does not lead to pathology) of a disease-associated expansion repeat protein in a cell in an individual. For example, in some cases, a method of the present disclosure provides for at a 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 100% (or 2-fold), at least 2.5-fold, at least 5-fold, at least 10-fold, or more than 10-fold, increased production of a normal form (e.g., a form comprising an expansion repeat that does not lead to pathology) of a disease-associated expansion repeat protein in a cell in an individual.

[0158] A method of the present disclosure can be used to treat any trinucleotide or hexanucleotide expansion repeat disease. Examples of diseases that can be treated using a method of the present disclosure include Huntington's Disease, spinal and bulbar muscular atrophy (SBMA), spinocerebellar ataxia (SCA) Type 1, SCA Type 2, SCA Type 3, SCA Type 6, SCA Type 7, SCA Type 8, SCA Type 12, SCA Type 17, Friedreich's ataxia, C9orf72 frontotemporal dementia, Fragile X syndrome, and myotonic dystrophy.

[0159] The present disclosure provides methods of treating a disease, the method comprising administering to an individual having such disease a hybrid RNA molecule of the present disclosure, or a DNA molecule comprising a nucleotide sequence encoding the hybrid RNA molecule, or a recombinant expression vector comprising the DNA molecule. A method of the present disclosure can be used to treat any genetic disease, e.g., where an RNA comprises a defective intron, a defective donor splice site, a defective acceptor splice site, a defective exon, and the like. FIG. 19 provides non-limiting examples of diseases that can be treated using a method of the present disclosure.

[0160] A target RNA can be present in a cell in vitro (e.g., a cell obtained from an individual in need of treatment), where the cell is modified in vitro using a method of the present disclosure, and the modified cell is introduced into the individual from whom the cell was obtained. A target cell can be present in an individual in vivo. Suitable target cells (where a target cell comprises a target RNA) include, but are not limited to, blood cells, lymphocytes, T cells, hematopoietic stem cells, neuronal stem cells, neurons, glial cells, spleen cells, lung cells (e.g., lung epithelial cells), bone marrow cells, stem cells, and the like.

[0161] FIG. 18 provides non-limiting examples of targeting region sequences that can be included in a hybrid RNA molecule of the present disclosure.

Rhett Syndrome

[0162] As one example, as set out in FIG. 18, an MECP2 target RNA can be targeted, for use in treating Rhett Syndrome. For example, the target sequence of a MECP2 target RNA can be intron 1, intron 2, or intron 3. A suitable donor sequence can include, e.g., the following sequence:

```
(SEQ ID NO: 33)
GGAAGAAAAGTCAGAAGACCAGGACCTCCAGGGCCTCAAGGACAA
ACCCCTCAAGTTTAAAAAGGTGAAGAAAGATAAGAAAGAAGAGAA
AGAGGGCAAGCATGAGCCCGTGCAGCCATCAGCCACCACCTCTGC
TGAGCCCGCAGAGGCAGGCAAGCAGAGACATCAGAAGGGTCAGG
CTCCGCCCCGGCTGTGCCGAAGCTTCTGCCTCCCCAACACAGCG
GCGCTCCATCATCCGTGACCGGGGACCCATGTATGATGACCCAC
CCTGCCTGAAGGCTGGACACGGAAGCTTAAGCAAAGGAAATCTGG
CCGCTCTGCTGGGAAGTATGATGTGTATTGTATCAAA.
```

Huntington's Disease

[0163] As another example, as set out in FIG. 18, a huntingtin-encoding target RNA (HTT target RNA) can be targeted, in connection with treating Huntington's disease. For example, the target sequence of an HTT target RNA can be intron 1. A suitable donor sequence can include, e.g., the following sequence:

```
(SEQ ID NO: 34)
GCTGCCGGGACGGGTCCAAGATGGACGGCCGCTCAGGTTCTGCTT
TTACCTGCGGCCAGAGCCCCATTTCATGCCCCGGTGCTGAGCGG
```

-continued

CGCCGCGAGTCGCGCCGAGGCTCCGGGACTGCCGTGCCGGGCG
 GGAGACCGCCATGGCGACCTGGAAAAGCTGATGAAGGCCCTTCGA
 GTCCCTCAAGTCCTTCAGCAGCAGCAGCAGCAGCAGCAGCAGCA
 GCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAACAGCCGCC
 ACCGCCGCCGCCGCCGCCGCCCTCCTCAGCTTCTCAGCCGCC
 GCGCGAGGCACAGCCGCTGCTGCCTCAGCCGACGCCGCCGCCGCC
 GCCGCCGCCGCCGCCGCCGCCGCCGCTGTGGCTGAGGAGCCGCT
 GCACCGACC.

Examples of Non-Limiting Aspects of the Disclosure

[0164] Aspects, including embodiments, of the present subject matter described above may be beneficial alone or in combination, with one or more other aspects or embodiments. Without limiting the foregoing description, certain non-limiting aspects of the disclosure are provided below. As will be apparent to those of skill in the art upon reading this disclosure, each of the individually numbered aspects may be used or combined with any of the preceding or following individually numbered aspects.

[0165] This is intended to provide support for all such combinations of aspects and is not limited to combinations of aspects explicitly provided below:

[0166] Aspect 1. A hybrid RNA molecule comprising

[0167] a) a targeting region that comprises a nucleotide sequence that is complementary to a target sequence of a target RNA; and

[0168] b) a donor sequence to said targeting region.

[0169] Aspect 2. The hybrid RNA molecule of aspect 1, further comprising one or more of:

[0170] a) a binding region that binds to an RNA binding or modifying polypeptide;

[0171] b) a protein-recruiting region;

[0172] c) a stabilization region.

[0173] Aspect 3. The hybrid RNA molecule of aspect 2, wherein the hybrid RNA molecule comprises a binding region that binds to an RNA binding or modifying polypeptide, and wherein the RNA binding or modifying polypeptide is an RNA-targeting CRISPR-Cas effector polypeptide.

[0174] Aspect 4. The hybrid RNA molecule of aspect 3, wherein the RNA-targeting CRISPR-Cas effector polypeptide is a Type II, a Type III, or a Type VI CRISPR-Cas effector polypeptide.

[0175] Aspect 5. The hybrid RNA molecule of aspect 4, wherein the Type II CRISPR-Cas effector polypeptide is a Cas9 polypeptide.

[0176] Aspect 6. The hybrid RNA molecule of aspect 4, wherein the Type VI CRISPR-Cas effector polypeptide is a Cas13a polypeptide, a Cas13b polypeptide, a Cas13c polypeptide, a Cas13d polypeptide, a Cas13e polypeptide, a Cas13f polypeptide, a Cas13X polypeptide, or a Cas13Y polypeptide.

[0177] Aspect 7. The hybrid molecule of aspect 4, wherein the Type III RNA-targeting CRISPR-Cas effector polypeptide is a Cas7-11 polypeptide.

[0178] Aspect 8. The hybrid RNA molecule of aspect 2, wherein the hybrid RNA molecule comprises a protein-recruiting region.

[0179] Aspect 9. The hybrid RNA molecule of aspect 8, wherein the protein-recruiting region comprises a nucleotide sequence that binds one or more spliceosome polypeptides.

[0180] Aspect 10. The hybrid RNA molecule of aspect 2, wherein the hybrid RNA molecule comprises a stabilization region.

[0181] Aspect 11. The hybrid RNA molecule of aspect 10, wherein the stabilization region is capable of forming a secondary structure.

[0182] Aspect 12. The hybrid RNA molecule of aspect 11, wherein the secondary structure comprises a pseudoknot, a stem-loop, or a tetraloop.

[0183] Aspect 13. The hybrid RNA molecule of any one of aspects 1-12, wherein the RNA donor comprises one or more of:

[0184] a) a coding region comprising a nucleotide sequence encoding all or a portion of a polypeptide;

[0185] b) an untranslated region (UTR);

[0186] c) a bar code;

[0187] d) an internal ribosomal entry site (IRES);

[0188] e) a poly(pyrimidine) tract;

[0189] f) an RNA response element;

[0190] g) an intron; and

[0191] h) an aptamer.

[0192] Aspect 14. The hybrid RNA molecule of any one of aspects 1-13, further comprising a splice function region comprising one or more of:

[0193] a) a splice modifying sequence;

[0194] b) a splice exclusion sequence;

[0195] c) an intron;

[0196] d) a branch point;

[0197] e) an intronic splice enhancer;

[0198] f) an intronic splice silencer; and

[0199] g) a poly(pyrimidine) tract.

[0200] Aspect 15. The hybrid DNA molecule of aspect 14, wherein the splice modifying sequence comprises a splice acceptor, a splice donor, a branchpoint, or a polypyrimidine tract.

[0201] Aspect 16. The hybrid RNA molecule of aspect 13, wherein the RNA donor comprises a nucleotide sequence encoding all or a portion of a polypeptide.

[0202] Aspect 17. The hybrid RNA molecule of aspect 16, wherein the RNA donor comprises a nucleotide sequence encoding a portion of a non-pathological form of a disease-associated polypeptide, wherein the disease-associated polypeptide comprises an amino acid expansion.

[0203] Aspect 18. The hybrid RNA molecule of aspect 17, wherein the disease-associated polypeptide comprises a poly (Gln) repeat region, and wherein the non-pathological form of the polypeptide comprises a (Gln)x repeat region, wherein x is less than the number of repeats associated with a disease.

[0204] Aspect 19. The hybrid RNA molecule of aspect 17, wherein the disease-associated polypeptide comprises a poly (Glu) repeat region, and wherein the non-pathological form of the polypeptide comprises a (Glu)x repeat region, wherein x is less than the number of repeats associated with a disease.

[0205] Aspect 20. The hybrid RNA molecule of aspect 13, wherein the RNA donor comprises an intron comprising a non-pathological number of repeats of GAA.

[0206] Aspect 21. The hybrid RNA molecule of aspect 13, wherein the RNA donor comprises a 5'-UTR comprising a non-pathological number of repeats of CGG, CCG, or GGC.

[0207] Aspect 22. The hybrid RNA molecule of aspect 13, wherein the RNA donor comprises a 3'-UTR comprising a non-pathological number of repeats of CTG.

[0208] Aspect 23. The hybrid RNA molecule of any one of aspects 1-22, comprising a SINE-derived nuclear RNA Localization (SIRLOIN) motif.

[0209] Aspect 24. The hybrid RNA molecule of any one of aspects 1-23, further comprising a second targeting region that comprises a nucleotide sequence that is complementary to a second target sequence of the target RNA and that interferes with cis splicing of the target RNA.

[0210] Aspect 25. The hybrid RNA molecule of any one of aspects 1-24, wherein the RNA donor has a length of from 2 nucleotides to 10,000 nucleotides.

[0211] Aspect 26. A DNA molecule comprising a nucleotide sequence encoding the hybrid RNA molecule of any one of aspects 1-25.

[0212] Aspect 27. The DNA molecule of aspect 26, wherein the nucleotide sequence is operably linked to a transcriptional control element.

[0213] Aspect 28. The DNA molecule of aspect 27, wherein the transcriptional control element comprises a promoter, and wherein the promoter is one or more of: a constitutive promoter, an inducible promoter, a cell type-specific promoter, and a tissue-specific promoter.

[0214] Aspect 29. A recombinant expression vector comprising the DNA molecule of any one of aspects 26-28.

[0215] Aspect 30. A composition comprising:

[0216] a) the hybrid RNA molecule of any one of aspects 1-25; and

[0217] b) one or more of a nanoparticle, a lipid, a buffer, and a nuclease inhibitor.

[0218] Aspect 31. A composition comprising:

[0219] a1) an RNA-targeting CRISPR-Cas effector polypeptide; and

[0220] b1) a hybrid RNA molecule of any one of aspects 1-25; or

[0221] a2) a fusion polypeptide comprising:

[0222] i) an RNA-targeting CRISPR-Cas effector polypeptide; and

[0223] ii) one or more heterologous polypeptides; and

[0224] b2) a hybrid RNA molecule of any one of aspects 1-25.

[0225] Aspect 32. The composition of aspect 31, wherein the RNA-targeting CRISPR-Cas effector polypeptide is a Type II, a Type III, or a Type VI CRISPR-Cas effector polypeptide.

[0226] Aspect 33. The composition of aspect 32, wherein the Type II CRISPR-Cas effector polypeptide is a Cas9 polypeptide.

[0227] Aspect 34. The composition of aspect 32, wherein the Type III CRISPR-Cas effector polypeptide is a Cas7-11 polypeptide.

[0228] Aspect 35. The composition of aspect 32, wherein the Type VI CRISPR-Cas effector polypeptide is a Cas13a polypeptide, a Cas13b polypeptide, a Cas13c polypeptide, a Cas13d polypeptide, a Cas13e polypeptide, a Cas13f polypeptide, a Cas13X polypeptide, or a Cas13Y polypeptide.

[0229] Aspect 36. The composition of aspect 31, wherein the composition comprises:

[0230] a) a fusion polypeptide comprising:

[0231] i) an RNA-targeting CRISPR-Cas effector polypeptide; and

[0232] ii) one or more heterologous polypeptides; and

[0233] b) a hybrid RNA molecule of any one of aspects 1-25.

[0234] Aspect 37. The composition of aspect 36, wherein at least one of the one or more heterologous polypeptides comprises an RNA Polymerase II-binding polypeptide.

[0235] Aspect 38. The composition of aspect 37, wherein the RNA Polymerase II-binding polypeptide comprises a C-terminal interacting domain (CID).

[0236] Aspect 39. The composition of aspect 36, wherein at least one of the one or more heterologous polypeptides comprises a polypeptide that provides for enhanced localization to an SR protein granule.

[0237] Aspect 40. The composition of aspect 39, wherein the polypeptide that provides for enhanced localization to an SR protein granule comprise an RS domain.

[0238] Aspect 41. The composition of any one of aspects 32-40, wherein the RNA-targeting CRISPR-Cas effector polypeptide exhibits reduced enzymatic activity compared to the enzymatic activity of a wild-type RNA-targeting Cas effector polypeptide.

[0239] Aspect 42. The composition of any one of aspects 32-41, wherein the RNA-targeting CRISPR-Cas effector polypeptide comprises a first higher eukaryotes and prokaryotes nucleotide-binding (HEPN) domain and a second HEPN domain.

[0240] Aspect 43. The composition of aspect 42, wherein the RNA-targeting CRISPR-Cas effector polypeptide comprises a mutation in the first HEPN domain and/or the second HEPN domain.

[0241] Aspect 44. The composition of any one of aspects 32-41, wherein the RNA-targeting CRISPR-Cas effector polypeptide is a variant Cas7-11 comprising a mutation in a catalytic domain.

[0242] Aspect 45. A method for modifying a target RNA, the method comprising contacting the target RNA with the hybrid RNA molecule of any one of aspects 1-25, wherein said contacting results in modification of the target RNA.

[0243] Aspect 46. The method of aspect 45, wherein the modification comprises inserting the RNA donor present in the hybrid RNA molecule into the target RNA.

[0244] Aspect 47. The method of aspect 45, wherein the modification comprises replacing one or more endogenous nucleotides in the target RNA with the RNA donor present in the hybrid RNA molecule.

[0245] Aspect 48. The method of any one of aspects 45-47, wherein the target RNA is in a eukaryotic cell.

[0246] Aspect 49. The method of aspect 48, wherein the eukaryotic cell is in vitro.

[0247] Aspect 50. The method of aspect 48, wherein the eukaryotic cell is in vivo.

[0248] Aspect 51. The method of any one of aspects 48-50, wherein the cell is a mammalian cell.

[0249] Aspect 52. The method of any one of aspects 45-51, wherein the target RNA is a pre-mRNA, a circular RNA, a partially spliced RNA, a non-coding RNA, a non-host cell RNA, a regulatory RNA, a coding RNA, a transfer RNA (tRNA), a pre-ribosomal RNA, a ribosomal RNA, a mature RNA with cryptic splice sites, or a long non-coding RNA (lncRNA).

[0250] Aspect 53. A method for modifying a target RNA, the method comprising contacting the target RNA with the composition of any one of aspects 30-44.

[0251] Aspect 54. The method of aspect 53, wherein the composition comprises an RNA-targeting CRISPR-Cas

guide RNA comprising a targeting region that comprises a nucleotide sequence that is complementary to a second target sequence of the target RNA, wherein the RNA-targeting CRISPR-Cas guide RNA inhibits cis-splicing of the target RNA.

[0252] Aspect 55. The method of aspect 53 or aspect 54, wherein the target RNA is a pre-mRNA, a circular RNA, a partially spliced RNA, a non-coding RNA, a non-host cell RNA, a regulatory RNA, a coding RNA, a transfer RNA (tRNA), a pre-ribosomal RNA, a ribosomal RNA, a mature RNA with cryptic splice sites, or a long non-coding RNA (lncRNA).

[0253] Aspect 56. A composition comprising:

[0254] a1) a nucleic acid comprising a nucleotide sequence encoding an RNA-targeting CRISPR-Cas effector polypeptide; and

[0255] b1) a hybrid RNA molecule of any one of aspects 1-25; or

[0256] a2) a nucleic acid comprising a nucleotide sequence encoding a CRISPR-Cas fusion polypeptide comprising:

[0257] i) an RNA-targeting CRISPR-Cas effector polypeptide; and

[0258] ii) one or more heterologous polypeptides; and

[0259] b2) a hybrid RNA molecule of any one of aspects 1-25.

[0260] Aspect 57. The composition of aspect 56, wherein the nucleic acid is present in a recombinant expression vector.

[0261] Aspect 58. The composition of aspect 56 or aspect 57, wherein the nucleotide sequence encoding the RNA-targeting CRISPR-Cas effector polypeptide or the nucleotide sequence encoding the CRISPR-Cas fusion polypeptide is operably linked to a transcriptional control element.

[0262] Aspect 59. The composition of aspect 58, wherein the transcriptional control element is a regulatable promoter.

[0263] Aspect 60. A composition comprising:

[0264] a) a DNA molecule comprising:

[0265] i) a nucleotide sequence encoding an RNA-targeting CRISPR-Cas effector polypeptide; and

[0266] ii) a nucleotide sequence encoding a hybrid RNA molecule of any one of aspects 1-25; or

[0267] b) a DNA molecule comprising:

[0268] i) a nucleotide sequence encoding a fusion polypeptide comprising an RNA-targeting CRISPR-Cas effector polypeptide; and one or more heterologous polypeptides; and

[0269] ii) a nucleotide sequence encoding a hybrid RNA molecule of any one of aspects 1-25.

[0270] Aspect 61. The composition of aspect 55, wherein one or more of the nucleotide sequences is operably linked to a transcriptional control element.

[0271] Aspect 62. A method for modifying a target RNA in a cell, the method comprising introducing into the cell a composition of any one of aspects 56-61.

[0272] Aspect 63. The method of aspect 62, wherein the target RNA is in a eukaryotic cell.

[0273] Aspect 64. The method of aspect 63, wherein the eukaryotic cell is in vitro.

[0274] Aspect 65. The method of aspect 58, wherein the eukaryotic cell is in vivo.

[0275] Aspect 66. The method of any one of aspects 62-65, wherein the cell is a mammalian cell.

[0276] Aspect 67. A method of treating a trinucleotide or hexanucleotide expansion disease in an individual, the method comprising introducing into a cell in the individual a hybrid RNA molecule of any one of aspects 1-25, or a DNA molecule comprising a nucleotide sequence encoding the hybrid RNA molecule of any one of aspects 1-25, wherein the hybrid RNA molecule provides for generation of an mRNA that does not include a pathological number of trinucleotide or hexanucleotide repeats.

[0277] Aspect 68. The method of aspect 67, wherein the trinucleotide or hexanucleotide expansion disease is Huntington Disease, spinal and bulbar muscular atrophy (SBMA), spinocerebellar ataxia (SCA) Type 1, SCA Type 2, SCA Type 3, SCA Type 6, SCA Type 7, SCA Type 8, SCA Type 12, SCA Type 17, Friedreich's ataxia, C9ORF72 frontotemporal dementia, Fragile X syndrome, or myotonic dystrophy.

[0278] Aspect 69. The method of aspect 67 or aspect 68, wherein the cell is a muscle cell or a neuron.

EXAMPLES

[0279] The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how to make and use the present invention, and are not intended to limit the scope of what the inventors regard as their invention nor are they intended to represent that the experiments below are all or the only experiments performed. Efforts have been made to ensure accuracy with respect to numbers used (e.g. amounts, temperature, etc.) but some experimental errors and deviations should be accounted for. Unless indicated otherwise, parts are parts by weight, molecular weight is weight average molecular weight, temperature is in degrees Celsius, and pressure is at or near atmospheric. Standard abbreviations may be used, e.g., bp, base pair(s); kb, kilobase(s); pl, picoliter(s); s or sec, second(s); min, minute(s); h or hr, hour(s); aa, amino acid(s); kb, kilobase(s); bp, base pair(s); nt, nucleotide(s); i.m., intramuscular(ly); i.p., intraperitoneal (ly); s.c., subcutaneous(ly); and the like.

Example 1

[0280] This example demonstrates 3' trans-splicing with a trans splicing molecule (TSM) fused to a guide RNA, using a split green fluorescent protein (GFP) reporter. A TSM fused to a guide RNA is referred to as a "gTSM." As illustrated schematically in FIG. 9A, the target transcript has 2 exons, one with the first 120 amino acids of GFP (5' GFP) and the second that is unrelated to GFP. The gTSM targets the intron between the exons on the target transcript and trans-splices in the rest of GFP (3' GFP) such that the trans-spliced transcript encodes active, fluorescent GFP.

[0281] A target molecule was constructed with the 5' half of msfGFP, RG6 intron 1, and exon 2 from the bichromatic reporter described in Orengo et al. (2006) Nucl. Acids Res. 34:e148. The gTSM constructs comprised an RfxCas13d direct repeat sequence, followed by a spacer targeting one of the locations shown in FIG. 10a or a non-targeting spacer, followed by a stuffer sequence, a branch point, a polypyrimidine tract (PPT), a splice acceptor (SA) and a cargo sequence, which in this case is the 3' half of msfGFP. The effector plasmid consists of catalytically inactive CasRx (dCasRx) linked to mCherry with a P2A linker. All constructs are driven by the Efla promoter.

(SEQ ID NO: 64)
 gttaatcgaattgaactgaaaggtattgattttaaggaggacggt
 aacatactggggcacaagttggagtacaactttaacagccataat
 gtgtatattaccgctgataagcagaaaaatgggataaaggccaac
 ttttaagatccgacataatgtcgaagatggtagtggtcaactggct
 gatcattaccaacaaaatacggccatcgagatggacctgtactc
 ttgctgacaatcattatctctccacgcaatcaaagctttccaag
 gacccaaacgaaaagagagatcacatggctcctctggaatttggtg
 actgccgcaggcatcactctcggtatggatgagctgtacaagtg
 a.

[0282] HEK293 cells were transfected with the target transcript, gTSM, and dCasRx or pUC19.

[0283] As illustrated schematically in FIG. 9B, the gTSM included 5 different spacers (targeting sequences): 5' Intron 1, Intron 1, Splice Acceptor (SA), Exon 2, or non-targeting (NT). The percent of GFP⁺ cells was determined.

[0284] The target and gTSM plasmids were transfected into HEK293 cells with and without dCasRx (FIG. 10B) at a 1:1:1 mass ratio. The results are shown in FIG. 10.

[0285] The data show strong trans-splicing activity with the gTSM targeting the 5' end of Intron 1 of the target molecule in the presence of dCasRx, and much lower trans-splicing efficiency without the effector. The other gTSM spacers also have some trans-splicing efficiency, which is boosted in the presence of dCasRx, and there seems to be very low trans-splicing efficiency with the NT gTSM. Overall, a spacer sequence dependent and effector dependent effect was observed on trans-splicing efficiency.

[0286] As shown in FIG. 10, the pUC19 control (without (w/o) dCasRx) did not provide for production of a GFP-encoding transcript. In the cells that included dCasRx, a low level of background trans-splicing was observed, as illustrated with the cells transfected with the NT gTSM. The best gTSM targeting location was at the beginning of the intron on the target transcript (5' Intron 1), which resulted in ~40% GFP⁺ cells.

[0287] This result was also observed when the effector was catalytically active Cas7-11 (FIG. 20). In this case, the absence of the effector abrogates trans-splicing efficiency. This points to the sensitivity of the gTSM itself to flanking regions, as the direct repeat sequencing in the gTSMs used with Cas7-11 is different from the CasRx direct repeat.

Example 2

[0288] A “gTSM screen” and a “gIM screen” were carried out. For the gTSM screen, multiple gTSMs targeting various locations within an intron of a transcript were screened; this was done using a 2-plasmid transfection (gTSM construct and effector construct). For the gIM screen, multiple gIMs targeting various locations within the intron of a transcript downstream of the gTSM, identified in the gTSM screen as having the highest activity, were screened; this was done using a 3-plasmid transfection (gTSM construct, gIM construct, effector construct).

[0289] The qPCR data indicated the levels of trans-splicing junctions relative to GAPDH expression. The closer to the x-axis indicated a higher level of trans-splicing. Droplet

digital PCR (ddPCR) directly quantified cis and trans junctions; efficiencies are reported as a percentage of total cis and trans junctions.

Methods

Cell Culture

[0290] The HEK293 cell line was purchased from ThermoFisher. HEK293 cells were cultured in DMEM, supplemented with fetal bovine serum (FBS) with 100 U/mL of Penicillin-Streptomycin.

Plasmid Transfections

[0291] All plasmids were prepared using the NucleoSpin Xtra Midi or Maxi EF kits. HEK293 cells were seeded at 14k density into 96 well plates (ThermoFisher 152037). Plasmids were transfected using Lipofectamine 2000 from Invitrogen according to the manufacturer's protocol.

Flow Cytometry

[0292] All flow cytometry was performed using an Attune NxT Flow Cytometer. Cells were trypsinized using TrypLE Express reagent according to the manufacturer's protocols 48 hours post transfection. Trypsinization was inhibited by adding Stain Buffer with 2% FBS (Catalog #: 554656). Cells were transferred to a clear, 96 well U bottom plate, spun for 5 minutes @500 rpm. Trypsin and stain buffer were removed, and 200 μ L of fresh stain buffer was added to each well and mixed. The cells were then run on the Attune according to manufacturer protocols.

qPCR Readout of Endogenous Trans-Splicing

[0293] 48 hours post-transfection, transfected cells were washed with DPBS (ThermoFisher, 14190094). Cells were then incubated and mixed for 8 minutes at room temperature with 50 μ L of RNA lysis buffer (9.6 mM Tris-HCl (pH 7.8), 0.5 mM MgCl₂, 0.44 mM CaCl₂, 10 M DTT, 0.1% (wt/vol) Triton X-114, 300 U ml⁻¹ DNase I, and 3 U ml⁻¹ proteinase K in UltraPure water). 30 μ L of lysed cells were then transferred into a PCR plate with 3 μ L of RNA lysis stop solution (1 mM Proteinase K inhibitor, 90 mM EGTA, 113 μ M DTT in UltraPure water) and incubated at room temperature for 2 minutes. 8 μ L of the lysis solution was then transferred into the RevertAid RT Reverse Transcription mastermix (Thermo Fisher Scientific, cat. No. K1691) and the reverse transcription reaction was incubated at 25° C. for 10 min, 37° C. for 60 min and then 95° C. for 5 min. The completed reverse transcription reaction was then carried into the qPCR reaction. qPCR primers were designed using PrimerBlast and probes were designed to bind across the junction of the trans or cis splice junctions. The RT reaction was then added as a template into a mastermix for four technical replicates using Applied Biosystems™ TaqMan™ Universal PCR Master Mix (Supplier: Applied Biosystems™ 4304437) and read out on a Roche LightCycler. For analysis, trans-splicing junction Cp values were normalized to the corresponding GAPDH Cp values for each condition to account for cell count.

ddPCR Readout of Endogenous Trans-Splicing

[0294] Transfected cells were lysed and reverse transcribed as described previously. qPCR primers were designed using PrimerBlast and probes were designed to bind across the junction of the trans or cis splice junctions. qPCR reactions were set up according to the manufacturer's

instructions using BioRad ddPCR SuperMix for Probes (#1863024). Droplets were generated with the BioRad Automated Digital Droplet Generator per manufacturer instructions and read out using the BioRad Droplet Reader. Trans-splicing efficiency was measured as a percentage of the sum of trans and cis splicing junctions.

Results

[0295] A guided trans-splicing molecule (gTSM) and a guided interfering molecule (gIM) were used for CRISPR-catalyzed trans-splicing, targeting an exon of various target genes.

[0296] The results are shown in FIG. 21 and FIG. 22.

[0297] FIG. 21A-21E. Δ Cp of trans-splicing junctions over a set of endogenous genes for testing gTSMs. The gTSM constructs were tested across a set of genes using dCasRx as the effector: TDP43 (TAR DNA-binding protein-43), SMARCA4 (SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily A, member 4), C9orf72 (chromosome 9 open reading frame 72), HIF1A (hypoxia inducible factor 1 subunit alpha), and ATXN10 (spinocerebellar ataxia type 10 protein). Each transfection consisted of two constructs, one construct expressed the gTSM and the other expressed the effector (dCasRx) FIG. 21A (targeting TDP43). Various sites on a TDP43 intron were targeted with a gTSM. FIG. 21B (targeting SMARCA4). Various sites on a SMARCA4 intron were targeted with a gTSM. FIG. 21C (targeting C9orf72). Various sites on a C9orf72 intron were targeted with a gTSM. FIG. 21D (targeting HIF1A). Various sites on a HIF1A intron were targeted with a gTSM.

[0298] FIG. 21E (targeting ATXN10). Various sites on an ATXN10 intron were targeted with a gTSM.

[0299] The best performing gTSM (from FIG. 21A-21E) was chosen for two of these genes (TDP43, SMARCA4) and were subsequently used to test the gIM concept. The results are shown in FIG. 22A-22B. gTSM-2 from the gTSM screen targeting TDP43 was paired with different gIMs targeting downstream of the gTSM on the same intron. gTSM-7 from the gTSM screen targeting SMARCA4 was paired with different gIMs targeting downstream of the gTSM on the same intron. These transfections consisted of three constructs: one construct expressed the gTSM, a second construct expressed the effector (dCasRx), and the third construct expressed Cas7-11 and its guide.

[0300] FIG. 22A-22B. Targeting TDP43 with hit gTSM and testing gIMs (FIG. 22A). Targeting SMARCA4 with hit gTSM and testing gIMs (FIG. 22B).

Example 3

[0301] This example demonstrates 3' trans-splicing of an endogenous transcript using a trans-splicing molecule (TSM) fused to a guide RNA, referred to as a "gTSM", and a guided interfering molecule, referred to as a "gIM". The gTSM comprises a trans-splicing molecule comprising a spacer targeting the intron intervening exons 7 and 8 of the ITGB1 (integrin subunit beta 1) transcript or a non-targeting spacer, a cargo sequence (i.e. donor RNA), and, optionally, an RNA-targeting CRISPR effector. The gIM comprises an

RNA-targeting CRISPR effector and an interfering guide that targets downstream of the gTSM for cleavage.

[0302] As illustrated schematically in FIG. 23, the gTSM and gIM target the intron between exon 7 and 8 of the target ITGB1 transcript and trans-splices in the gTSM cargo sequence, resulting in a trans-spliced ITGB1 transcript with exon 7 fused to the cargo and no exon 8.

[0303] Four approaches for trans-splicing of the ITGB1 transcript were tested: trans-splicing with only a gTSM comprising a targeting spacer (targeting gTSM; no gIM), trans-splicing with both a gTSM comprising a targeting spacer and the gIM (targeting gTSM; with gIM), trans-splicing with both a gTSM comprising a non-targeting spacer and the gIM (NT gTSM; with gIM), and a gTSM comprising a non-targeting spacer only (NT gTSM; no gIM) as a control. Each of these approaches was also tested with or without an RNA-targeting CRISPR effector for the gTSM. Trans-splicing in ITGB1 (integrin subunit beta 1) was measured with ddPCR to quantitatively assess trans-splicing efficiencies. The results are shown in FIG. 24.

[0304] As shown in FIG. 24, the condition with both a targeting gTSM and gIM (targeting gTSM; with gIM) resulted in ~20% trans-splicing. The results demonstrate ~2% trans-splicing efficiency without gIM (targeting gTSM; no gIM).

[0305] FIG. 24. Percent trans-splicing efficiency for ITGB1 measured by ddPCR.

[0306] The Trans-splicing efficiencies for the ITGB1 transcript of four different RNA-targeting CRISPR effectors were tested: trans-splicing with dCasRx, trans-splicing with dCasRx fused to a double-stranded RNA-binding domain (dCASRx-RBD), trans-splicing with an enhanced dCasRx (HifiCasRx), and trans-splicing with an enhanced dCasRx fused to a double-stranded RNA-binding domain (Hifi-CasRx-RBD). Each RNA-targeting CRISPR effector was tested with the combination of a targeting gTSM with a gIM or the combination of a non-targeting gTSM (NT gTSM) with a gIM. Trans-splicing in ITGB1 was measured with ddPCR to quantitatively assess trans-splicing efficiencies. The results are shown in FIG. 25.

[0307] As shown in FIG. 25, each RNA-targeting CRISPR effector, in combination with a targeting gTSM and gIM, demonstrated ~15%-20% trans-splicing of ITGB1 transcripts.

[0308] FIG. 25. Percent trans-splicing efficiency for ITGB1 with various RNA-targeting CRISPR effectors measured by ddPCR.

[0309] While the present invention has been described with reference to the specific embodiments thereof, it should be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the true spirit and scope of the invention. In addition, many modifications may be made to adapt a particular situation, material, composition of matter, process, process step or steps, to the objective, spirit and scope of the present invention. All such modifications are intended to be within the scope of the claims appended hereto.

SEQUENCE LISTING

Sequence total quantity: 64

SEQ ID NO: 1 moltype = AA length = 1049
 FEATURE Location/Qualifiers
 source 1..1049
 mol_type = protein
 organism = Listeria portnoyi

SEQUENCE: 1

```

MKITKMRVDG RTIVMERTSK EGQLVYEGID GNKTTEIIFD KKKESFYKSI LNKTVRKLDE 60
KVKDRRQRAI NKAINKEIAE LMLAVLHQEV PRLKLHNLKS LSTESLTKLF KPKFQNMISY 120
PPSKDAEHVQ FCLTDIAVPA IQGLDNIKPD WGVFFEKLKP YTDWVESYIH FKQTTIQRSI 180
EQNKIQSPDS PRKVVLQKYV AAFLNGEPLG LDLVAKKYKL ADLAESFKLV DLNEDKSANY 240
KIKACLQQHQ RNILDELKED PELNQYGIEV KKYIQRYFPI KRAPNRSKHA RADFLKKELI 300
ESIVEQQFKN AVYHYVLEQG KMEAYELTNP KTKDLQDIRS GEAFSFKFIN ACAFASNNLK 360
TILNPECEKD ILGKGEPKKS LLNSTTRS DV VKKMIPFFSD EIQNVNPFDEA IWAIRGSIQQ 420
IRNEVYHCKK HSWERILKIK DFEPNKMKYE DSDMQKLMDK DIAKIPDFIE EKLKSSGVIR 480
FYSHDKLQSI WEMKQGFSL TTNAPFVPSF KRVIYAKGHDY QTSKNRYDDL GLTTFDILEY 540
GEEDFRARYF LTKLVYYQQF MPWFTADNNA FRDAANFVLR LNKNRQQDAK AFINIREVEE 600
GEMPRDYMGY VQGQIAIHED STEDTPNHFE KFISQVFIKG FDSHMSADL KFIKNPRNQG 660
LEQSEIEGMS FDIKVEPSFL KNKDDYIAFW TFCMLDARH LSELRNEMIK YDGHLTGEQE 720
IIGLALLGVD SRENDWKQFF SSEWEYKIM KGYVGEELYQ CEPYRQSDGK TPILFRGVEQ 780
ARKYGTETVI QRLFDASPEF KVKSCNITEW ERQKETIET IRRKELHNE WAKNPEKPQN 840
NAFFKEYKEC CDAIDAYNWH KNKTTLVYVN ELHLLIEIL GRYVGYVAIA DRDFQCMANQ 900
YFKHSGITER VEYWDNRLLK SIKKLDTFK KEGLFVSEKN ARNHIAHLNY LSLKSECTLL 960
YLSERLREIF KYDRKLKNAV SKSLIDILDR HGMSVVFANL KENKHLRVIK SLEPKKLRLH 1020
GEKKIDNGYI ETNQVSEEYC GIVKRLLEI 1049

```

SEQ ID NO: 2 moltype = AA length = 1159
 FEATURE Location/Qualifiers
 source 1..1159
 mol_type = protein
 organism = Leptotrichia buccalis

SEQUENCE: 2

```

MKVTVKGGIS HKKYTSEGR L VKSESEENRT DERLSALLNM RLDMYIKNPS STETKENQKR 60
IGLKKKFFSN KMVYLKDNLT SLKNGKKENI DREYSETDIL ESDVRDKKNF AVLKKIYLYNE 120
NVNSEELVVF ENDIKKLLNK INSLKYSFEK NKANYQKINE NNIKKVEGKS KRNIYDYR 180
ESAKRLDAYVS NVKEAFKLY KEEDIAKLVL ELENLTKEK YKIREFYHEI IGRKNDKENF 240
AKIIEEIQN VNNMKELIEK VPDMSSELKS QVIFYKYLDK EELNDKNIKY AFCHFVEIEM 300
SQLLKNYVYK RLSNISNDKI KRIFEYQNLK KLIENKLLNK LDTYVRNCGK YNYLQDGEI 360
ATSDFIARNR QNEAFLRNII GVSSVAYFSL RNILETENEN DITGRMRGKT VKNNKGEEKY 420
VSGEVDKIYN ENKKNEVKEN LKMFYSYDFN MDNKEIEDF PANIDEAISS IRHGIVHFNL 480
ELEKGDIFAF KNIAPSEISK KMFQNEINEK KLKLIKFRQL NSANVFRYLE KYKILNYLKR 540
TRFEFVNKNI PFVPSFTKLY SRIDDLKNSL GIYWKTPKTN DDNKTKEIID AQIYLLKNIY 600
YGEFLNPFMS NNGNFFEISK EIIELNKNDK RNLKTGFYKL QKFEDIQEKI PKEYLANIQS 660
LYMINAGNQD EEEKDTYIDF IQKIFLKGFM TYLANNGRLS LIYIGSDEET NTSLEKKQE 720
FDKFLKKYEQ NNNIKIPYEI NEFLREIKLG NILKYTERLN MFYLLILKLLN HKELTNLKG 780
LEKYQSANKE EAFSDQLELI NLLNLDNMRV TEDFELEADE IGKFLDFNGN KVKDNKELKK 840
PDTNKIYFPG ENIKHRAFY NIKKYGMLNL LEKIADKAGY KISIEELKKY SNKKNEITEK 900
HKMQENLHRK YARPRKDEKF TDDEYESYKQ ALENIEEYTH LKNKVEFNEL NLLQGLLLRI 960
LHRLVGYSI WERDLRFLK GEPPENQYIE EIFNFENKKN VKYKGGQIVE KYIKFYKELH 1020
QNDEVKINKY SSANIKYKQ EKKDLYIRNY IAHFNYIPHA EISLLEVLN LRKLLSYDRK 1080
LKNVAVKSVV DILKEYGVA TFKIGADKKI GIQTLESEKI VHLKNLKKKK LMTDRNSEEL 1140
CKLVKIMFEY KMEEKXSEN 1159

```

SEQ ID NO: 3 moltype = length =
 SEQUENCE: 3
 000

SEQ ID NO: 4 moltype = AA length = 1127
 FEATURE Location/Qualifiers
 source 1..1127
 mol_type = protein
 organism = Prevotella buccae

SEQUENCE: 4

```

MQKQDKLFVD RKKNAIFAPP KYITIMENKE KPPIIYELT DKHFWAAFLN LARHNVYTTI 60
NHINRRLETA ELKDDGYMMG IKGSWNEQAK KLDKKVRLRD LIMKHFPFLE AAAYEMTNSK 120
SPNNKEQREK EQSEALSLNN LKNVLFIFLE KLQVLNRYYS HYKYSEESP KPIFETSLKN 180
MYKVFANVR LVKRDYMHHE NIDMQDRDFT LNRKKQVGR KNIIDSPNFH YHFADKEGNN 240
TIAGLLFFVS LFLDKKDAIW MQKKLKGFKD GRNLRQMTN EVFCRSRIS PKLLENVQT 300
KDWMLQDMLN ELVRCPSLY ERLREKDRES FKVPFDIFSD DYNAEEEPFK NTLVRHQDRF 360
PYFVLRFDL NEIFEQLRFQ IDLGTYHFSI YNKRIGEDE VRHLTHHLYG FARIQDFAPQ 420
NQPEEWRKLV KDLDFHETSQ EPIYISKTAP YHLENEKIGI KFCSAHNNLF PSLQTDKTCN 480
GRSKFNLGTQ FTAEAFSLVH ELLPMMFYVL LLTKDYSRKE SADKVEGIIR KEISNIYAIY 540
DAFANNEINS QADLTRRLQN TNILQGHLPK QMISILKGRQ KDMGKEAERK IGEMIDDTQR 600
RLDLLCKQTN IKRIGKRNA GLLKSGKIAD WLVDNMMRFQ PVQKDQNNIP INNSKANSTE 660
YRMLQRALAL PGSENFRLKA YFNQMLVGN DNPHPFLAET QWEHQTNILS FYRNYLEARK 720

```

-continued

KYLKGLKPQN	WKQYQHFLIL	KVQKTNRNVL	VTGWKNSFNL	PRGIFTQPIR	EWFEKHNNNSK	780
RIYDQILSFD	RVGFVAKAIP	LYFAEEYKDN	VQPFYDYDPN	IGNRLKPKKR	QFLDKKERVE	840
LWQKNKELFK	NYPSEKKKTD	LAYLDPLSWK	KFERELRLIK	NQDIVTWLMF	KELFNMATVE	900
GLKIGEIHLR	DIDTNTANEE	SNNILNRIMP	MKLVPKTYET	DNKGNILKER	PLATFYIEET	960
ETKVLKQGNF	KALVKDRRLN	GLFSFAETTD	LNLEEHPISK	LSVDLELIKY	QTTRISIFEM	1020
TLGLEKKLID	KYSTLPTDSF	RNMLERWLQC	KANRPELKNT	VNSLIAVRNA	FSHNQYPMYD	1080
ATLFAEVKKF	TLFPSVDTKK	IELNIAPQLL	EIVGKAIKEI	EKSENKN		1127

SEQ ID NO: 5 moltype = AA length = 966
 FEATURE Location/Qualifiers
 source 1..966
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 5

IEKKKSFAKG	MGVKSTLVSG	SKVYMTTFAE	GSDARLEKIV	EGDSIRSUNE	GEAFSAEMAD	60
KNAGYKIGNA	KFSHPKGYAV	VANNPLYTGP	VQQDMLGLKE	TLEKRYFGES	ADGNDNICIQ	120
VIHNILDIEK	ILAEYITNAA	YAVNNISGLD	KDIIGFGKFS	TVVYTYDEFKD	PEHHRAAFNN	180
NDKLINAIKA	QYDEFDNFLD	NPRLGYFGQA	FFSKEGRNYI	INYGNECYDI	LALLSGLRHW	240
VVHNNEEESR	ISRTWLNVLD	KNLDNEYIST	LNLYYDRITN	ELTNSFSKNS	AANVNVIAET	300
LGINPAEFAE	QYFRFSIMKE	QKNLGFNITK	LREVMLDRKD	MSEIRKNHKV	FDSIRTKVYT	360
MMDFVIYRYI	IEEDAKVAAA	NKSLPDNEKS	LSEKDIFVIN	LRGSFNDDQK	DALYYDEANR	420
IWRKLENIMH	NIKEFRGNKT	REYKKKDAPR	LPRILPAGRD	VSASFSLMYA	LTMFLDGKEI	480
NDLLTTLINK	FDNIQSFLKV	MPLIGVNAKF	VEEYAFFKDS	AKIADELRLI	KSFARMGEPI	540
ADARRAMYID	AIRILGTNLS	YDELKALADT	FSLDENGNKL	KKGKHGMRNF	IINNVISNKR	600
PHYLIRYGD	AHLHEIAKNE	AVVKFVLGRI	ADIQKKQCGN	GKNQIDRYYE	TCIGKDKGKS	660
VSEKVDALTK	IITGMNYDQF	DKKRSVIEDT	GRENAEREKF	KKIISLYLTV	IYHILKNIVN	720
INARVVGIFH	CVERDAQLYK	EKGYDINLKK	LEEKGFSSVT	KLCAGIDETA	PDKRKDVEKE	780
MAERAKESID	SLESANPKLY	ANYIKYSDEK	KAEFPTRQIN	REKAKTALNA	YLRNTKWNVI	840
IREDLLRIDN	KTCTLFANKA	VALEVARYVH	AYINDIAEVN	SYFQLYHYIM	QRIIMNERYE	900
KSSGKVSEYF	DAVNDEKKYN	DRLLKLLCVP	FGYCIPRFKN	LSIEALFDRN	EAAKFDKEKK	960
KVSGNS						966

SEQ ID NO: 6 moltype = AA length = 966
 FEATURE Location/Qualifiers
 source 1..966
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 6

IEKKKSFAKG	MGVKSTLVSG	SKVYMTTFAE	GSDARLEKIV	EGDSIRSUNE	GEAFSAEMAD	60
KNAGYKIGNA	KFSHPKGYAV	VANNPLYTGP	VQQDMLGLKE	TLEKRYFGES	ADGNDNICIQ	120
VIHNILDIEK	ILAEYITNAA	YAVNNISGLD	KDIIGFGKFS	TVVYTYDEFKD	PEHHRAAFNN	180
NDKLINAIKA	QYDEFDNFLD	NPRLGYFGQA	FFSKEGRNYI	INYGNECYDI	LALLSGLRHW	240
VVHNNEEESR	ISRTWLNVLD	KNLDNEYIST	LNLYYDRITN	ELTNSFSKNS	AANVNVIAET	300
LGINPAEFAE	QYFRFSIMKE	QKNLGFNITK	LREVMLDRKD	MSEIRKNHKV	FDSIRTKVYT	360
MMDFVIYRYI	IEEDAKVAAA	NKSLPDNEKS	LSEKDIFVIN	LRGSFNDDQK	DALYYDEANR	420
IWRKLENIMH	NIKEFRGNKT	REYKKKDAPR	LPRILPAGRD	VSASFSLMYA	LTMFLDGKEI	480
NDLLTTLINK	FDNIQSFLKV	MPLIGVNAKF	VEEYAFFKDS	AKIADELRLI	KSFARMGEPI	540
ADARRAMYID	AIRILGTNLS	YDELKALADT	FSLDENGNKL	KKGKHGMRNF	IINNVISNKR	600
PHYLIRYGD	AHLHEIAKNE	AVVKFVLGRI	ADIQKKQCGN	GKNQIDRYYE	TCIGKDKGKS	660
VSEKVDALTK	IITGMNYDQF	DKKRSVIEDT	GRENAEREKF	KKIISLYLTV	IYHILKNIVN	720
INARVVGIFH	CVERDAQLYK	EKGYDINLKK	LEEKGFSSVT	KLCAGIDETA	PDKRKDVEKE	780
MAERAKESID	SLESANPKLY	ANYIKYSDEK	KAEFPTRQIN	REKAKTALNA	YLRNTKWNVI	840
IREDLLRIDN	KTCTLFANKA	VALEVARYVH	AYINDIAEVN	SYFQLYHYIM	QRIIMNERYE	900
KSSGKVSEYF	DAVNDEKKYN	DRLLKLLCVP	FGYCIPRFKN	LSIEALFDRN	EAAKFDKEKK	960
KVSGNS						966

SEQ ID NO: 7 moltype = AA length = 1082
 FEATURE Location/Qualifiers
 source 1..1082
 mol_type = protein
 organism = Prevotella sp.

SEQUENCE: 7

NQKKYFGTYS	VMAMLNAQTV	LDHIQKVADI	EGEQNENNEN	LWFHPVMSHL	YNAKNGYDKQ	60
PEKTMPIIER	LQSYFPFLKI	MAENQREYSN	GKYKQNRVEV	NSNDIFEVLK	RAFGLVKMYR	120
DLTNHYKTYE	EKLNDGCEFL	TSTEQPLSGM	INNYTVALR	NMNERYGKKT	EDLAFIQDKR	180
FKFVKDAYGK	KKSQVNTGFF	LSLQDYNGDT	QKKLHLSGVG	IALLICLFLD	KQYINIFLSR	240
LPIFSSYNAQ	SEERRIIRS	FGINSIKLPK	DRIHSEKSNK	SVAMDMLENE	KRCPDELFTT	300
LSAEKQSRFR	IISDDHNEVL	MKRSSDRFVP	LLLQYIDYVK	LFDHIRPHVN	MGLRLYLKKA	360
DKTCIDGQTR	VRVIEQPLNG	FGRLEEAEATM	RKQENGTFGN	SGIRIRDFEN	MKRDDANPAN	420
YPYIVDTYTH	YILENNKVEM	FINDKEDSAP	LLPVIEDDRY	VVKTIPTSCRM	STLEIPAMAF	480
HMFLEFGSKT	EKLIVDVHNR	YKRLFQAMQK	EEVTAENIAS	FGIAESDLPO	KILDLSISGNA	540
HGKDVDAFIR	LTVDDMLTDT	ERRIKRPFKD	RKSIRSADNK	MGKRGFKQIS	TGKLADFLAK	600
DIVLFQPSVN	DGENKITGLN	YRIMQSAIAV	YDSGDDYEAK	QQFKLMFEKA	RLIGKGTTEP	660
HPFLYKVFAR	SIPANAVEFY	ERYLIERKFY	LTGLSNEIKK	GNRVDVPFIR	RDQNKWKTPA	720
MKTLGRIYSE	SLPVELPRQM	FDNEIKSHLK	SLPQMEGIDF	NNANVTYLIA	EYMKRVLDD	780
FQTFYQWNRN	YRYMDMLKGE	YDRKGSLLQC	FTSVEEREGL	WKERASRTER	YRKQASNKIR	840

-continued

```

SNRQMRNASS EEIETILDKR LSNRNEYQK SEKVIRRYRV QDALLFLAK KTLTELADFD 900
GERFKLKEIM PDAEKGILSE IMPMSPTFEK GKKYTTITSE GMKLKNYGDF FVLASDKRIG 960
NLLLELVGSDI VSKEDIMEEF NKYDQCRPEI SSIVFNLEKW AFDTYPELSA RVDREEKVDF 1020
KSILKILLNN KNINKEQSDI LRKIRNAFDH NNYPDKGVVE IKALPEIAMS IKKAFGEYAI 1080
MK 1082

```

```

SEQ ID NO: 8      moltype = AA length = 1082
FEATURE          Location/Qualifiers
source           1..1082
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 8
NQKKYFGTYS VMAMLNAQTV LDHIQKVADI EGEQNENNEN LWFHPVMSHL YNAKNGYDKQ 60
PEKTMFIIER LQSYFPFLKI MAENQREYSN GKYKQNRVEV NSNDIFEVLK RAFGVLKMYR 120
DLTNAYKTYE EKLNDGCEFL TSTEQPLSGM INNYTVALR NMNERYGKYT EDLAFIQDKR 180
FKFVKDAYGK KKSQVNTGFF LSLQDVNGDT QKKLHLSGVG IALLICFLD QYINIFLSR 240
LPIFSSYNAQ SEERRIIRRS FGINSIKLPK DRIHSEKSNK SVAMDMLNEV KRCPDELFTT 300
LSAEKQSRFR IISDDHNEVL MKRSSDRFVP LLLQYIDYK LFDHIRFHVN MGKLRVLLKA 360
DKTCIDGQTR VRVIEQPLNG FGRLEEAETM RKQENGTFGN SGIRIRDFEN MKRDDANPAN 420
YPYIVDTYTH YILENNKVM FINDKEDSAP LLPVIEDDRY VVKTIPTSCRM STLEIPAMAF 480
HMFPLGSKKT EKLIVDVHNR YKRLFQAMQK EEVTAENIAS FGIAESDLPO KILDLSIGNA 540
HGKDVDADFIR LTVDDMLTDT ERRIKRFKDD RKSIRSADNK MGKRGFKQIS TGKLADFLAK 600
DIVLPQPSVN DGENKITGLN YRIMQSAIAV YDSGDDYEAK QQFKLMFEKA RLIGKGTTEP 660
HPFLYKVFAR SIPANAVEFY ERYLIERKFY LTGLSNEIKK GNRVDVPFIR RDQNKWKTPA 720
MKTGLRIYSE DLPVELPRQM FDNEIKSHLK SLPQMEGIDF NNANVTYLIA EYMKRVLDDD 780
PQTFYQWNRN YRYMDMLKGE YDRKGSLOHC FTSVEEREGL WKERASRTER YRKQASNKIR 840
SNRQMRNASS EEIETILDKR LSNRNEYQK SEKVIRRYRV QDALLFLAK KTLTELADFD 900
GERFKLKEIM PDAEKGILSE IMPMSPTFEK GKKYTTITSE GMKLKNYGDF FVLASDKRIG 960
NLLLELVGSDI VSKEDIMEEF NKYDQCRPEI SSIVFNLEKW AFDTYPELSA RVDREEKVDF 1020
KSILKILLNN KNINKEQSDI LRKIRNAFDA NNYPDKGVVE IKALPEIAMS IKKAFGEYAI 1080
MK 1082

```

```

SEQ ID NO: 9      moltype = AA length = 367
FEATURE          Location/Qualifiers
source           1..367
                  mol_type = protein
                  organism = Homo sapiens

```

```

SEQUENCE: 9
SPTSPAYEPR SPGGYTQSP SYSPTSPSYS PTSPSYSPTS PNYSTSPSY SPTSPSYSPT 60
SPSYSPTSPTS YSPTSPSYSP SYSPTSPSYS PTSPSYSPTS PSYSPTSPTS 120
SPTSPSYSPT SPSYSPTSPTS YSPTSPSYSP TSPNYSTSP NYTPTSPSYS PTSPSYSPTS 180
PNYTPTSPNY SPTSPSYSP SPSYSPTSPTS YSPSSPRYTP QSPTYTPSSP SYSPSSPSYS 240
PTSPKYTPTS PSYSPTSPTS TPTSPKYSPT SPKYSPTSPT YSPTSPTYSP TTPKYSPTSPT 300
TYSPTSPTYT PTSPKYSPTS PTYSPTSPTKY SPTSPTYTSP SPKGSTYSPT SPGYSPTSPT 360
YSLTSPA 367

```

```

SEQ ID NO: 10     moltype = RNA length = 419
FEATURE          Location/Qualifiers
source           1..419
                  mol_type = genomic RNA
                  organism = Encephalomyocarditis virus

```

```

SEQUENCE: 10
gtctctcttga cgagcattcc taggggtctt tccctctctg ccaaaggaat gcaaggtctg 60
ttgaatgtcg tgaaggaagc agttctctctg gaagcttctt gaagacaaac aacgtctgta 120
gcgacccttt gcaggcagcg gaacccccca cctggcgaca ggtgcctctg cgccaaaag 180
ccacgtgtat aagatacacc tgcaaaaggc gcacaacccc agtgccacgt tgtgagttgg 240
atagttgtgg aaagagtc aaatggctctc tcaagcgtat tcaacaaggg gctgaaggat 300
gccagaaggg tacccattg tatgggatct gatctggggc ctcgggtgcac atgctttaca 360
tgtgtttagt cgagggttaa aaacgtctag gcccccga ccacggggac gtggttttc 419

```

```

SEQ ID NO: 11     moltype = RNA length = 174
FEATURE          Location/Qualifiers
source           1..174
                  mol_type = genomic RNA
                  organism = Foot-and-mouth disease virus SAT 2

```

```

SEQUENCE: 11
cttggtttct ggagacagcg taaggatgcc cttcaggtac cccgaggtaa caagcgacac 60
tcgggatctg agaaggggat tggggcttct atccaagcg ccaatttaaa aagcttctat 120
gcctgaatag gtgaccggag gccggcacct tttcttttac ccacaactta cttt 174

```

```

SEQ ID NO: 12     moltype = RNA length = 432
FEATURE          Location/Qualifiers
source           1..432
                  mol_type = genomic RNA
                  organism = Foot-and-mouth disease virus A

```

```

SEQUENCE: 12

```

-continued

```

gtgcaatctg aaactccgcc tggctcttcc aggtctagag gggtgacact ttgtactgtg 60
catgactcca cggttogatcc actggcgagt gctagtagta gtactgctgt ttctagcgcg 120
aacacgcagg ccgtgggact ccccccttgg taacaaggac ccgcggggcc aaaagccacg 180
tctctgaccc gtcgtgtgtg caaaccacgc acagcagctt tactgcgaaa accactttaa 240
ggtgacattg atactggtag tcacgcactg gtgacaggct aaggatgcc ttcaggtagc 300
ccgaggcaac aagcgacact cgggactctg gaaggggact ggggcttctg taaaagcgcc 360
cagtttaaaa agcttctatg cctgaatagg tgacgggagg ccggcacctt tccctctact 420
accactgact ct                                     432

```

```

SEQ ID NO: 13      moltype = RNA length = 353
FEATURE           Location/Qualifiers
source            1..353
                  mol_type = genomic RNA
                  organism = Cricket paralysis virus

```

```

SEQUENCE: 13
ttcgatacca agagctgggtg tctgagggat tacgactcgc gtggtcggac agcacatgtt 60
agtcatgtat tatggtgttt cccaacacgt atctttgctc tagtattgaa cttagtgat 120
gagtgtactt ggcattgtaca cctgattagt tagcccttgt ggatacgcca ctgccgtca 180
ggtacagtag gacatacccc aaatattggg ttccactcgt agagaagaaa attagtggcg 240
aaacgtacgg tttatgtcaa aggagaaggt gattttaatt gcacctgaag caaccctagg 300
tcttcccttt gatgtgagct aataaggagt aaccgattct tacaatgtga tca 353

```

```

SEQ ID NO: 14      moltype = RNA length = 174
FEATURE           Location/Qualifiers
source            1..174
                  mol_type = mRNA
                  organism = Homo sapiens

```

```

SEQUENCE: 14
ccgggaaccc ctaccccatc cccttattca gcacatgaaa taaacaaggg gcatccaaat 60
cttgcggaac cgccccggg acatgcatcg tccctggac tctctcaac cccttatccc 120
tctggacaga atgcaggctc aaccacgctg gtataccctc aaaccctca gaca 174

```

```

SEQ ID NO: 15      moltype = RNA length = 94
FEATURE           Location/Qualifiers
source            1..94
                  mol_type = mRNA
                  organism = Homo sapiens

```

```

SEQUENCE: 15
cgggccccctg ccgcctgcaa gtcggaaatt gcgctgtgct cctgtgtctac ggctgtggc 60
tggaactgctt gctgtgccc aactggctgg caag 94

```

```

SEQ ID NO: 16      moltype = RNA length = 294
FEATURE           Location/Qualifiers
source            1..294
                  mol_type = mRNA
                  organism = Homo sapiens

```

```

SEQUENCE: 16
gctagctcgg gccgggagga gccgcagccg gaggaggggg aggaggaaga agagaaggaa 60
gaggagaggg ggcgcagtg ggcactcggc gctcggaagc cgggctcatg gacgggtgag 120
ggcgcggtgt gcgcagacag tgetccagcc gcgcgcgctc cccaggccct ggccccggcc 180
tcgggcccgg gaggaagagt agctcgccga ggcgcagagg agagcgggcc gccccacagc 240
ccgagccgga gagggagcgc gagccgcgcc ggcgccggtc gggcctccga aacc 294

```

```

SEQ ID NO: 17      moltype = AA length = 1601
FEATURE           Location/Qualifiers
source            1..1601
                  mol_type = protein
                  organism = Desulfonema ishimotonii

```

```

SEQUENCE: 17
MTTMMKISIE FLEPFRMTKW QESTRRNKNN KEFVRGQAPA RWHRNKKDNT KGRPYITGTL 60
LRSVIRSAE NLLTSLSDGKI SEKTCPCGKF DTEDKDRLLQ LRQRSTLRWT DKNPCPDNAE 120
TYCPFCCELLG RSGNDGKAE KKDWRFRIFH GNLSLPGKPD FDGPKAIGSQ RVLNRVDFKS 180
GKAHDFFKAY EVDHTRFRFR EGEITIDNKV SAEARKLLCD SLKFTDRLCG ALCVIRFDEY 240
TPAADSGKQT ENVQAEPNAN LAEKTAEQII SILDDNKTE YTRLADAIR SLRRSSKLVA 300
GLPKDHDGKD NGLDWDIGKK KKDENVVIR QILTTSDATK ELKNAGKWE FCEKLGEALY 360
LKSMDMSGGL KITRRLGDA EFHGKPDRL EKSRSVIGSV LKETVVGEL VAKTPFFFGA 420
IDEDAKQTDL QVLLTPDNKY RLPRSAVRGI LRRDLQTYFD SPCNAELGGR PCMCKTCRIM 480
RGITVMDARS EYNAPPEIRH RTRINPFTGT VAEGALFNME VAPEGIVFPF QLRYSGEDG 540
LPDALKTVLK WWAEGQAFMS GAATGKGRF RMENAKYETL DLSDENQRND YLKNWGWDE 600
KGLEELKKRL NSGLPEPGNY RDPKWHEINV SIEMASPPIN GDPIRAADV KRGTDVVTFK 660
YKAEGEAEKP VCAYKAESFR GVIRSAVARI HMDGVPSTE LTHSDCECLL CQIFGSEYEA 720
GKIRFEDLVF ESDPEPVTFD HVAIDRFTGG AADKKKFDSD PLPGSPARPL MLKGSFWIRR 780
DVLEDEEYCK ALGKALADV NGLYPLGGKS AIGYGQVKS LGIKDDKRIS RLMNPAFDET 840
DVAVPEPKPT DAEVRIEAEK VYYPHYFVEP HKKVEREEKP CGHQKPFHEGR LTGKIRCKLI 900
TKTPLIVPT NDDFFFPAD KEARKEKDEY HKSIAFFRLH KQIMIPGSEL RGMVSSVYET 960
VTNSCFRIFD ETKRLSWRMD ADHQNVLDQF LPRGVATDGK HIQKFSETAR VPPYDKTQKH 1020

```

-continued

FDILDEQEIA	GEKPVRMVVK	RFIKRLSLVD	PAKHPQKKQD	NKWKRRKEGI	ATFIEQKNGS	1080
YYFNVVTNNG	CTSFHLWHKP	DNFDQEKLEG	IQNGEKLDGW	VRDSRYQKAF	QEIPENDPDG	1140
WECKEGYLHV	VGPSKVEFSD	KKGDVINNFQ	GTLPSPVNDW	KTIRTNDFKN	RKRKNPEVFC	1200
CEDDKGNYIT	MAKYCETFFF	DLKENEEYEI	PEKARIKYKE	LLRVYNNNPQ	AVPESVFQSR	1260
VARENVEKLG	SGDLVYFKHN	EKYVEDIVPV	RISRTVDDRM	IGKRMSADLR	PCHGDWVEDG	1320
DLALNAYPE	KRLLLRHPKG	LCPACRLFGT	GSYKGRVRFG	FASLENDPEW	LIPGKNPGDP	1380
FHGGPVMLSL	LERPRPTWSI	PGSDNKFKVP	GRKFYVHHHA	WKTIKDGNHP	TTGKAIEQSP	1440
NNRTVEALAG	GNSFSFEIAP	ENLKEWELGL	LIHSLQLEKG	LAHKLGMMAKS	MGFGSVEIDV	1500
ESVRLRKDWK	QWRNGNSEIP	NWLKGKFAKL	KEWFRDELDF	IENLKKLLWF	PEGDQAPRVC	1560
YPMRLKKDDP	NGNSGYEELK	DGEFKKEDRQ	KKLTPWTPW	A		1601

SEQ ID NO: 18 moltype = AA length = 1601
 FEATURE Location/Qualifiers
 source 1..1601
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 18

MTTMMKISIE	FLEPFRMTKW	QESTRRNKNN	KEFVRGQAF	RWHRNKKDNT	KGRPYITGTL	60
LRSAVIRSAE	NLLTSLDGGKI	SEKTCPCPKF	DTEDKDRLLQ	LRQRSTLRWT	DNPCPDNAE	120
TYCPFCELLG	RSNGDGGKAE	KKDWRPRIHF	GNLSLPGKPD	FDGPKAIGSQ	RVLNRVDFKS	180
GKAHDFFKAY	EVDHTRFPRF	EGEITIDNKV	SAEARKLLCD	SLKFTDRLCG	ALCVIRFDEY	240
TPAADSGKQT	ENVQAEPNAN	LAEKTAEQII	SILDDNKKTE	YTRLLADAIR	SLRRSSKLVA	300
GLPKDHDGKD	DHYLWDIGKK	KKDENSVTIR	QILTTSADTK	ELKNAGKWRE	FCEKLGEALY	360
LKSKDMSGGL	KITRILGDA	EFHGKPDRL	KRSVSIGSV	LKETVVCGEL	VAKTPFFFGA	420
IDEDAKQTAL	QVLLTPDNKY	RLPRSARVGI	LRRDLQTYFD	SPCNAELGGR	PCMCKTCRIM	480
RGITVMDARS	EYNAPPEIRH	RTRINPFTGT	VAEGALFNME	VAPEGIVPFF	QLRYRGSEDG	540
LPDAKTVLTK	WVAEGQAFMS	GAASTGKGRF	RMENAKYETL	DLSDENQRND	YLNKGWRDE	600
KGLEELKKRL	NSGLPEPGNY	RDPKWHEINV	SIEMASPFIN	GDPIRAAVDK	RGTAUVTFVK	660
YKAEGEAEAP	VCAKAESEFR	GVIRSAVARI	HMEDGVPLTE	LTHSDCECLL	CQIFGSEYEA	720
GKIRFEDLVF	ESDPEPVTFD	HVAIDRFTGG	AADKKKFDD	PLPGSPARPL	MLKGSFWIRR	780
DVLEDEEYCK	ALGKALADYN	NGLYPLGGKS	AIGYGQVKSL	GKGGDKKRIS	RLMNPADFET	840
DVAVPEKPKT	DAEVRIEAEK	VYYPHYFVEP	HKKVEREEKP	CGHQKFHEGR	LTGKIRCKLI	900
TKTPLIVPDT	SNDDFFRPAD	KEARKEKDEY	HKSIAFFRLH	KQIMIPGSEL	RGMVSSVYET	960
VTNSCFRIED	ETKRLSWRMD	ADHQNVLQDF	LPGRVTADGK	HIQKFSETAR	VPPYDKTQKH	1020
FDILDEQEIA	GEKPVRMVVK	RFIKRLSLVD	PAKHPQKKQD	NKWKRRKEGI	ATFIEQKNGS	1080
YYFNVVTNNG	CTSFHLWHKP	DNFDQEKLEG	IQNGEKLDGW	VRDSRYQKAF	QEIPENDPDG	1140
WECKEGYLHV	VGPSKVEFSD	KKGDVINNFQ	GTLPSPVNDW	KTIRTNDFKN	RKRKNPEVFC	1200
CEDDKGNYIT	MAKYCETFFF	DLKENEEYEI	PEKARIKYKE	LLRVYNNNPQ	AVPESVFQSR	1260
VARENVEKLG	SGDLVYFKHN	EKYVEDIVPV	RISRTVDDRM	IGKRMSADLR	PCHGDWVEDG	1320
DLALNAYPE	KRLLLRHPKG	LCPACRLFGT	GSYKGRVRFG	FASLENDPEW	LIPGKNPGDP	1380
FHGGPVMLSL	LERPRPTWSI	PGSDNKFKVP	GRKFYVHHHA	WKTIKDGNHP	TTGKAIEQSP	1440
NNRTVEALAG	GNSFSFEIAP	ENLKEWELGL	LIHSLQLEKG	LAHKLGMMAKS	MGFGSVEIDV	1500
ESVRLRKDWK	QWRNGNSEIP	NWLKGKFAKL	KEWFRDELDF	IENLKKLLWF	PEGDQAPRVC	1560
YPMRLKKDDP	NGNSGYEELK	DGEFKKEDRQ	KKLTPWTPW	A		1601

SEQ ID NO: 19 moltype = RNA length = 30
 FEATURE Location/Qualifiers
 source 1..30
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 19
 ttctgaattc ttccggatat cgccctggatc 30

SEQ ID NO: 20 moltype = RNA length = 30
 FEATURE Location/Qualifiers
 source 1..30
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 20
 aatggccttt tctgctcagt ctgcccatgg 30

SEQ ID NO: 21 moltype = RNA length = 30
 FEATURE Location/Qualifiers
 source 1..30
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 21
 tgccctaaag ctgtggggca gagctacggg 30

SEQ ID NO: 22 moltype = RNA length = 30
 FEATURE Location/Qualifiers
 source 1..30
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 22
 gatatgcctt ggatcctgag ccaggaggga 30

-continued

SEQ ID NO: 23	moltype = RNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 23		
gctctccctg ggcagggtgg cttcgaggct		30
SEQ ID NO: 24	moltype = RNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 24		
gtgctcagcc ccaatgctgc ctctccata		30
SEQ ID NO: 25	moltype = RNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 25		
ggcccactag gttcccaccc gaggaccagt		30
SEQ ID NO: 26	moltype = RNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 26		
ctgcagagag caggggtgca aggtcagccc		30
SEQ ID NO: 27	moltype = RNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 27		
tgccgggggg cggggggtcag gggcgggacg		30
SEQ ID NO: 28	moltype = RNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 28		
acagtataaa agtaccceaa agactaaaga		30
SEQ ID NO: 29	moltype = RNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 29		
ccatgaggga tccttgctcc tgcctccct		30
SEQ ID NO: 30	moltype = RNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 30		
cttagttcta cacctcattc attcattcag		30
SEQ ID NO: 31	moltype = RNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 31		
tactccatgt tgcgggcaga atggggatct		30
SEQ ID NO: 32	moltype = RNA length = 30	
FEATURE	Location/Qualifiers	
source	1..30	

-continued

```

mol_type = other RNA
organism = synthetic construct

SEQUENCE: 32
gggatctgga caggaagca cagggcacga 30

SEQ ID NO: 33      moltype = RNA length = 350
FEATURE           Location/Qualifiers
source            1..350
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 33
ggaagaaaag tcagaagacc aggacctcca gggcctcaag gacaaacccc tcaagtttaa 60
aaaggtgaag aaagataaga aagaagagaa agagggcaag catgagcccg tgcagccatc 120
agcccaccac tctgctgagc ccgcagaggc aggcaaagca gagacatcag aagggtcagg 180
ctcccctccg gctgtgccgg aagcttctgc ctcccccaaa cagcggcgct ccatcatccg 240
tgaccgggga cccatgtatg atgacccac cctgcctgaa ggctggacac ggaagcttaa 300
gcaaaggaaa tctggccgct ctgctgggaa gtatgatgtg tatttgatca 350

SEQ ID NO: 34      moltype = RNA length = 414
FEATURE           Location/Qualifiers
source            1..414
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 34
gctgccggga cgggtccaag atggacggcc gctcaggttc tgcttttacc tgcggcccg 60
agccccattc attgcccagg tgctgagcgg ccgcgcgagt cggcccgagg cctccgggga 120
ctgccgtgac gggcgggaga ccgccatggc gacctggaa aagctgatga aggccttcga 180
gtccctcaag tccttcagc agcagcagca gcagcagcag cagcagcagc agcagcagca 240
gcagcagcag cagcagcagc aacagccgcc accgcgcgcg ccgcgcgcgc cgctcctca 300
gcttcctcag ccgcgcgcgc aggcacagcc gctgctgcct cagccgcagc cgccccgcgc 360
gccgcccccg ccgccaccgc gcccggtgt ggctgaggag ccgctgcacc gacc 414

SEQ ID NO: 35      moltype = RNA length = 16
FEATURE           Location/Qualifiers
source            1..16
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 35
tcttcttttt tttctc 16

SEQ ID NO: 36      moltype = RNA length = 15
FEATURE           Location/Qualifiers
source            1..15
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 36
ttccctctct tttccc 15

SEQ ID NO: 37      moltype = RNA length = 39
FEATURE           Location/Qualifiers
source            1..39
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 37
ggagcaggcg ccggatccgg cgcaggagcc ggcgccacc 39

SEQ ID NO: 38      moltype = RNA length = 30
FEATURE           Location/Qualifiers
source            1..30
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 38
gactacctct atatgaaaga ggactaaaac 30

SEQ ID NO: 39      moltype = RNA length = 31
FEATURE           Location/Qualifiers
source            1..31
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 39
gaccacccca aaaatgaagg ggactaaaac a 31

SEQ ID NO: 40      moltype = RNA length = 36
FEATURE           Location/Qualifiers
source            1..36
                  mol_type = other RNA

```

-continued

```

                                organism = synthetic construct
SEQUENCE: 40
caagtaaacc cctaccaact ggtcgggggtt tgaaac                               36

SEQ ID NO: 41      moltype = RNA length = 36
FEATURE           Location/Qualifiers
source            1..36
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 41
gttgtggaag gtccagtttt gaggggctat tacaac                               36

SEQ ID NO: 42      moltype = RNA length = 16
FEATURE           Location/Qualifiers
source            1..16
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 42
gttgatgtca cggaac                                                       16

SEQ ID NO: 43      moltype = RNA length = 36
FEATURE           Location/Qualifiers
source            1..36
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 43
caactacaac cccgtaaaaa tacgggggttc tgaaac                               36

SEQ ID NO: 44      moltype = RNA length = 36
FEATURE           Location/Qualifiers
source            1..36
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 44
gaccaacacc tctgcaaaac tgcaggggtc taaaac                               36

SEQ ID NO: 45      moltype = RNA length = 42
FEATURE           Location/Qualifiers
source            1..42
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 45
ttgacgcggt tctatctagt tacgcgttaa accaactaga aa                        42

SEQ ID NO: 46      moltype = RNA length = 51
FEATURE           Location/Qualifiers
source            1..51
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 46
gggtcaggag cccccccct gaaccagga taacctcaa agtcgggggg c                51

SEQ ID NO: 47      moltype = RNA length = 60
FEATURE           Location/Qualifiers
source            1..60
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 47
gtcaggggtca ggagcccccc cctgaaccc aggataaccc tcaaagtcgg ggggcaaccc    60

SEQ ID NO: 48      moltype = RNA length = 56
FEATURE           Location/Qualifiers
source            1..56
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 48
gggtcaggag cccccccct gaaccagga aaacctcaa agtcgggggg caaccc          56

SEQ ID NO: 49      moltype = RNA length = 56
FEATURE           Location/Qualifiers
source            1..56
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 49
gggtcaggag cccccccct gcaccagga aaacctcaa agtcgggggg caaccc          56

```

-continued

SEQ ID NO: 50	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 50		
taattttctac taagtgtaga t		21
SEQ ID NO: 51	moltype = RNA length = 42	
FEATURE	Location/Qualifiers	
source	1..42	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 51		
ttgacgcgggt tctatctagt tacgcgttaa accaactaga aa		42
SEQ ID NO: 52	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 52		
taagtctcca tagaatggag g		21
SEQ ID NO: 53	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 53		
taattttctac tcttgtagat		20
SEQ ID NO: 54	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 54		
gggccctgaa gaagggccc		19
SEQ ID NO: 55	moltype = RNA length = 63	
FEATURE	Location/Qualifiers	
source	1..63	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 55		
ggaggtcagg gtcaggagcc cccccctgaa cccaggataa ccctcaaagt cgggggggcaa	60	
ccc	63	
SEQ ID NO: 56	moltype = RNA length = 55	
FEATURE	Location/Qualifiers	
source	1..55	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 56		
gggtcaggag cccccccctg aaccctcaggat aaccctcaaa gtcgggggggc aacc	55	
SEQ ID NO: 57	moltype = RNA length = 34	
FEATURE	Location/Qualifiers	
source	1..34	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 57		
ggccaacatg aggatcaccc atgtctgcag ggcc	34	
SEQ ID NO: 58	moltype = RNA length = 29	
FEATURE	Location/Qualifiers	
source	1..29	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 58		
cgggagcaga cgatatggcg tcgctccg	29	
SEQ ID NO: 59	moltype = RNA length = 26	
FEATURE	Location/Qualifiers	
source	1..26	

-continued

	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 59		
ccaaaggctc ttctcagagc caccca		26
SEQ ID NO: 60	moltype = RNA length = 31	
FEATURE	Location/Qualifiers	
source	1..31	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 60		
ggccagatct gagcctggga gctctctggc c		31
SEQ ID NO: 61	moltype = RNA length = 36	
FEATURE	Location/Qualifiers	
source	1..36	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 61		
atataacctt caccattagg ttcaaataat ggtaat		36
SEQ ID NO: 62	moltype = RNA length = 42	
FEATURE	Location/Qualifiers	
source	1..42	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 62		
cgctcccg gttcaagcga ttctctgcc tcagcctccc ga		42
SEQ ID NO: 63	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 63		
YSPTSPS		7
SEQ ID NO: 64	moltype = DNA length = 360	
FEATURE	Location/Qualifiers	
source	1..360	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 64		
gttaatcgaa ttgaactgaa aggtattgat tttaaggagg acggtaacat actggggcac	60	
aagttggagt acaactttaa cagccataat gtgtatatta ccgctgataa gcagaaaaat	120	
gggataaagg ccaactttaa gatccgacat aatgtcgaag atggtagtgt tcaactggct	180	
gatcattacc aacaaaatac gcccatcgga gatggacctg tactcttgcc tgacaatcat	240	
tatctctcca cgcaatcaaa gctttccaag gacccaaacg aaaagagaga tcacatggct	300	
cttctggaat ttgtgactgc cgcaggcatc actctcggta tggatgagct gtacaagtga	360	

What is claimed is:

1. A hybrid RNA molecule comprising
 - a) a targeting region that comprises a nucleotide sequence that is complementary to a target sequence of a target RNA; and
 - b) a donor sequence to said targeting region.
2. The hybrid RNA molecule of claim 1, further comprising one or more of:
 - a) a binding region that binds to an RNA binding or modifying polypeptide;
 - b) a protein-recruiting region;
 - c) a stabilization region.
3. The hybrid RNA molecule of claim 2, wherein the hybrid RNA molecule comprises a binding region that binds to an RNA binding or modifying polypeptide, and wherein the RNA binding or modifying polypeptide is an RNA-targeting CRISPR-Cas effector polypeptide.
4. The hybrid RNA molecule of claim 3, wherein the RNA-targeting CRISPR-Cas effector polypeptide is a Type II, a Type III, or a Type VI CRISPR-Cas effector polypeptide.

5. The hybrid RNA molecule of claim 4, wherein the Type II CRISPR-Cas effector polypeptide is a Cas9 polypeptide.

6. The hybrid RNA molecule of claim 4, wherein the Type VI CRISPR-Cas effector polypeptide is a Cas13a polypeptide, a Cas13b polypeptide, a Cas13c polypeptide, a Cas13d polypeptide, a Cas13e polypeptide, a Cas13f polypeptide, a Cas13X polypeptide, or a Cas13Y polypeptide.

7. The hybrid molecule of claim 4, wherein the Type III RNA-targeting CRISPR-Cas effector polypeptide is a Cas7-11 polypeptide.

8. The hybrid RNA molecule of claim 2, wherein the hybrid RNA molecule comprises a protein-recruiting region.

9. The hybrid RNA molecule of claim 8, wherein the protein-recruiting region comprises a nucleotide sequence that binds one or more spliceosome polypeptides.

10. The hybrid RNA molecule of claim 2, wherein the hybrid RNA molecule comprises a stabilization region.

11. The hybrid RNA molecule of claim 10, wherein the stabilization region is capable of forming a secondary structure.

12. The hybrid RNA molecule of claim **11**, wherein the secondary structure comprises a pseudoknot, a stem-loop, or a tetraloop.

13. The hybrid RNA molecule of any one of claims **1-12**, wherein the RNA donor comprises one or more of:

- a) a coding region comprising a nucleotide sequence encoding all or a portion of a polypeptide;
- b) an untranslated region (UTR);
- c) a bar code;
- d) an internal ribosomal entry site (IRES);
- e) a poly(pyrimidine) tract;
- f) an RNA response element;
- g) an intron; and
- h) an aptamer.

14. The hybrid RNA molecule of any one of claims **1-13**, further comprising a splice function region comprising one or more of:

- a) a splice modifying sequence;
- b) a splice exclusion sequence;
- c) an intron;
- d) a branch point;
- e) an intronic splice enhancer;
- f) an intronic splice silencer; and
- g) a poly(pyrimidine tract).

15. The hybrid DNA molecule of claim **14**, wherein the splice modifying sequence comprises a splice acceptor, a splice donor, a branchpoint, or a polypyrimidine tract.

16. The hybrid RNA molecule of claim **13**, wherein the RNA donor comprises a nucleotide sequence encoding all or a portion of a polypeptide.

17. The hybrid RNA molecule of claim **16**, wherein the RNA donor comprises a nucleotide sequence encoding a portion of a non-pathological form of a disease-associated polypeptide, wherein the disease-associated polypeptide comprises an amino acid expansion.

18. The hybrid RNA molecule of claim **17**, wherein the disease-associated polypeptide comprises a poly(Gln) repeat region, and wherein the non-pathological form of the polypeptide comprises a (Gln) x repeat region, wherein x is less than the number of repeats associated with a disease.

19. The hybrid RNA molecule of claim **17**, wherein the disease-associated polypeptide comprises a poly(Glu) repeat region, and wherein the non-pathological form of the polypeptide comprises a (Glu) x repeat region, wherein x is less than the number of repeats associated with a disease.

20. The hybrid RNA molecule of claim **13**, wherein the RNA donor comprises an intron comprising a non-pathological number of repeats of GAA.

21. The hybrid RNA molecule of claim **13**, wherein the RNA donor comprises a 5'-UTR comprising a non-pathological number of repeats of CCG, CCG, or GGC.

22. The hybrid RNA molecule of claim **13**, wherein the RNA donor comprises a 3'-UTR comprising a non-pathological number of repeats of CTG.

23. The hybrid RNA molecule of any one of claims **1-22**, comprising a SINE-derived nuclear RNA Localization (SIR-LOIN) motif.

24. The hybrid RNA molecule of any one of claims **1-23**, further comprising a second targeting region that comprises a nucleotide sequence that is complementary to a second target sequence of the target RNA and that interferes with cis splicing of the target RNA.

25. The hybrid RNA molecule of any one of claims **1-24**, wherein the RNA donor has a length of from 2 nucleotides to 10,000 nucleotides.

26. A DNA molecule comprising a nucleotide sequence encoding the hybrid RNA molecule of any one of claims **1-25**.

27. The DNA molecule of claim **26**, wherein the nucleotide sequence is operably linked to a transcriptional control element.

28. The DNA molecule of claim **27**, wherein the transcriptional control element comprises a promoter, and wherein the promoter is one or more of: a constitutive promoter, an inducible promoter, a cell type-specific promoter, and a tissue-specific promoter.

29. A recombinant expression vector comprising the DNA molecule of any one of claims **26-28**.

30. A composition comprising:

- a) the hybrid RNA molecule of any one of claims **1-25**; and
- b) one or more of a nanoparticle, a lipid, a buffer, and a nuclease inhibitor.

31. A composition comprising:

- al) an RNA-targeting CRISPR-Cas effector polypeptide; and
- b1) a hybrid RNA molecule of any one of claims **1-25**; or
- a2) a fusion polypeptide comprising:
 - i) an RNA-targeting CRISPR-Cas effector polypeptide; and
 - ii) one or more heterologous polypeptides; and
 - b2) a hybrid RNA molecule of any one of claims **1-25**.

32. The composition of claim **31**, wherein the RNA-targeting CRISPR-Cas effector polypeptide is a Type II, a Type III, or a Type VI CRISPR-Cas effector polypeptide.

33. The composition of claim **32**, wherein the Type II CRISPR-Cas effector polypeptide is a Cas9 polypeptide.

34. The composition of claim **32**, wherein the Type III CRISPR-Cas effector polypeptide is a Cas7-11 polypeptide.

35. The composition of claim **32**, wherein the Type VI CRISPR-Cas effector polypeptide is a Cas13a polypeptide, a Cas13b polypeptide, a Cas13c polypeptide, a Cas13d polypeptide, a Cas13e polypeptide, a Cas13f polypeptide, a Cas13X polypeptide, or a Cas13Y polypeptide.

36. The composition of claim **31**, wherein the composition comprises:

- a) a fusion polypeptide comprising:
 - i) an RNA-targeting CRISPR-Cas effector polypeptide; and
 - ii) one or more heterologous polypeptides; and
 - b) a hybrid RNA molecule of any one of claims **1-25**.

37. The composition of claim **36**, wherein at least one of the one or more heterologous polypeptides comprises an RNA Polymerase II-binding polypeptide.

38. The composition of claim **37**, wherein the RNA Polymerase II-binding polypeptide comprises a C-terminal interacting domain (CID).

39. The composition of claim **36**, wherein at least one of the one or more heterologous polypeptides comprises a polypeptide that provides for enhanced localization to an SR protein granule.

40. The composition of claim **39**, wherein the polypeptide that provides for enhanced localization to an SR protein granule comprise an RS domain.

41. The composition of any one of claims **32-40**, wherein the RNA-targeting CRISPR-Cas effector polypeptide exhib-

its reduced enzymatic activity compared to the enzymatic activity of a wild-type RNA-targeting Cas effector polypeptide.

42. The composition of any one of claims 32-41, wherein the RNA-targeting CRISPR-Cas effector polypeptide comprises a first higher eukaryotes and prokaryotes nucleotide-binding (HEPN) domain and a second HEPN domain.

43. The composition of claim 42, wherein the RNA-targeting CRISPR-Cas effector polypeptide comprises a mutation in the first HEPN domain and/or the second HEPN domain.

44. The composition of any one of claims 32-41, wherein the RNA-targeting CRISPR-Cas effector polypeptide is a variant Cas7-11 comprising a mutation in a catalytic domain.

45. A method for modifying a target RNA, the method comprising contacting the target RNA with the hybrid RNA molecule of any one of claims 1-25, wherein said contacting results in modification of the target RNA.

46. The method of claim 45, wherein the modification comprises inserting the RNA donor present in the hybrid RNA molecule into the target RNA.

47. The method of claim 45, wherein the modification comprises replacing one or more endogenous nucleotides in the target RNA with the RNA donor present in the hybrid RNA molecule.

48. The method of any one of claims 45-47, wherein the target RNA is in a eukaryotic cell.

49. The method of claim 48, wherein the eukaryotic cell is in vitro.

50. The method of claim 48, wherein the eukaryotic cell is in vivo.

51. The method of any one of claims 48-50, wherein the cell is a mammalian cell.

52. The method of any one of claims 45-51, wherein the target RNA is a pre-mRNA, a circular RNA, a partially spliced RNA, a non-coding RNA, a non-host cell RNA, a regulatory RNA, a coding RNA, a transfer RNA (tRNA), a pre-ribosomal RNA, a ribosomal RNA, a mature RNA with cryptic splice sites, or a long non-coding RNA (lncRNA).

53. A method for modifying a target RNA, the method comprising contacting the target RNA with the composition of any one of claims 30-44.

54. The method of claim 53, wherein the composition comprises an RNA-targeting CRISPR-Cas guide RNA comprising a targeting region that comprises a nucleotide sequence that is complementary to a second target sequence of the target RNA, wherein the RNA-targeting CRISPR-Cas guide RNA inhibits cis-splicing of the target RNA.

55. The method of claim 53 or claim 54, wherein the target RNA is a pre-mRNA, a circular RNA, a partially spliced RNA, a non-coding RNA, a non-host cell RNA, a regulatory RNA, a coding RNA, a transfer RNA (tRNA), a pre-ribosomal RNA, a ribosomal RNA, a mature RNA with cryptic splice sites, or a long non-coding RNA (lncRNA).

56. A composition comprising:

- a) a nucleic acid comprising a nucleotide sequence encoding an RNA-targeting CRISPR-Cas effector polypeptide; and
- b) a hybrid RNA molecule of any one of claims 1-25; or
- a) a nucleic acid comprising a nucleotide sequence encoding a CRISPR-Cas fusion polypeptide comprising:

- i) an RNA-targeting CRISPR-Cas effector polypeptide; and

- ii) one or more heterologous polypeptides; and

- b2) a hybrid RNA molecule of any one of claims 1-25.

57. The composition of claim 56, wherein the nucleic acid is present in a recombinant expression vector.

58. The composition of claim 56 or claim 57, wherein the nucleotide sequence encoding the RNA-targeting CRISPR-Cas effector polypeptide or the nucleotide sequence encoding the CRISPR-Cas fusion polypeptide is operably linked to a transcriptional control element.

59. The composition of claim 58, wherein the transcriptional control element is a regulatable promoter.

60. A composition comprising:

- a) a DNA molecule comprising:

- i) a nucleotide sequence encoding an RNA-targeting CRISPR-Cas effector polypeptide; and

- ii) a nucleotide sequence encoding a hybrid RNA molecule of any one of claims 1-25; or

- b) a DNA molecule comprising:

- i) a nucleotide sequence encoding a fusion polypeptide comprising an RNA-targeting CRISPR-Cas effector polypeptide; and one or more heterologous polypeptides; and

- ii) a nucleotide sequence encoding a hybrid RNA molecule of any one of claims 1-25.

61. The composition of claim 55, wherein one or more of the nucleotide sequences is operably linked to a transcriptional control element.

62. A method for modifying a target RNA in a cell, the method comprising introducing into the cell a composition of any one of claims 56-61.

63. The method of claim 62, wherein the target RNA is in a eukaryotic cell.

64. The method of claim 63, wherein the eukaryotic cell is in vitro.

65. The method of claim 58, wherein the eukaryotic cell is in vivo.

66. The method of any one of claims 62-65, wherein the cell is a mammalian cell.

67. A method of treating a trinucleotide or hexanucleotide expansion disease in an individual, the method comprising introducing into a neuron in the individual a hybrid RNA molecule of any one of claims 1-25, or a DNA molecule comprising a nucleotide sequence encoding the hybrid RNA molecule of any one of claims 1-25, wherein the hybrid RNA molecule provides for generation of an mRNA that does not include a pathological number of trinucleotide or hexanucleotide repeats.

68. The method of claim 67, wherein the trinucleotide or hexanucleotide expansion disease is Huntington Disease, spinal and bulbar muscular atrophy (SBMA), spinocerebellar ataxia (SCA) Type 1, SCA Type 2, SCA Type 3, SCA Type 6, SCA Type 7, SCA Type 8, SCA Type 12, SCA Type 17, Friedreich's ataxia, C90RF72 frontotemporal dementia, Fragile X syndrome, or myotonic dystrophy.

69. The hybrid RNA molecule of claim 16, wherein the RNA donor comprises a nucleotide sequence encoding a fluorescent protein.

70. The hybrid RNA molecule of claim 16, wherein the RNA donor comprises a nucleotide sequence encoding an epitope tag.